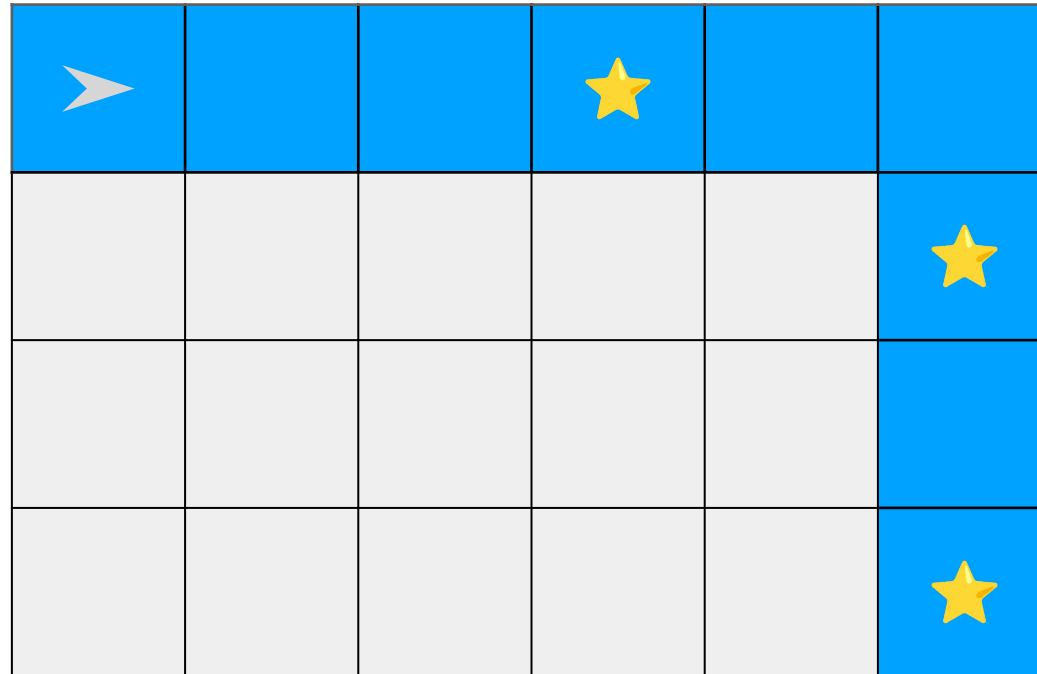


Programming with Robots

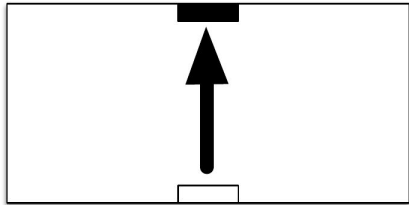
Mål: nå alle stjerner

➤ Robot

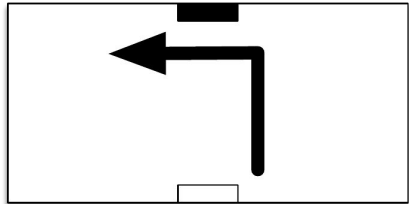
★ Stjerne



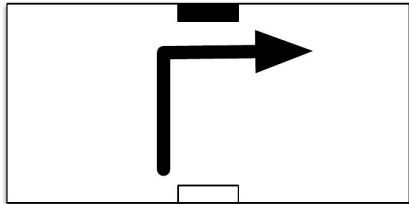
Kommandoer



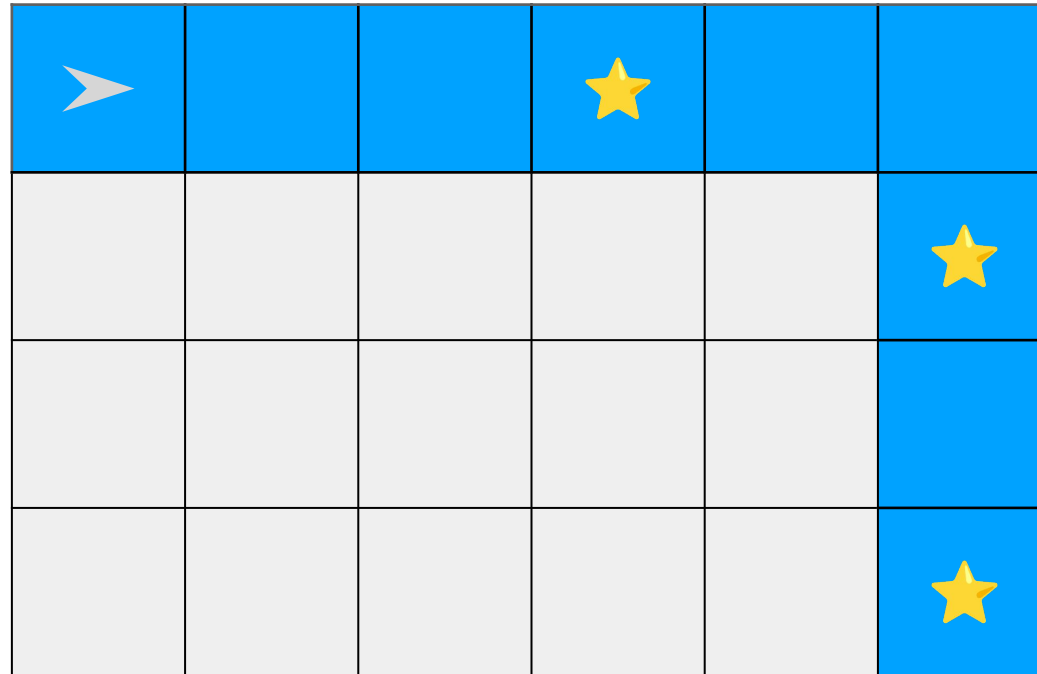
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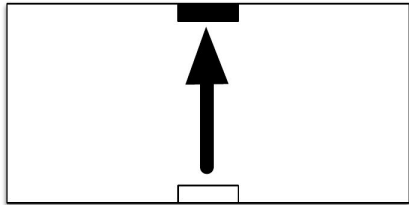
drej til venstre



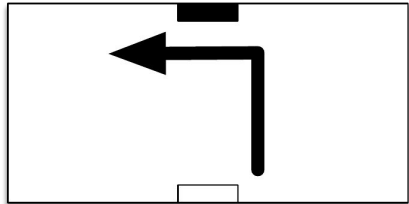
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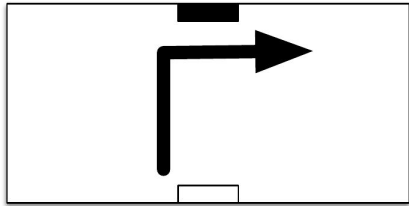
Program



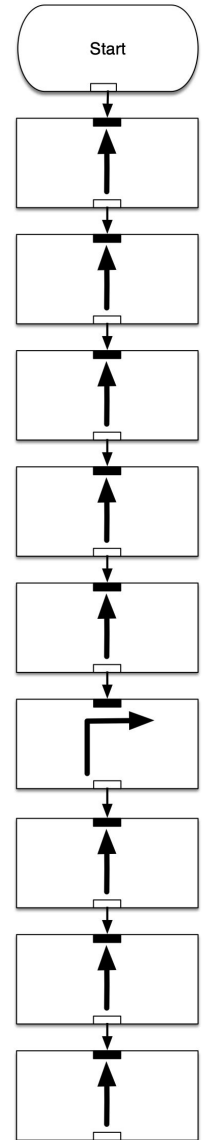
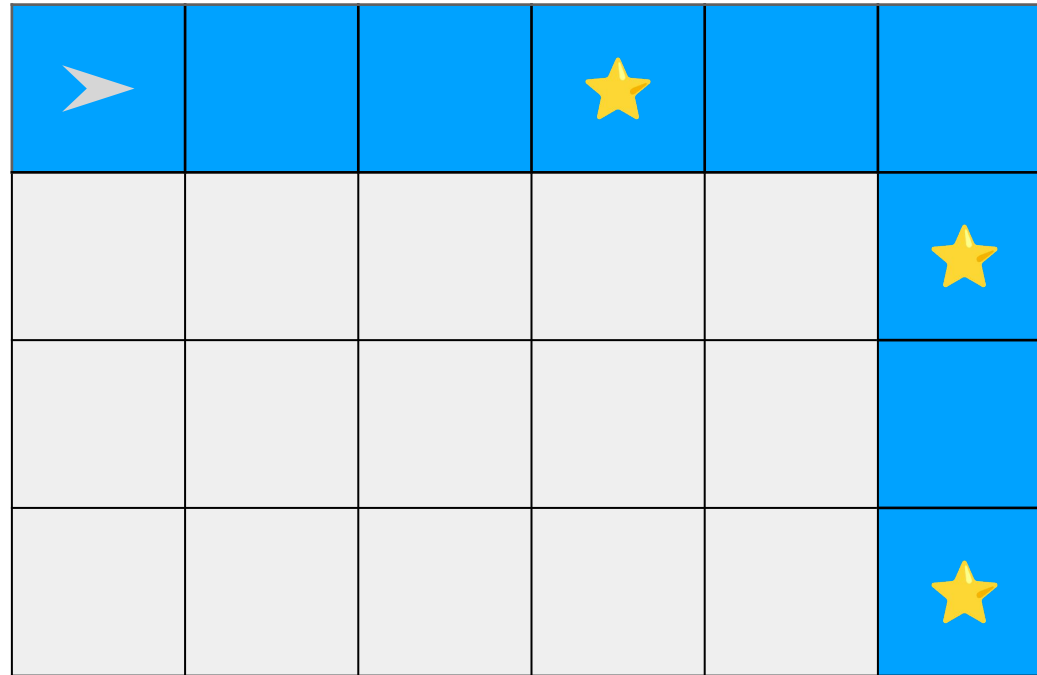
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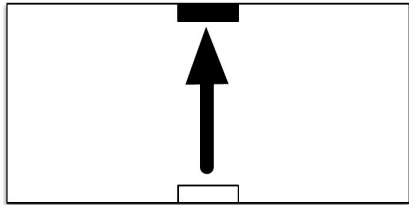
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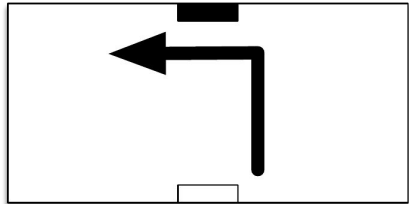
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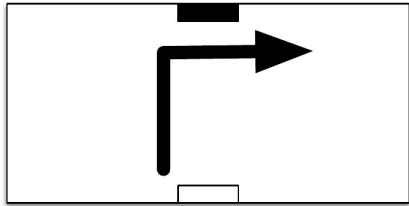
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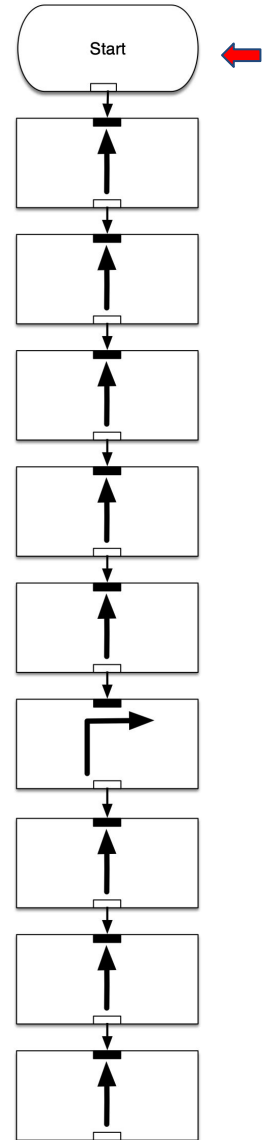
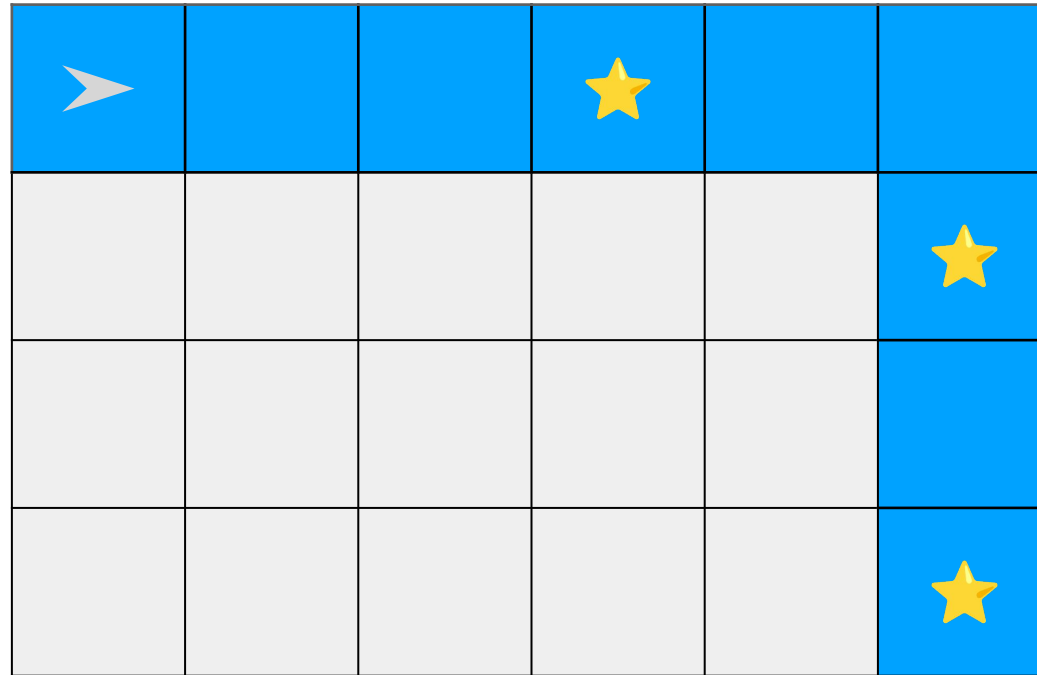
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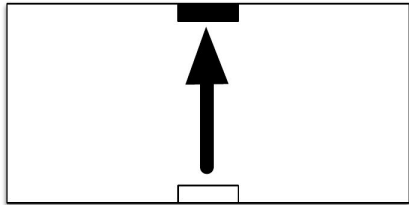
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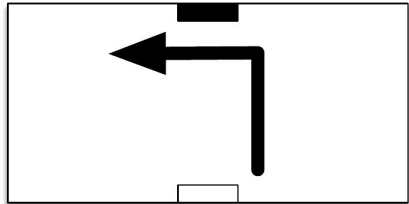
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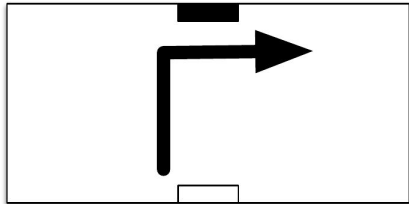
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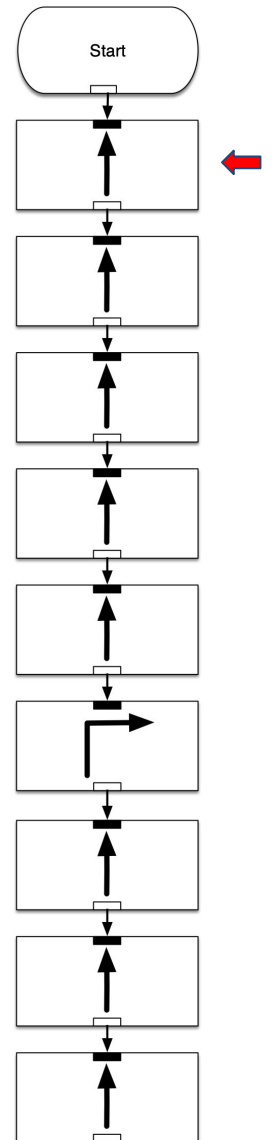
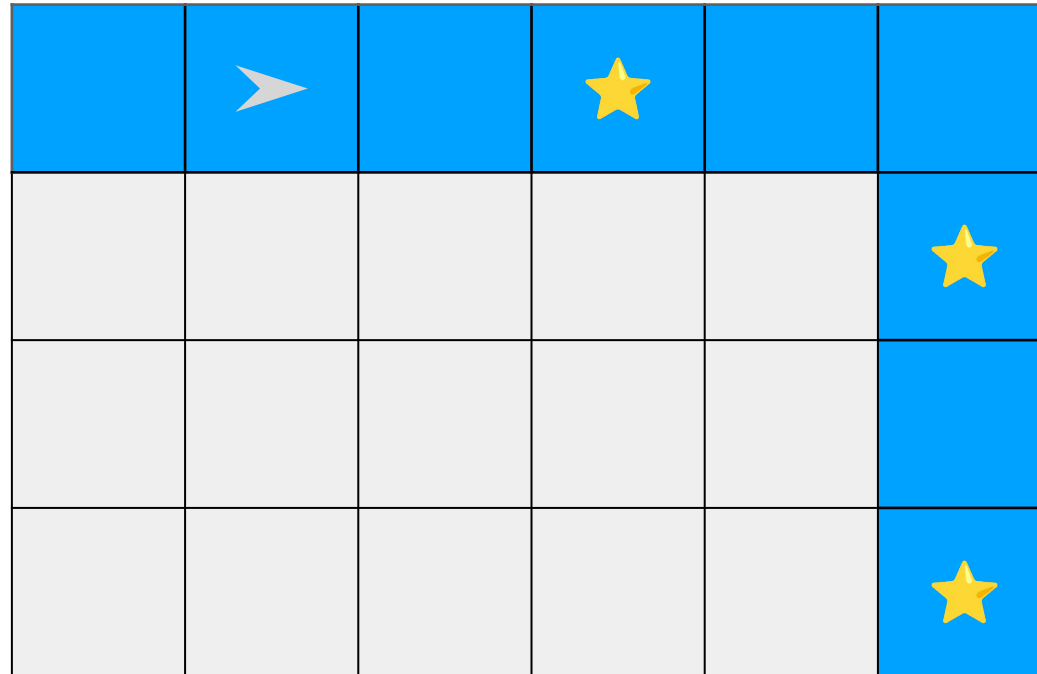
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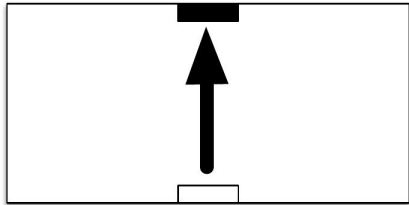
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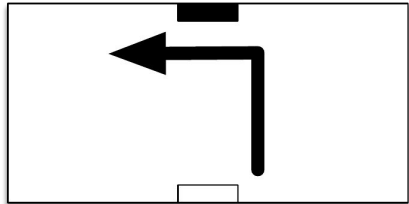
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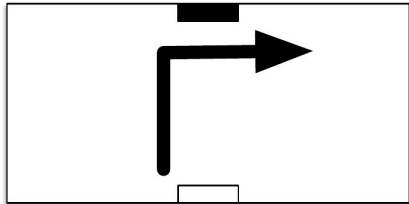
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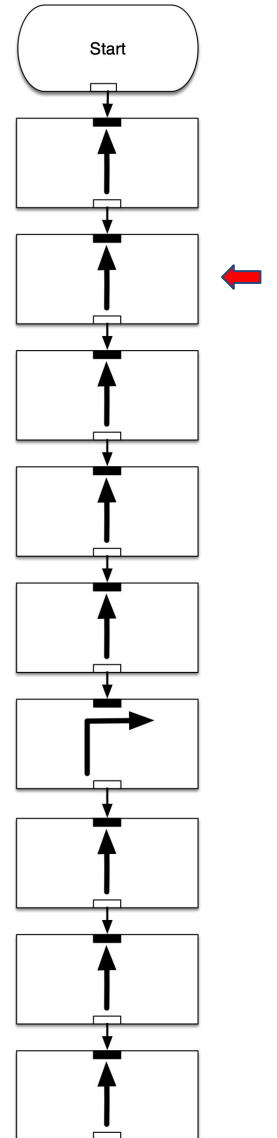
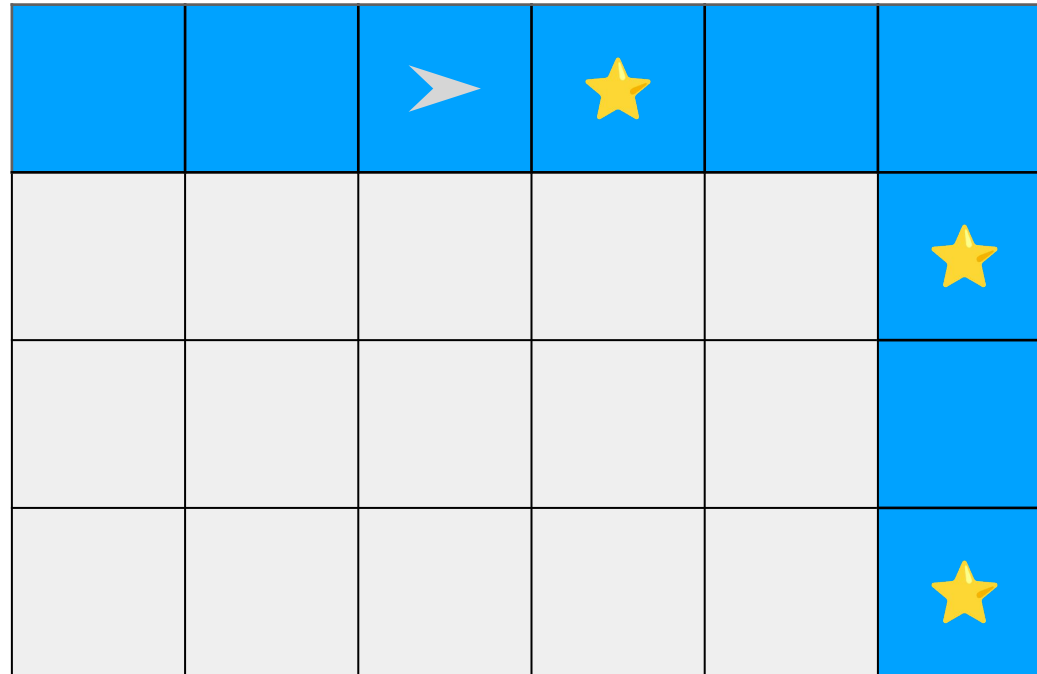
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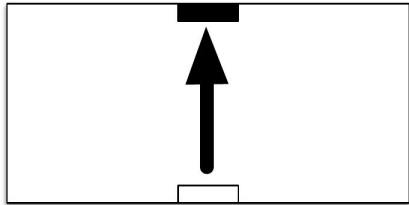
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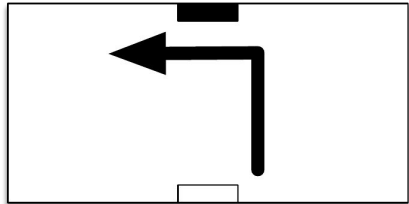
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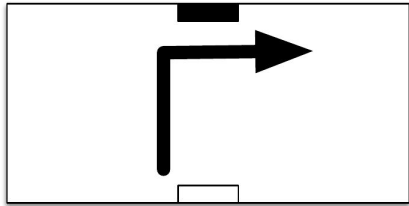
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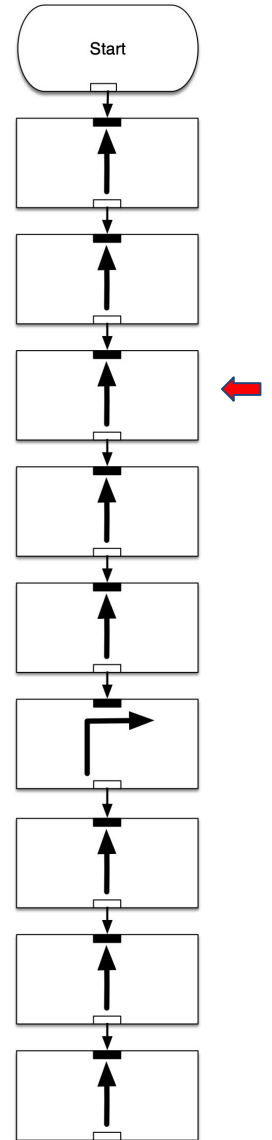
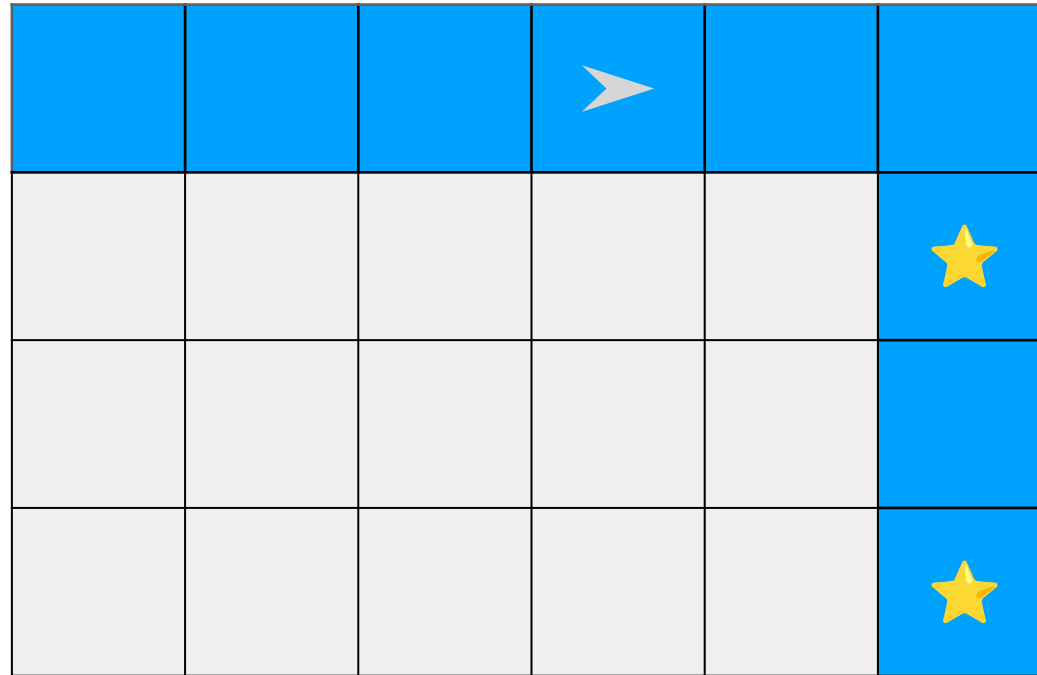
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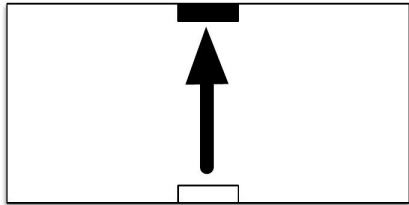
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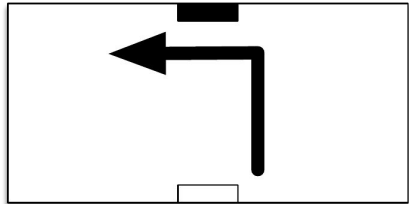
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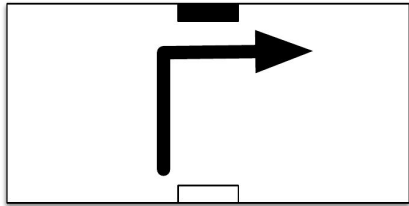
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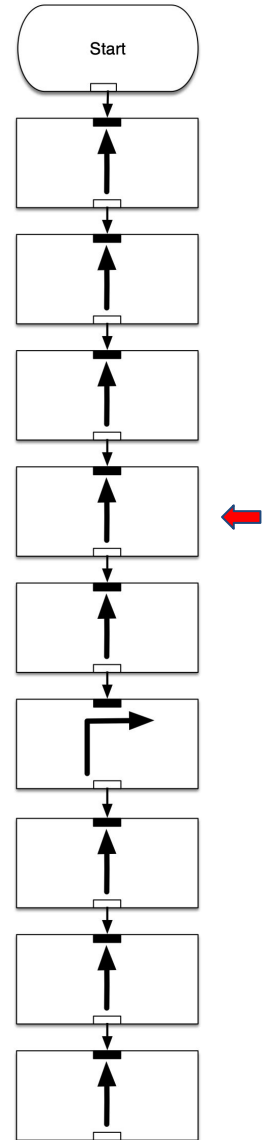
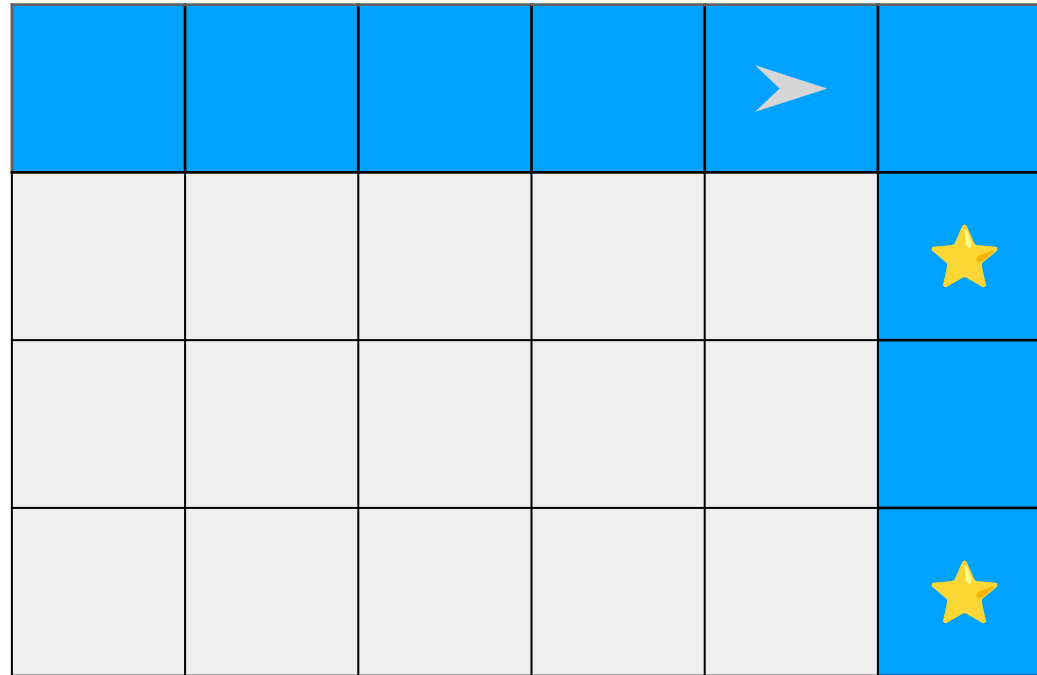
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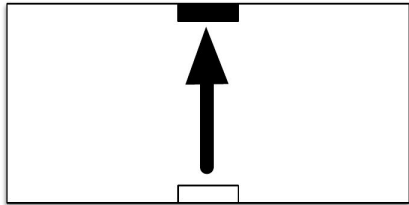
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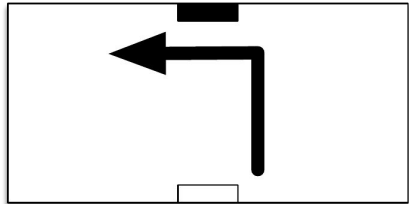
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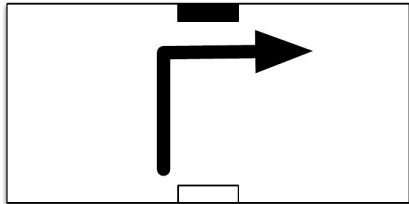
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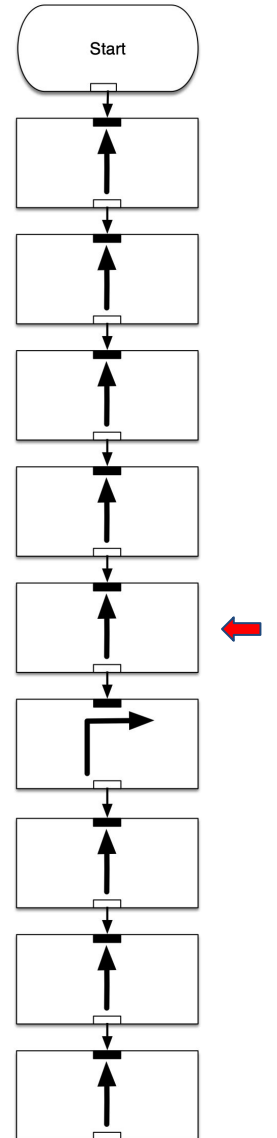
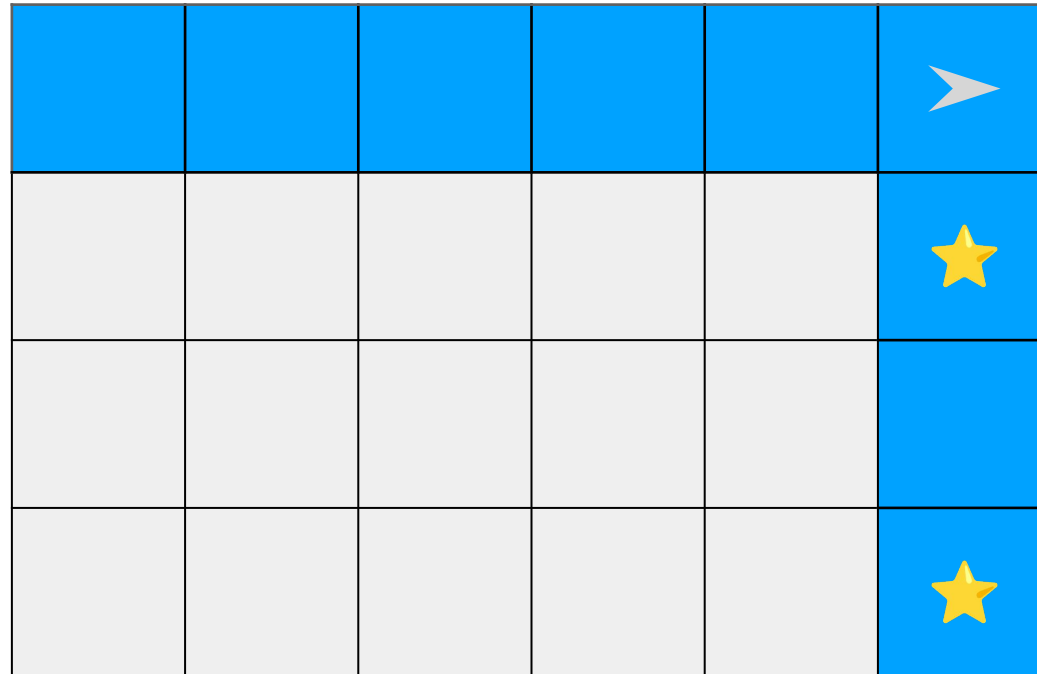
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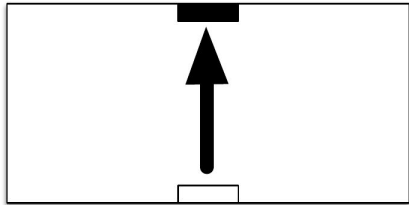
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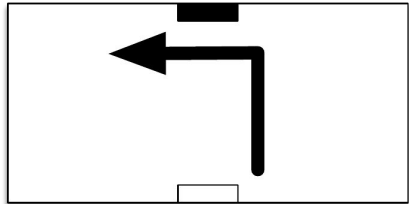
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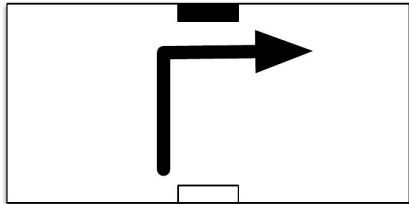
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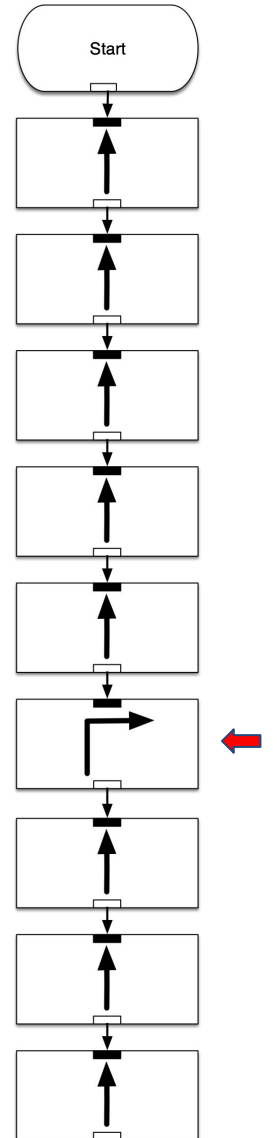
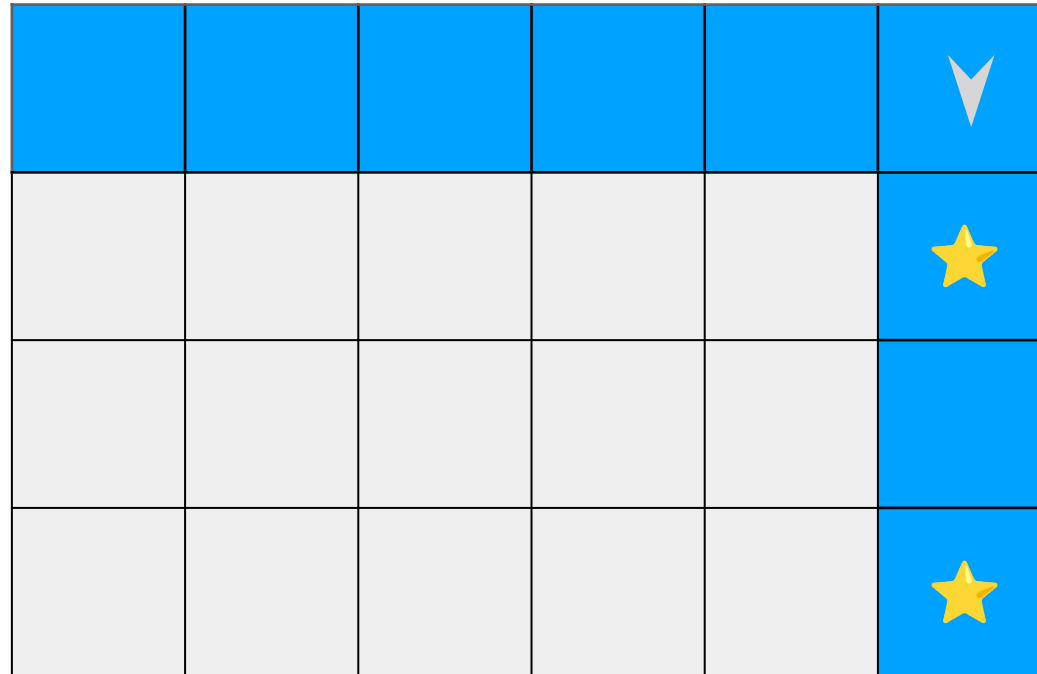
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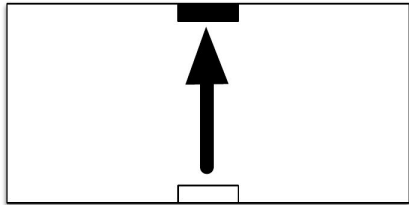
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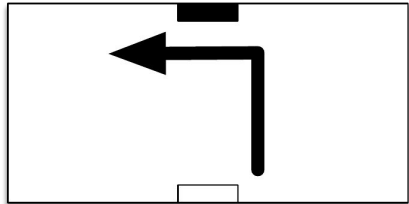
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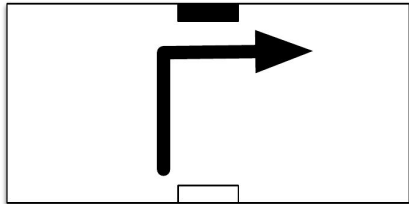
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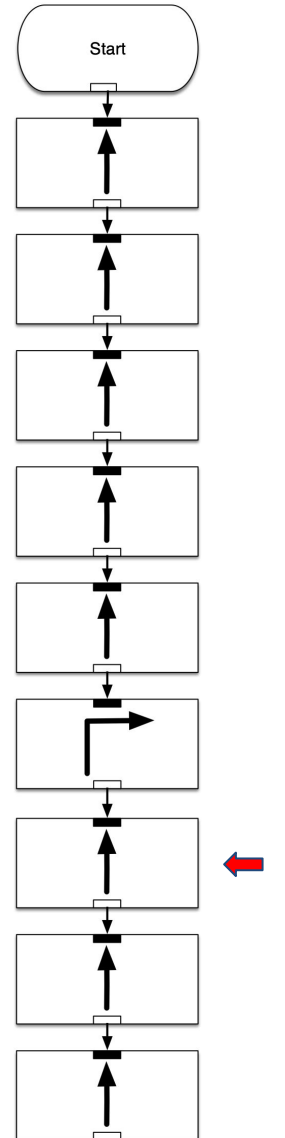
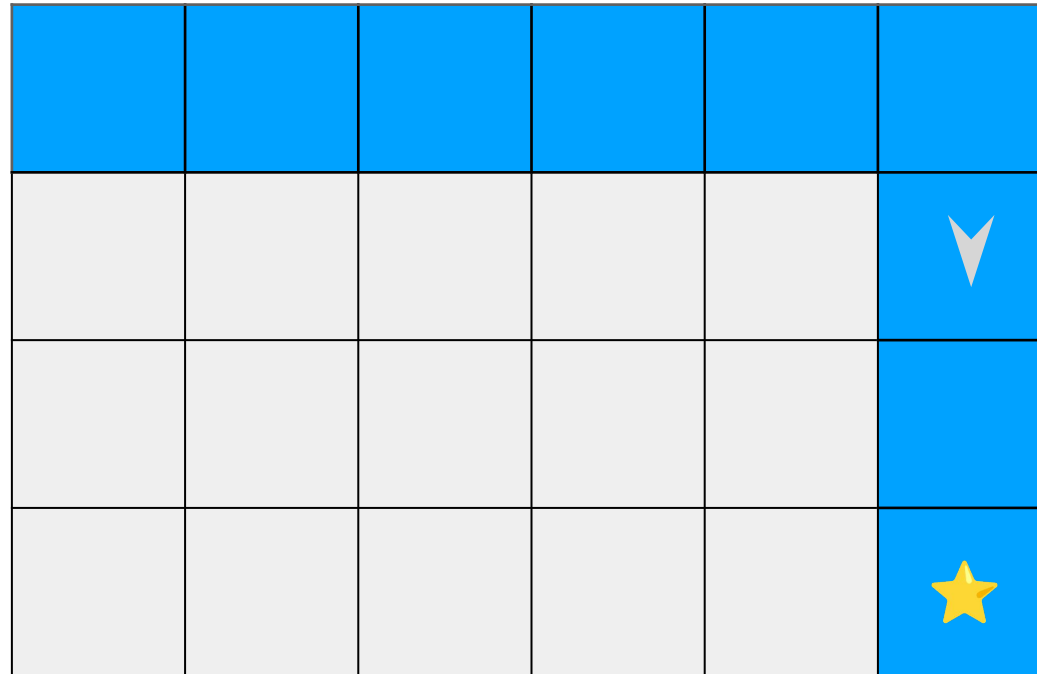
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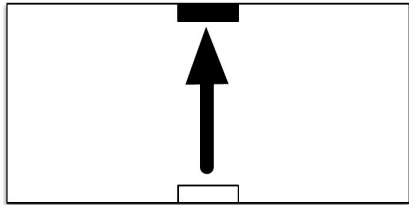
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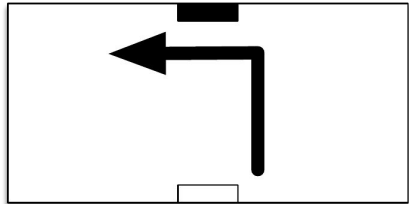
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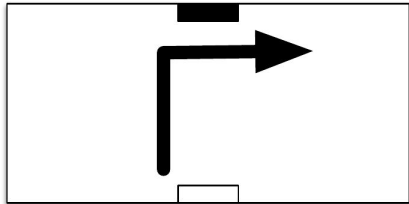
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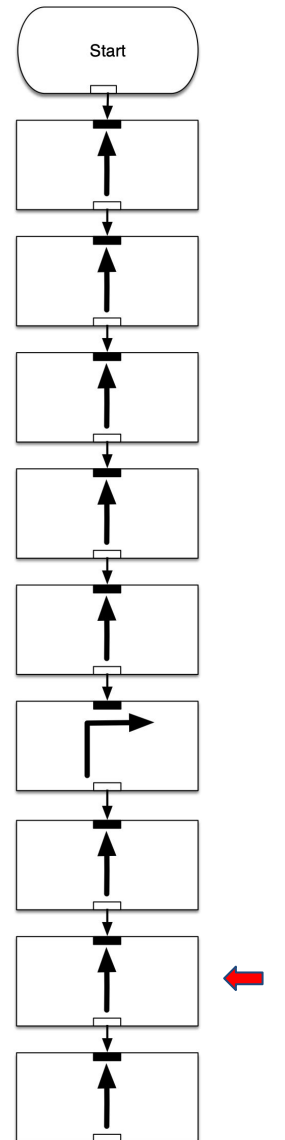
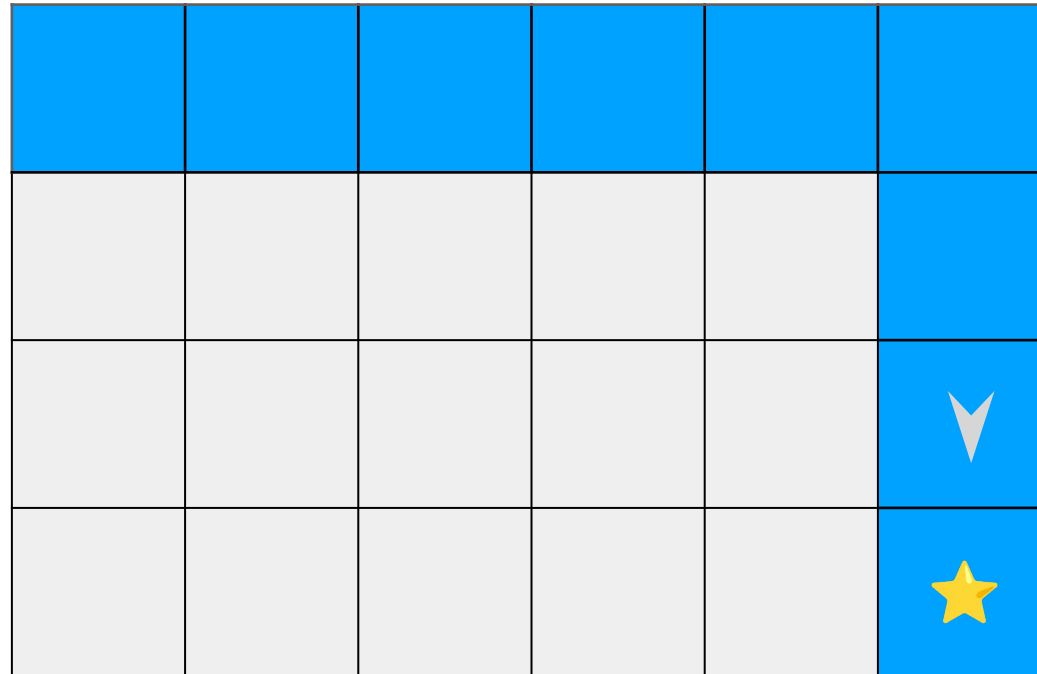
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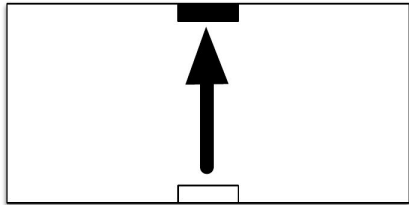
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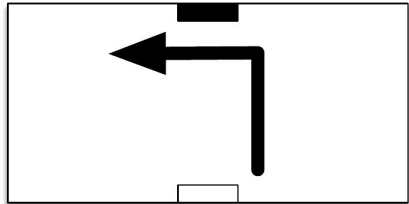
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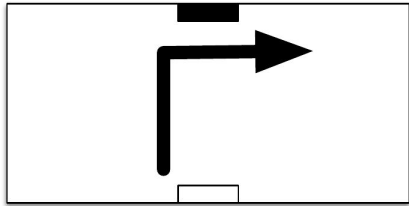
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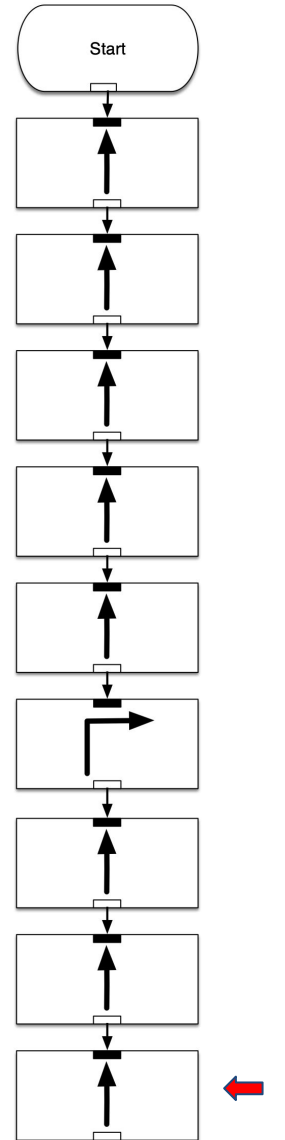
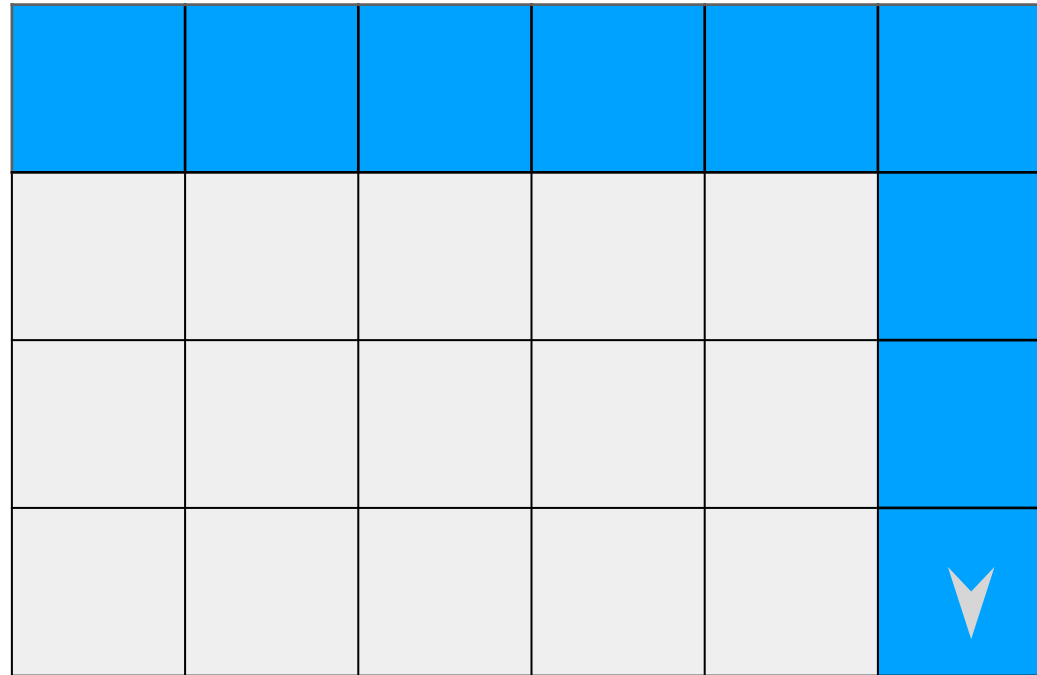
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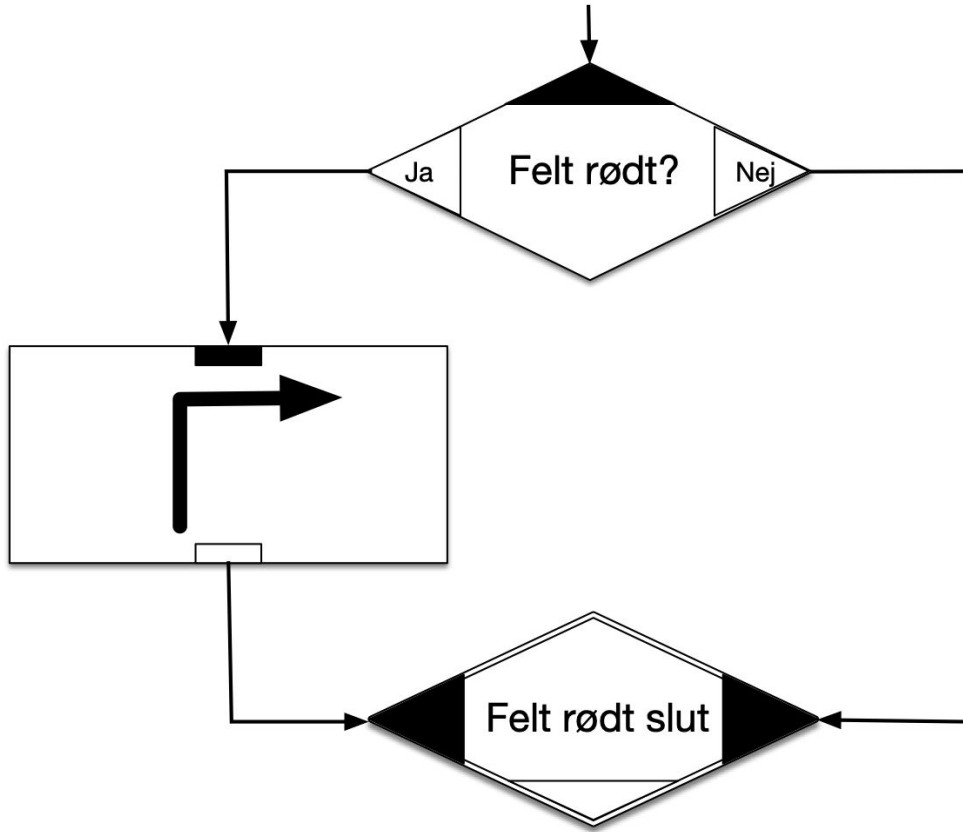
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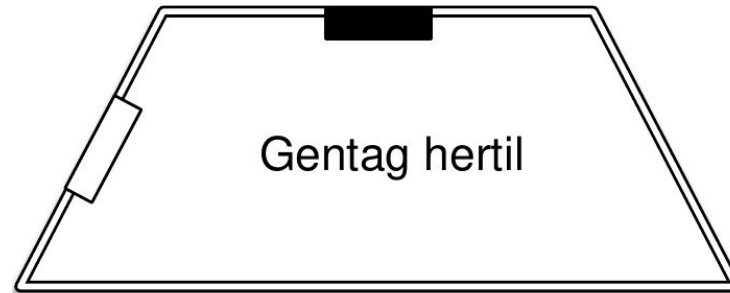
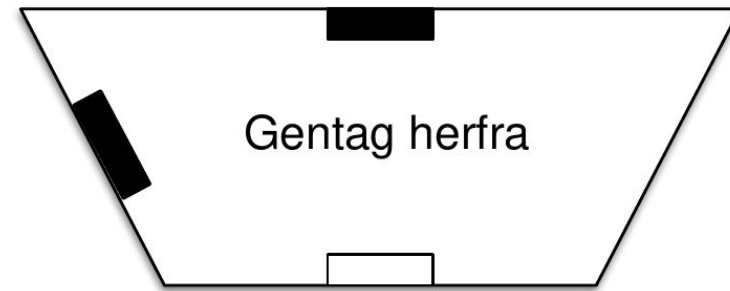
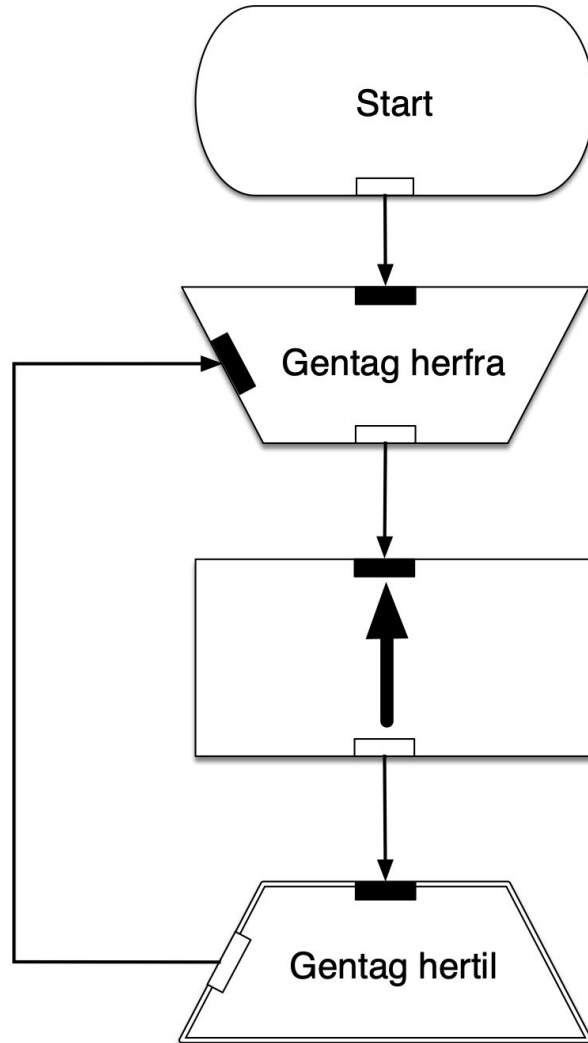
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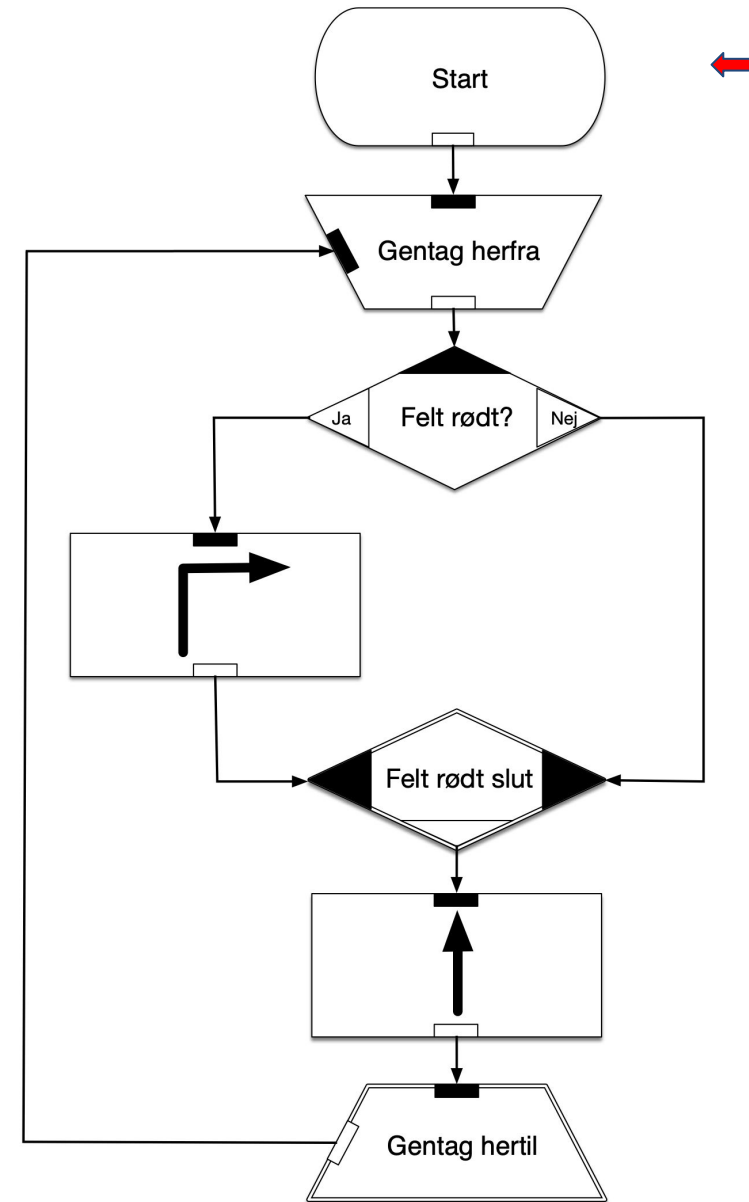
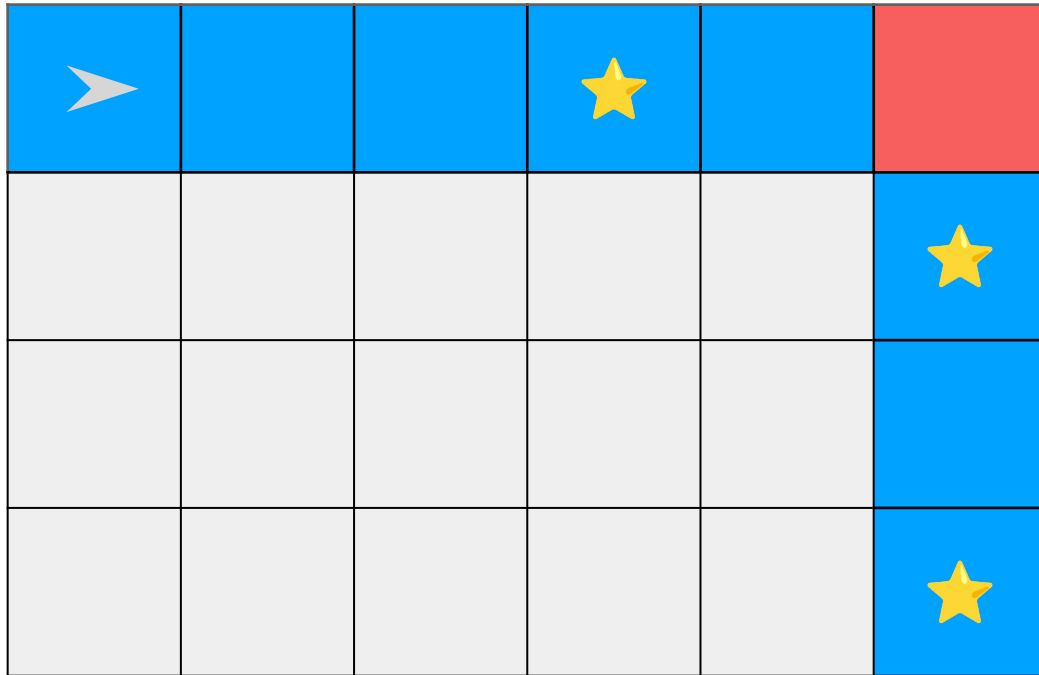
Betingelser



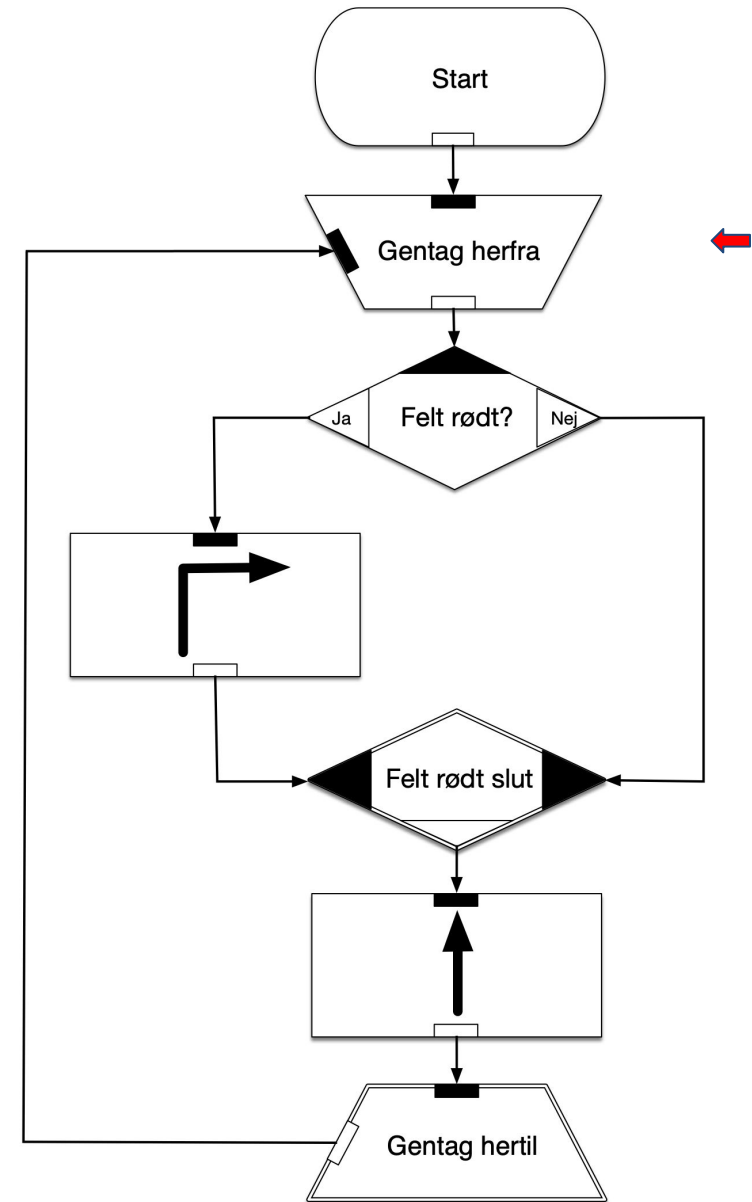
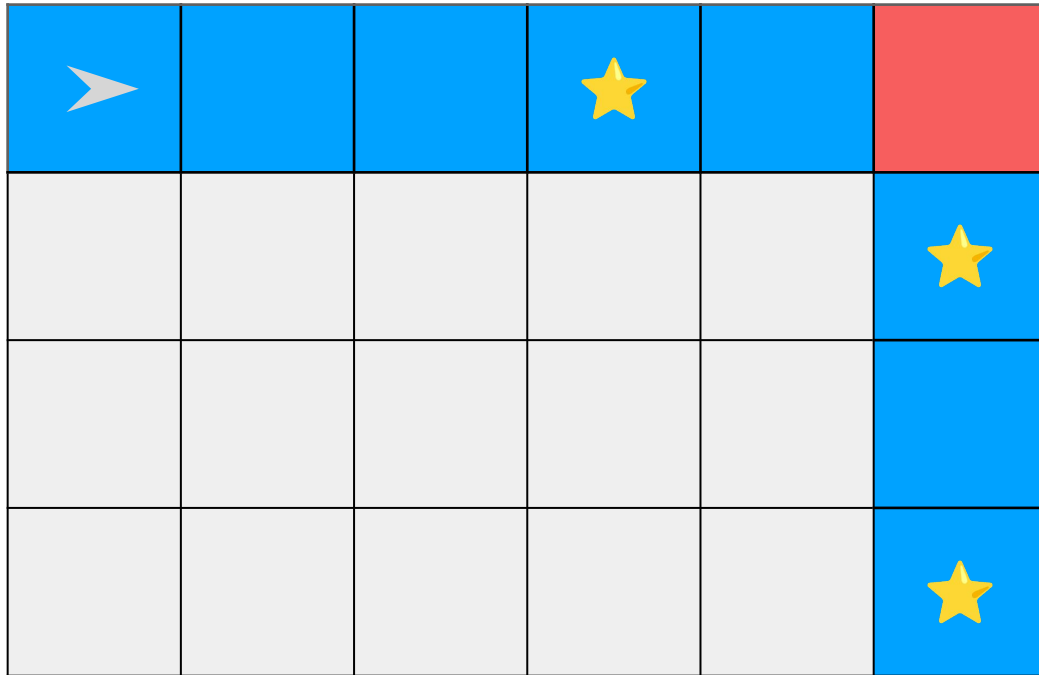
Løkker



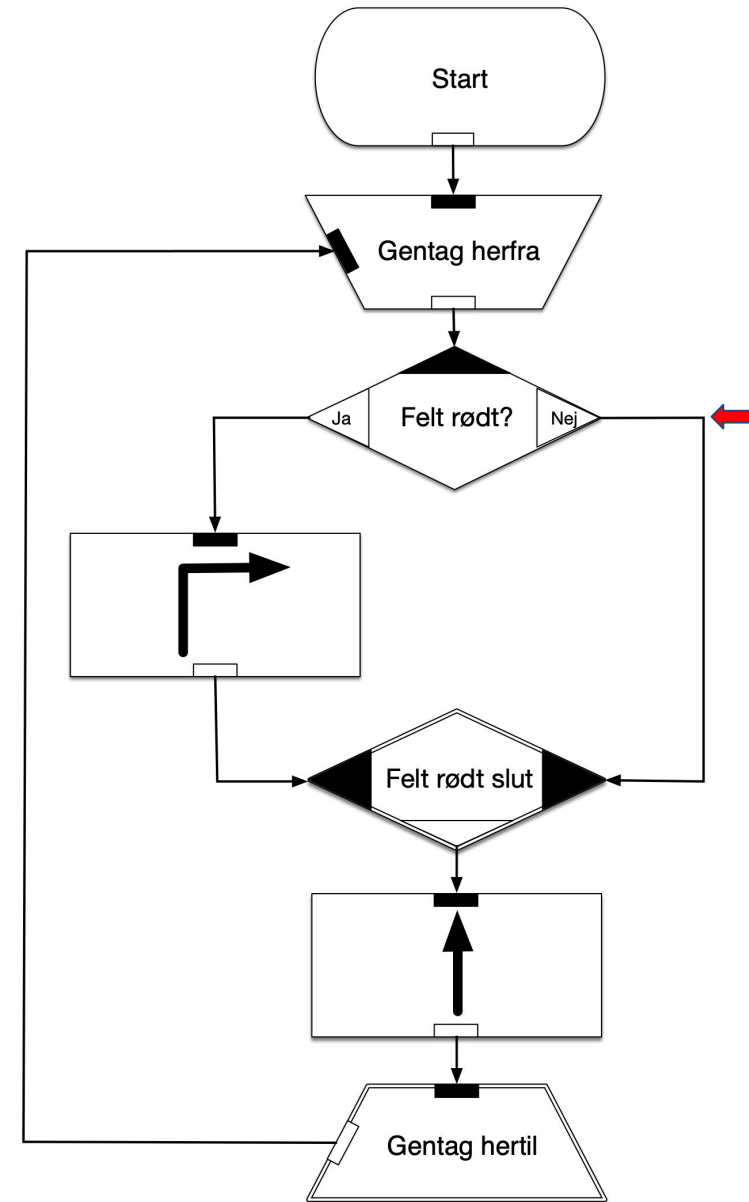
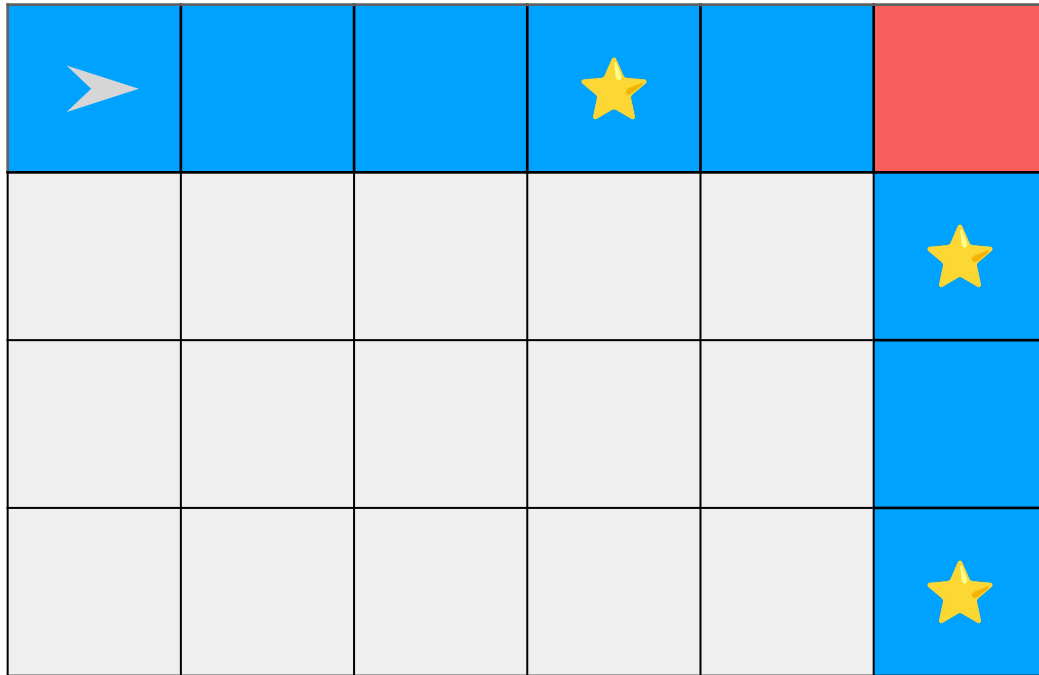
Program med løkker



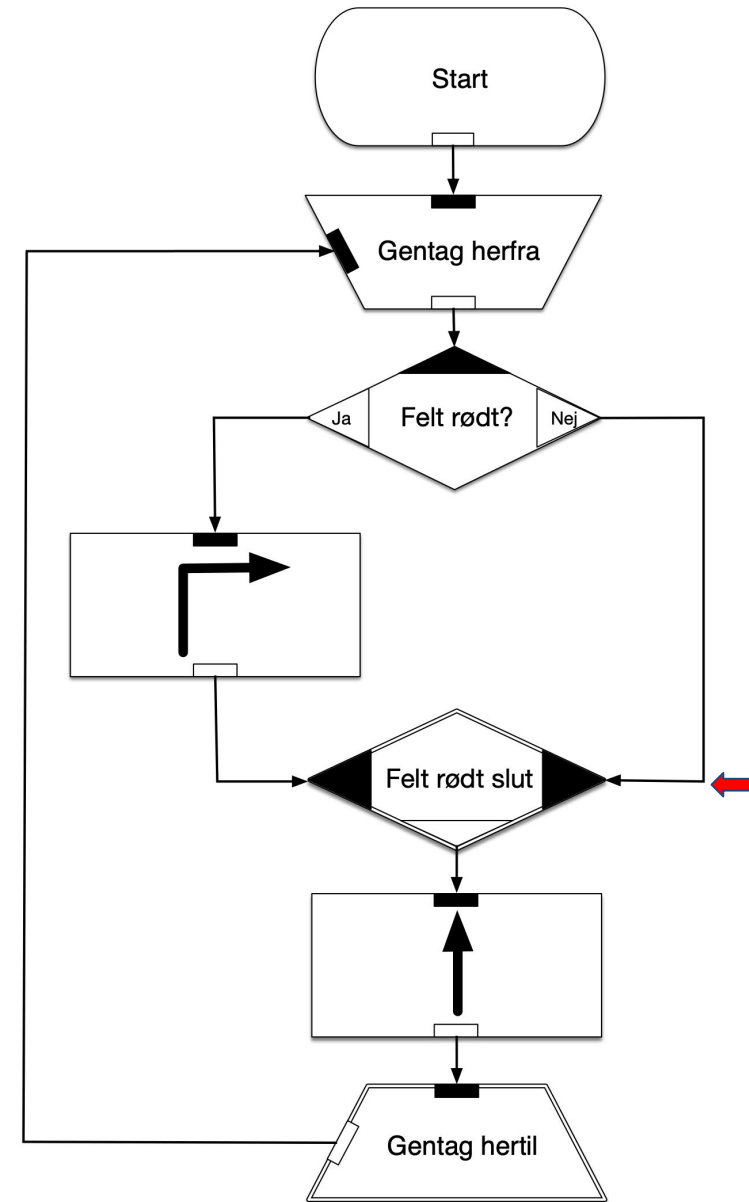
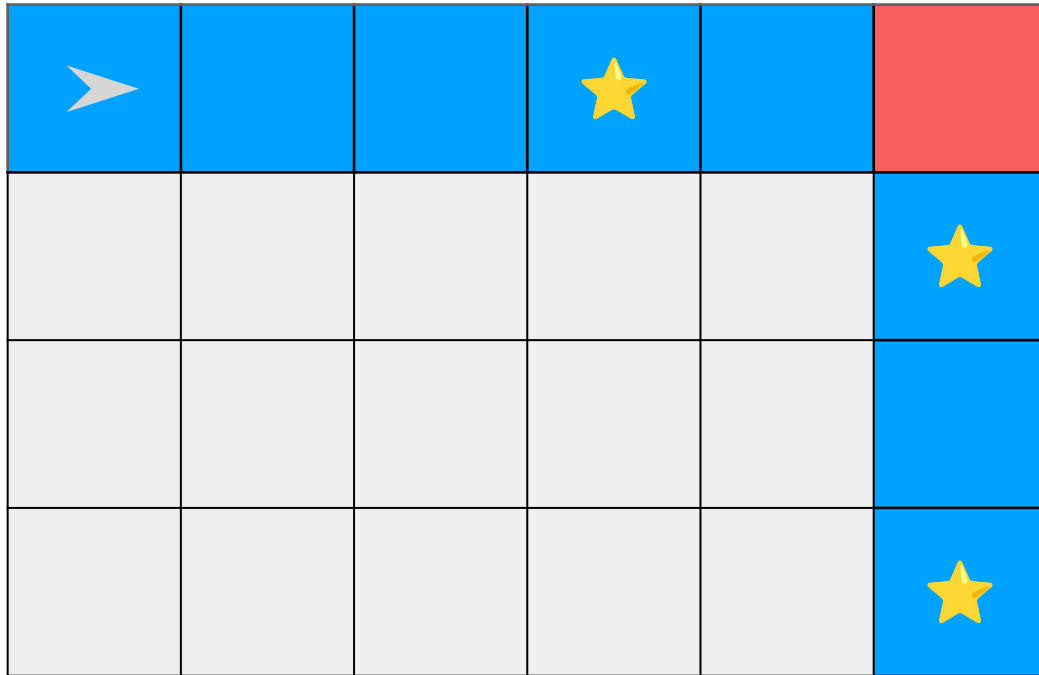
Program med løkker



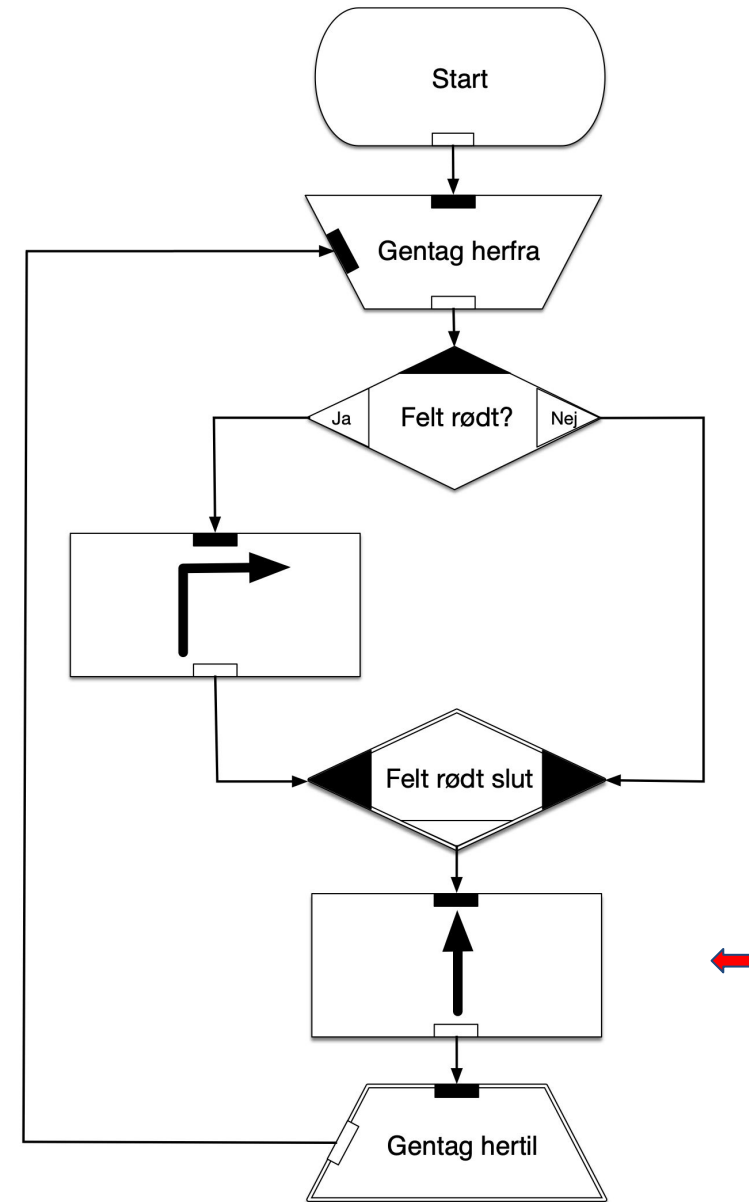
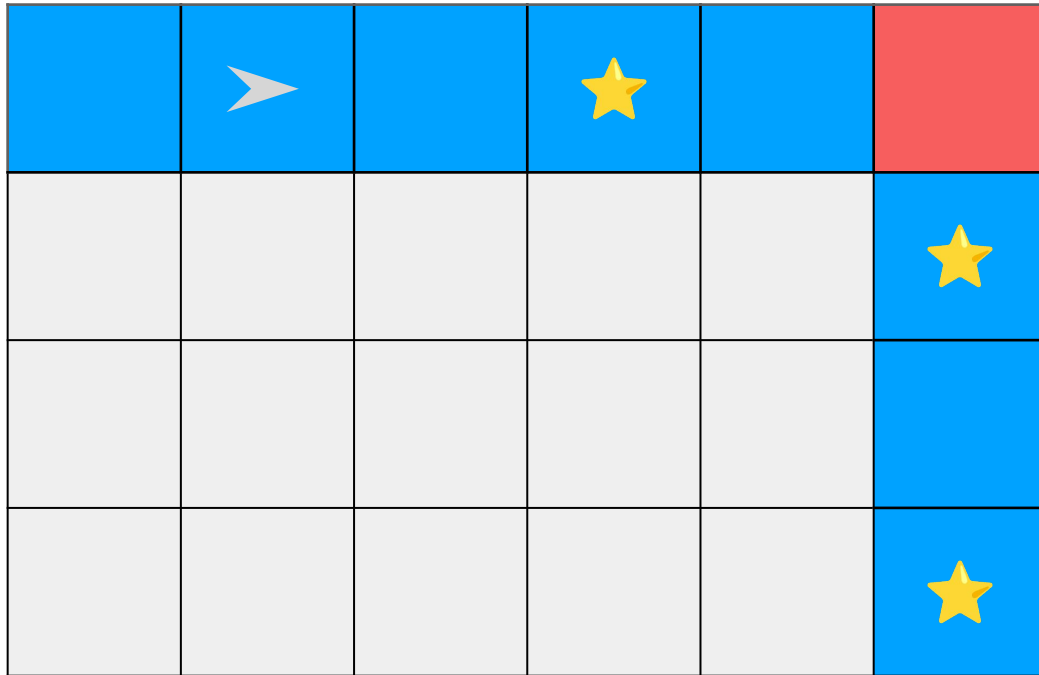
Program med løkker



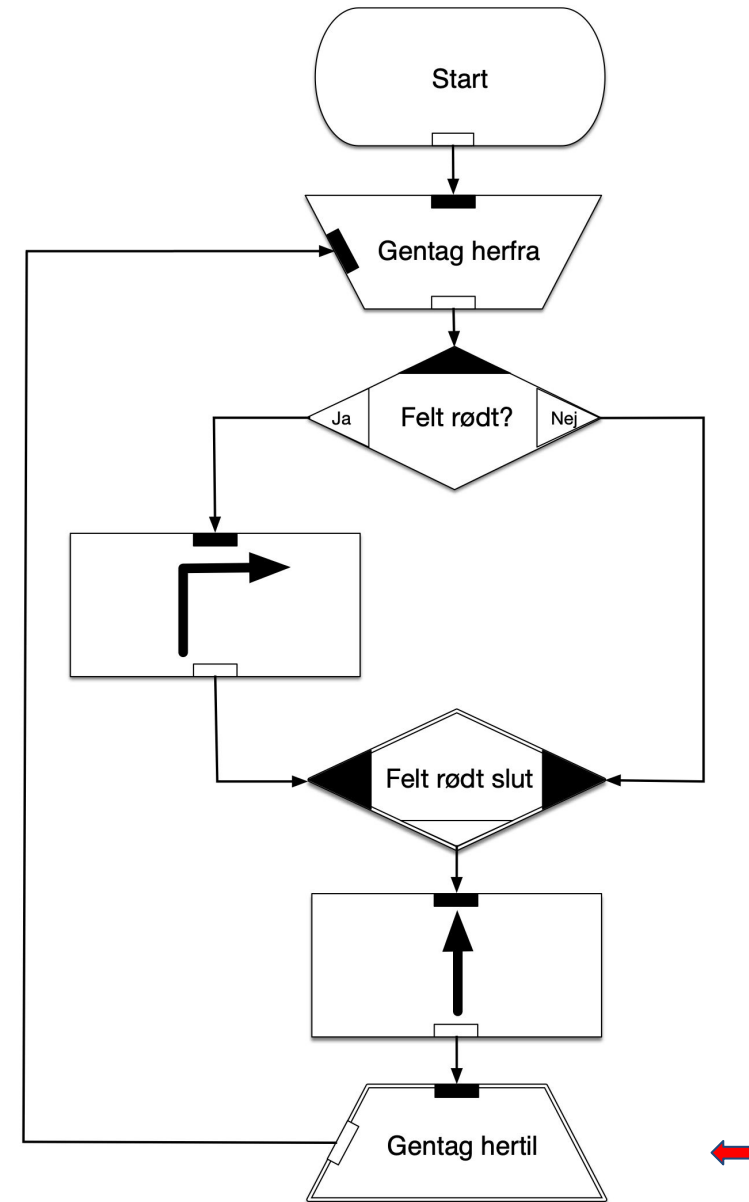
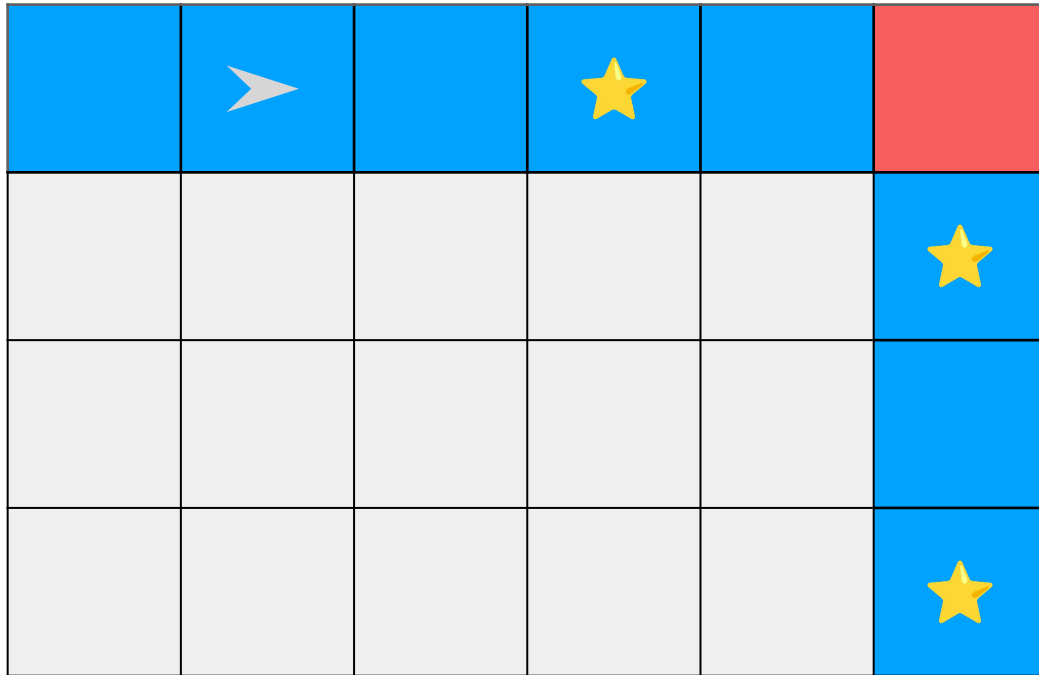
Program med løkker



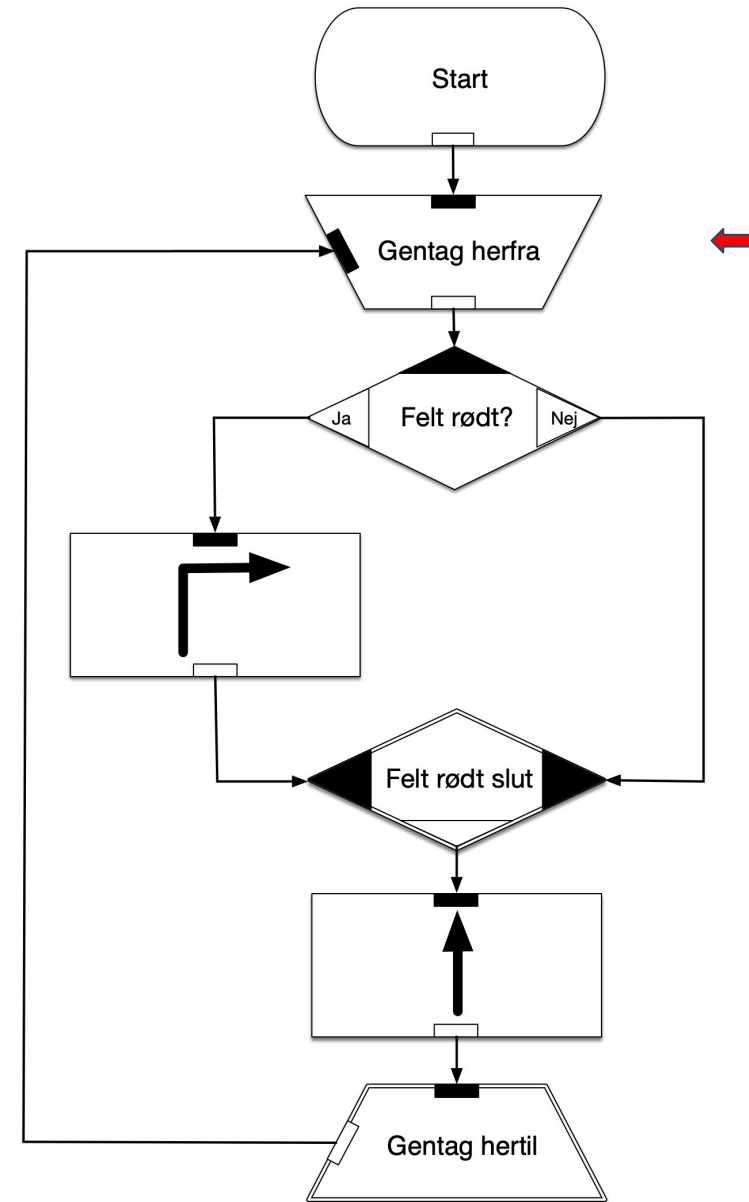
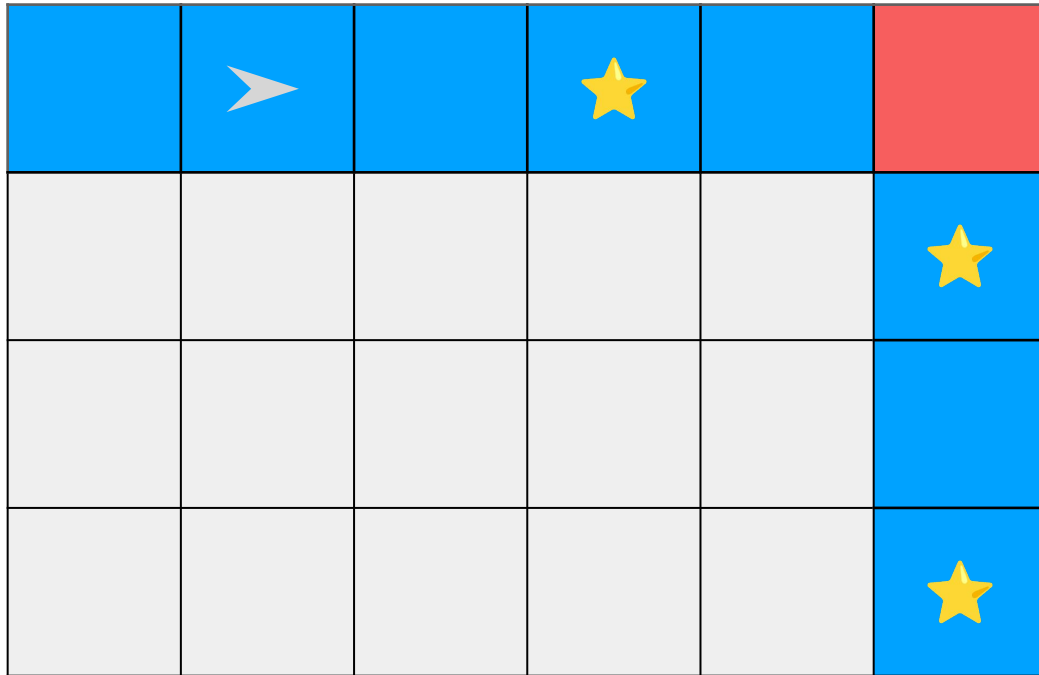
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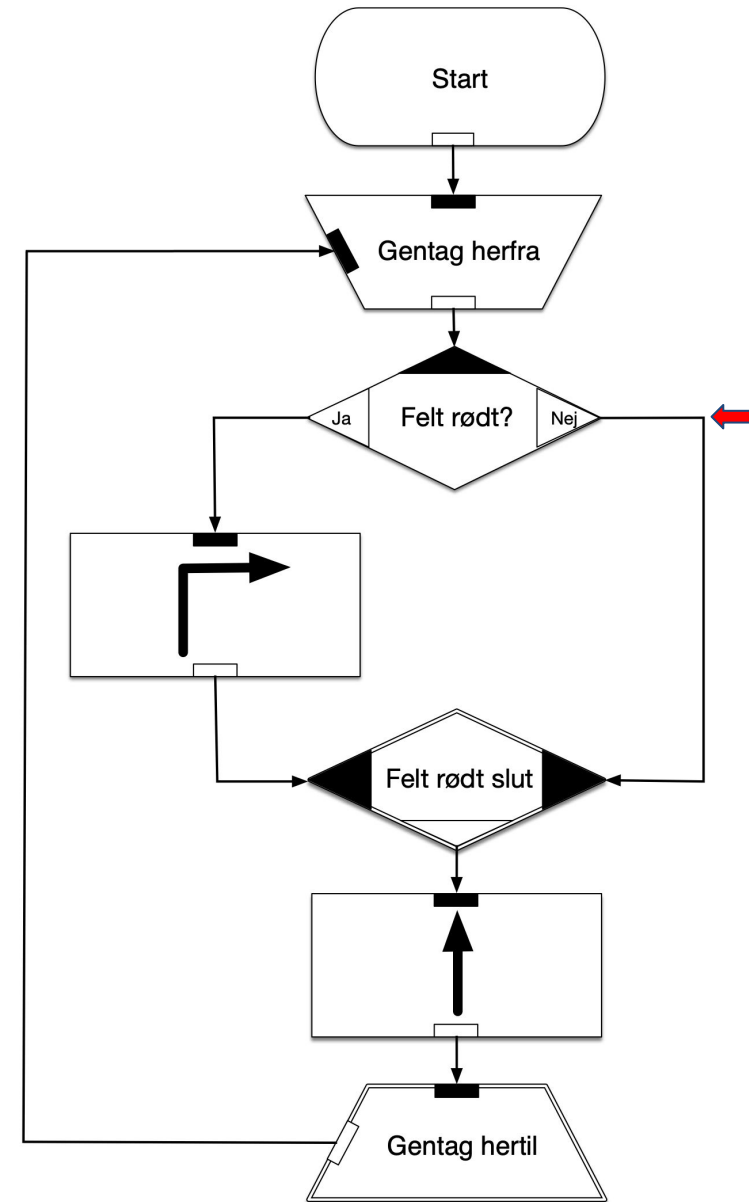
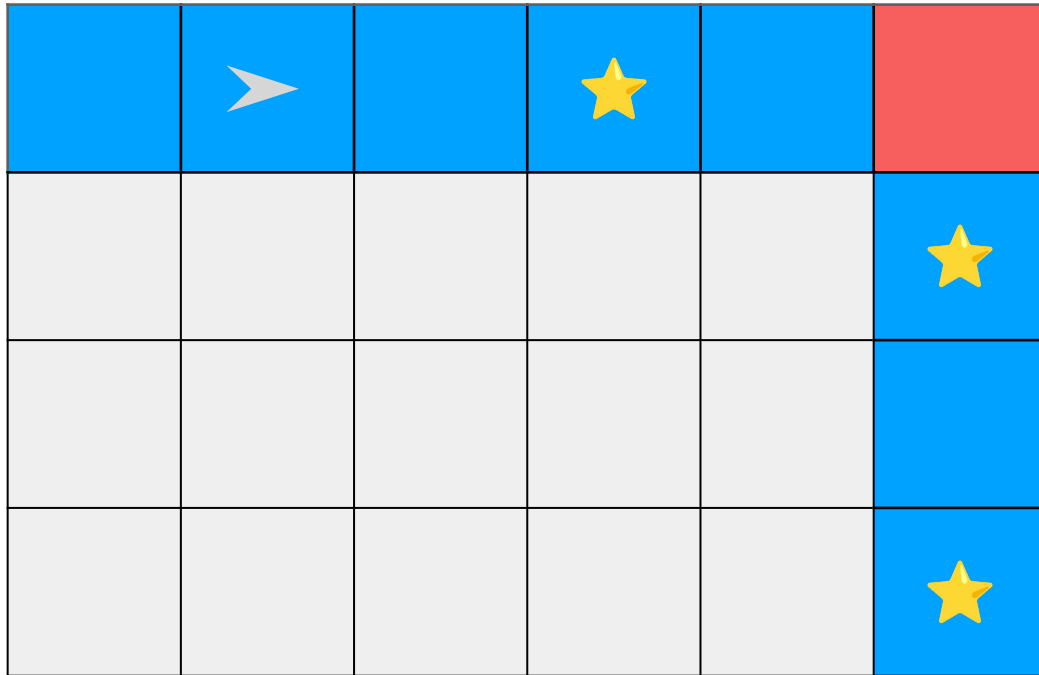
Program med løkker



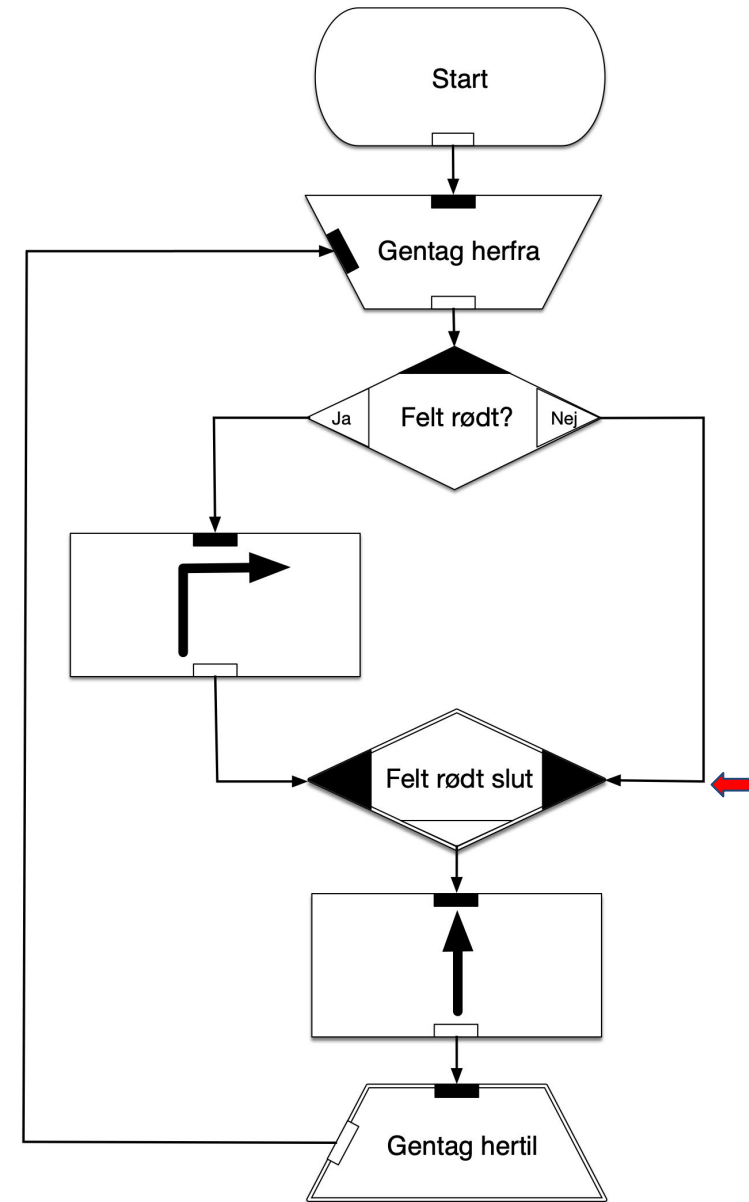
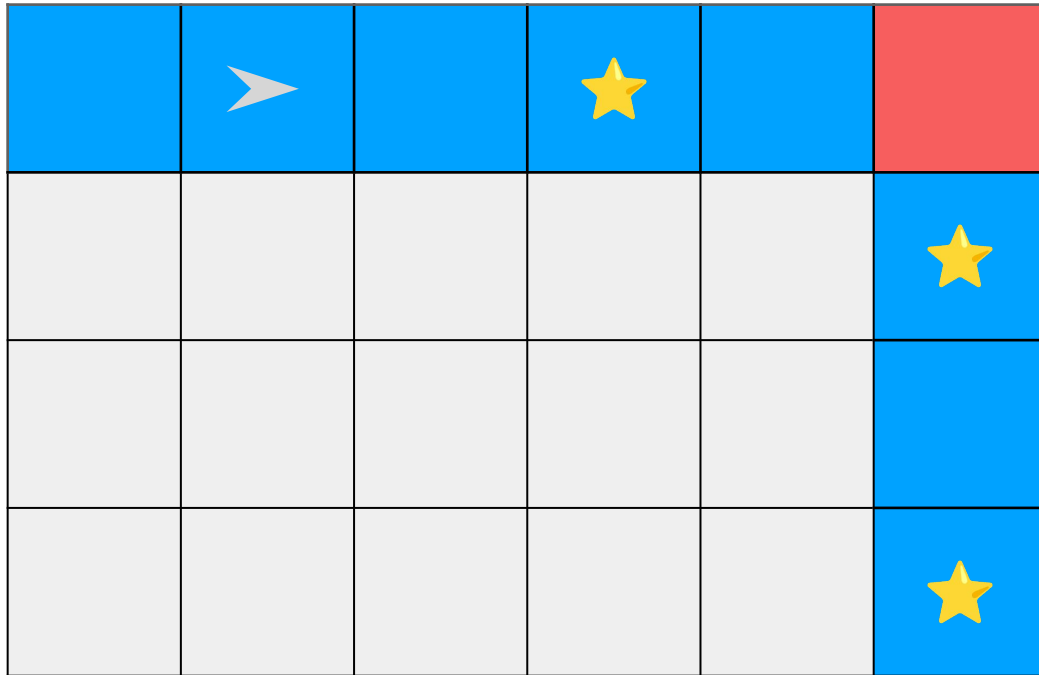
Program med løkker



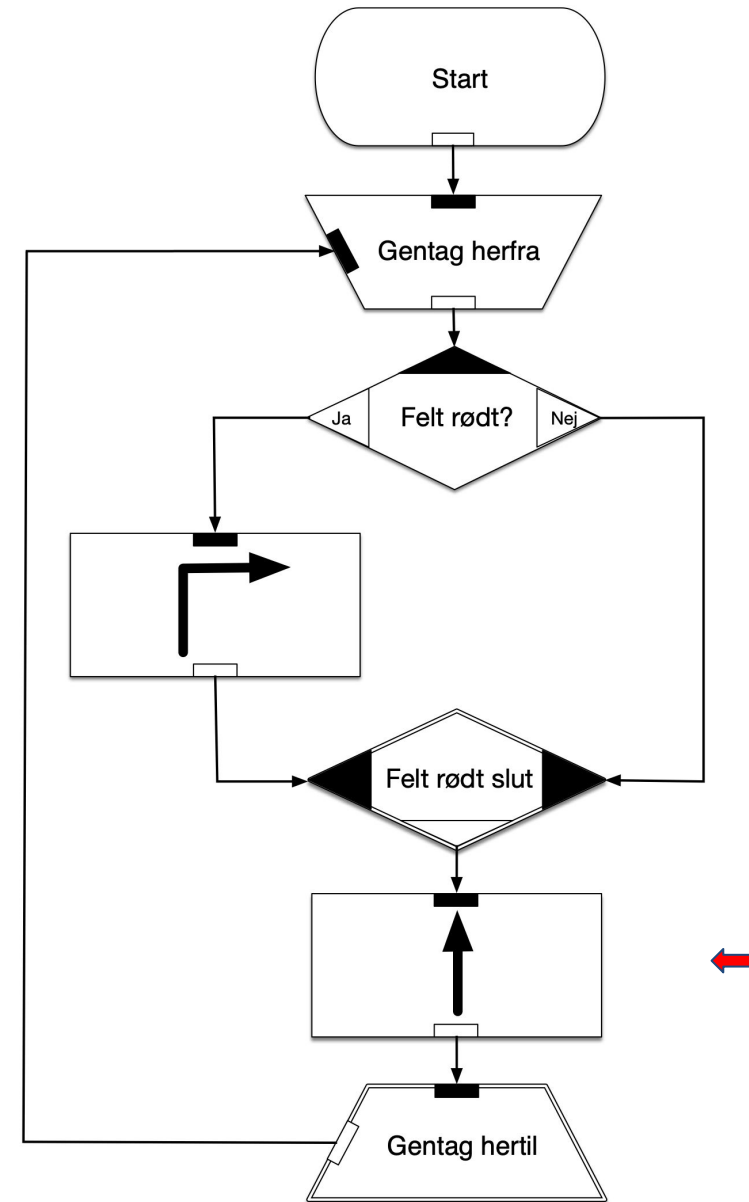
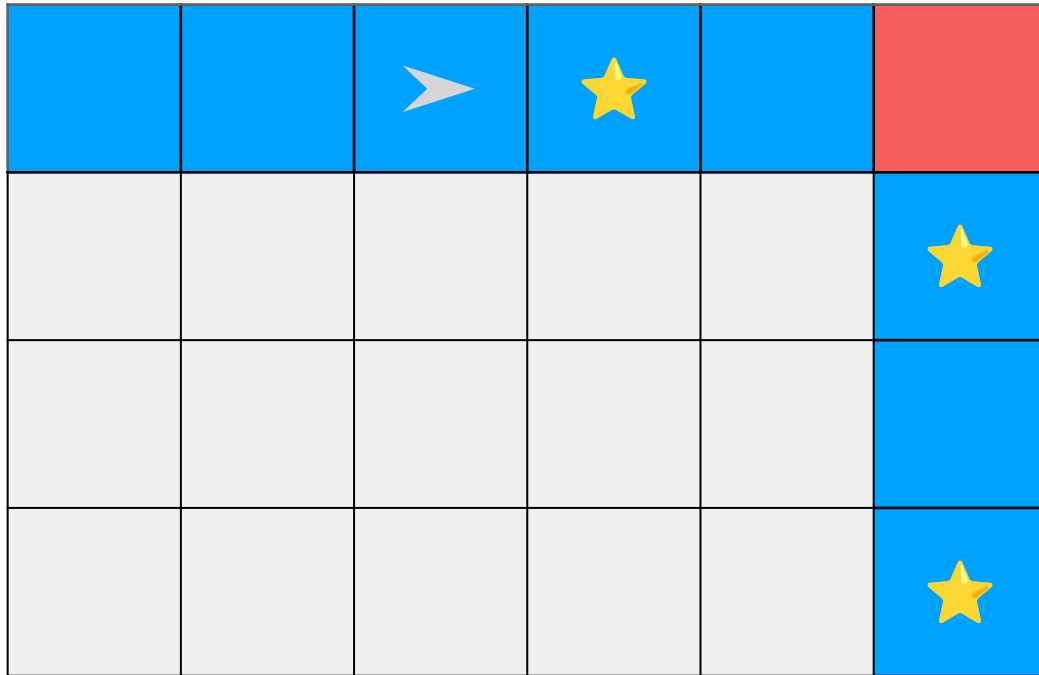
Program med løkker



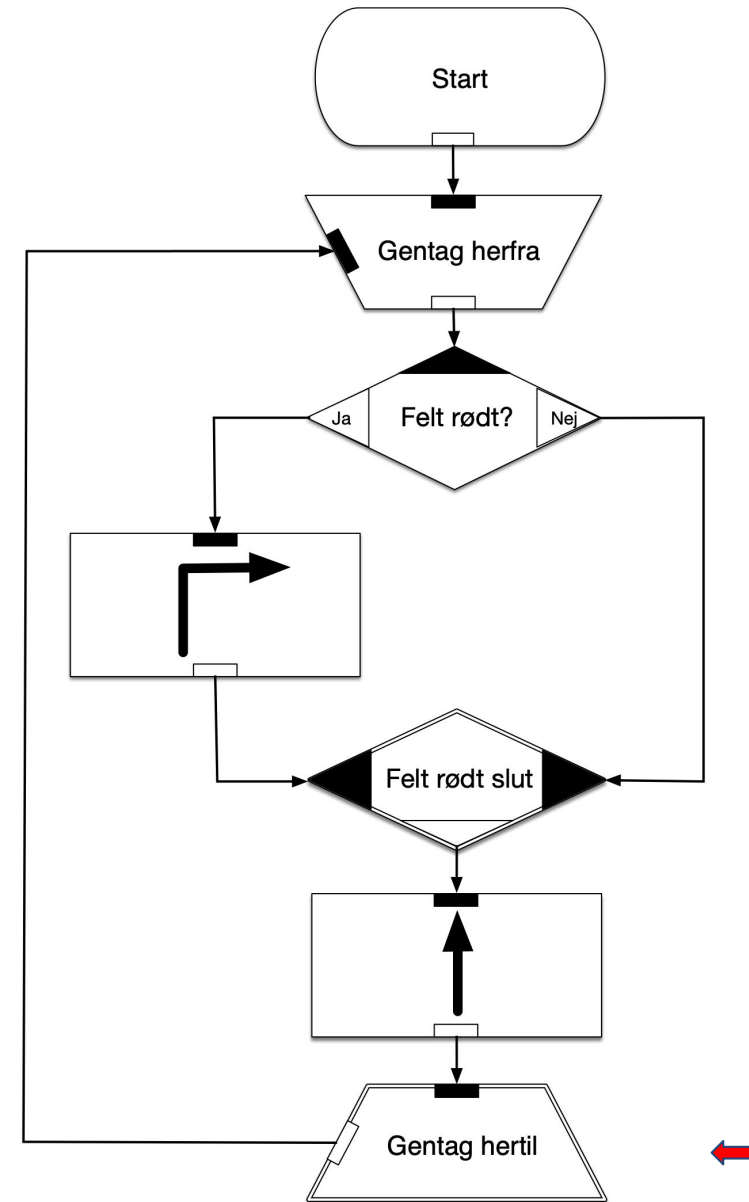
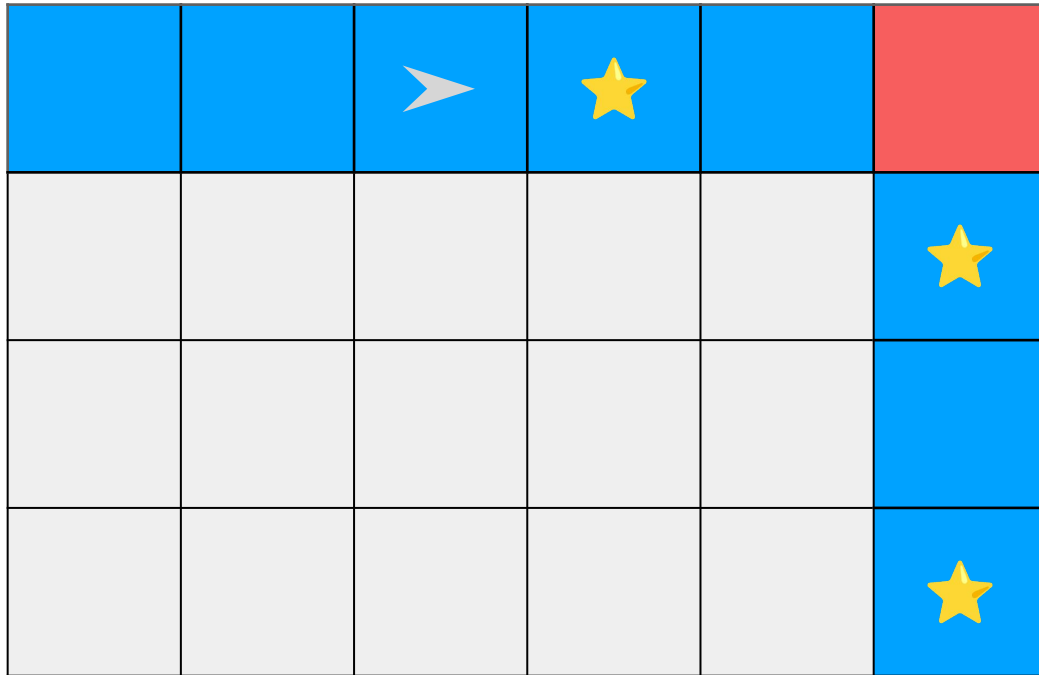
Program med løkker



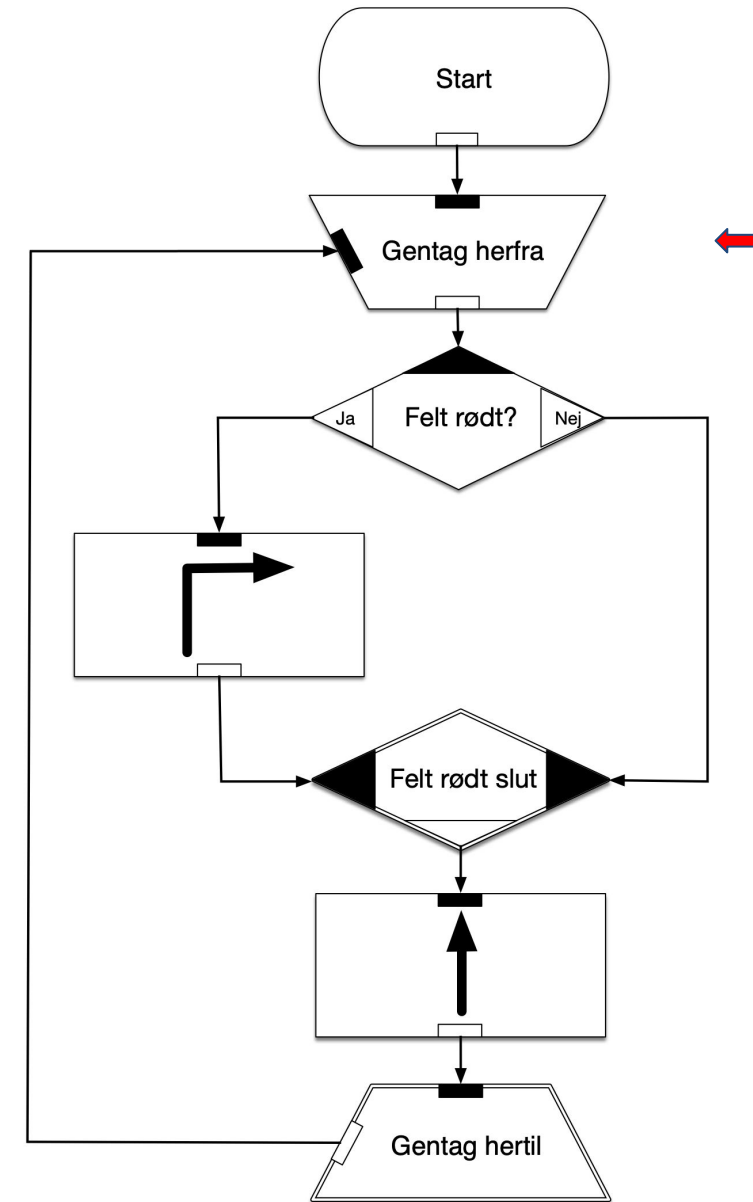
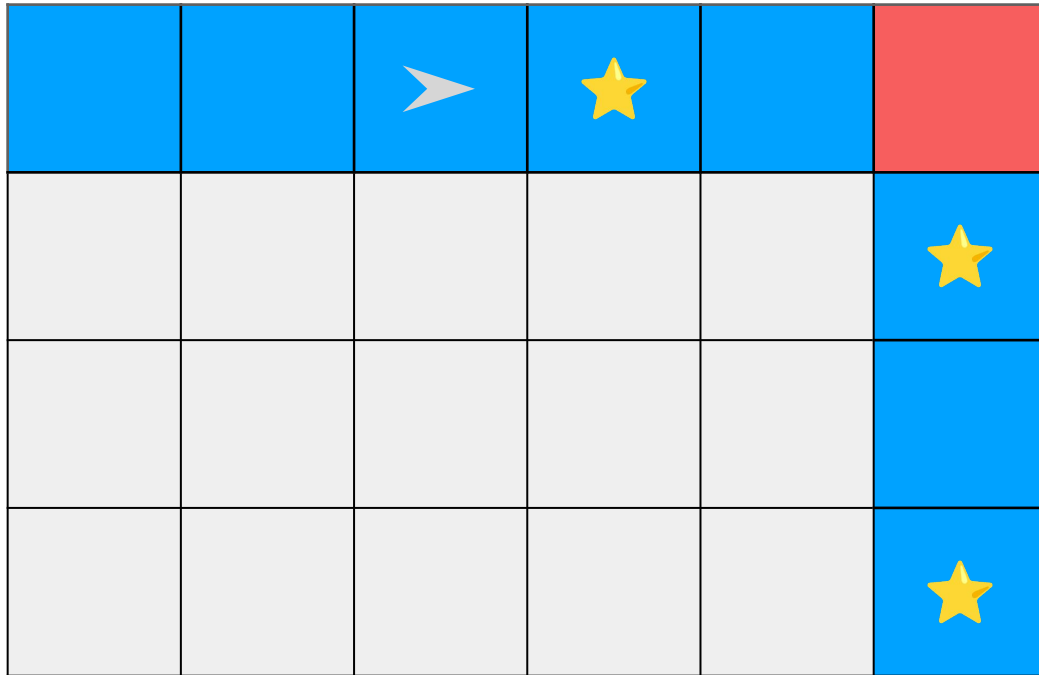
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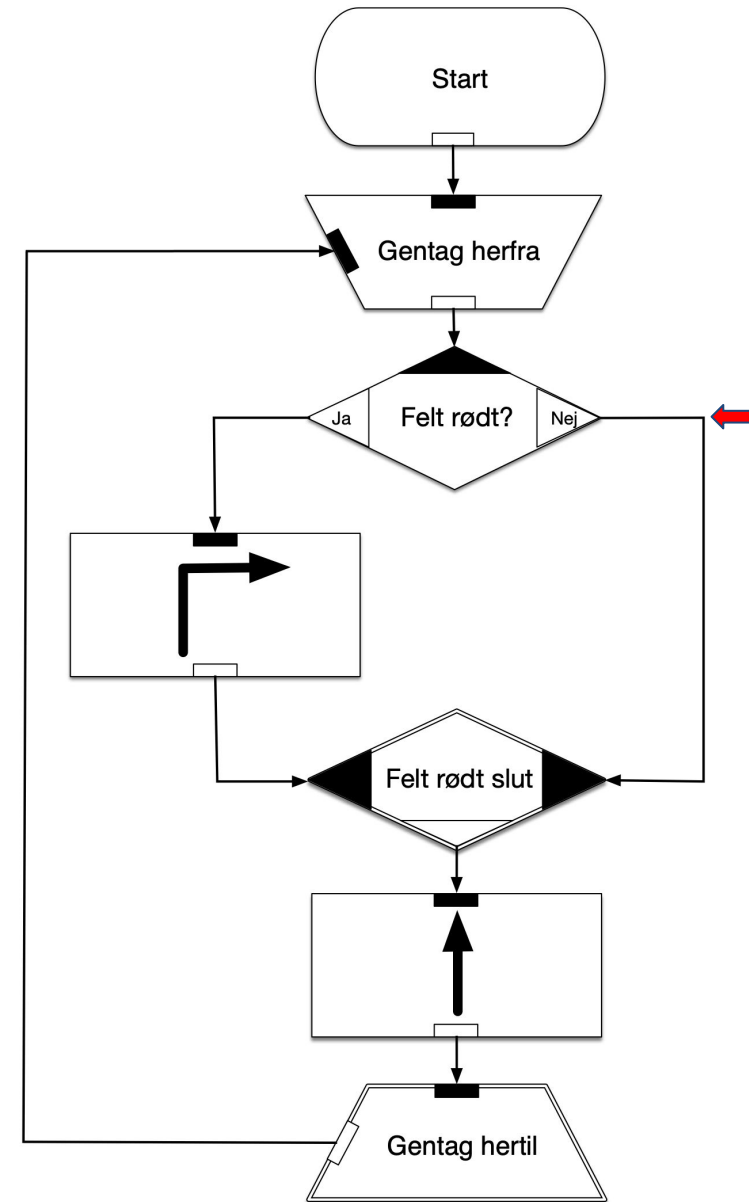
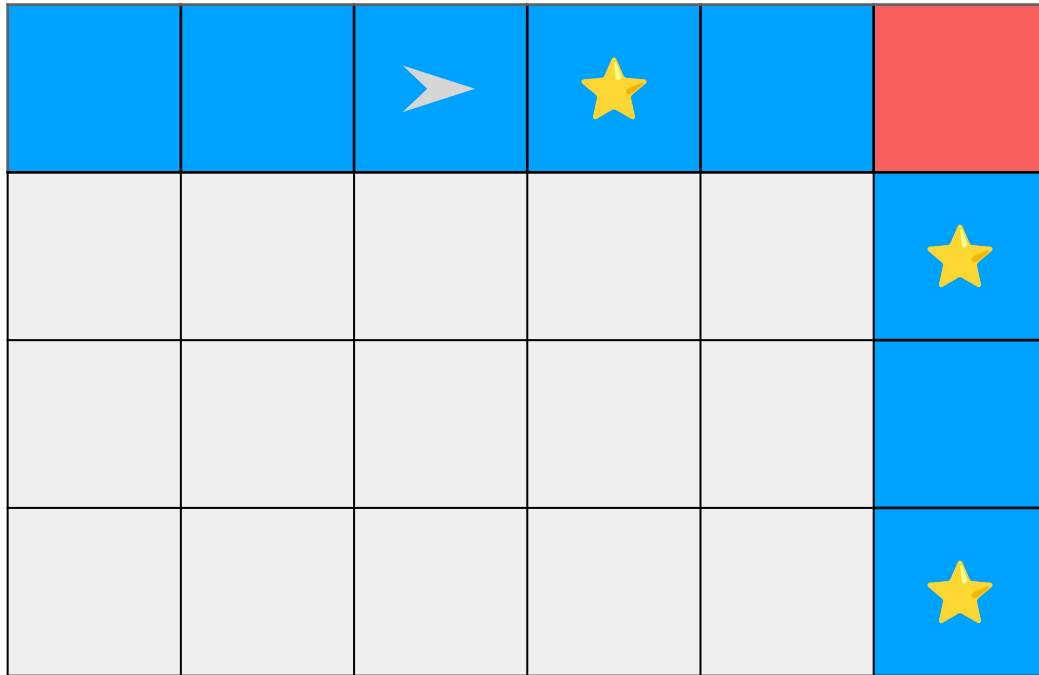
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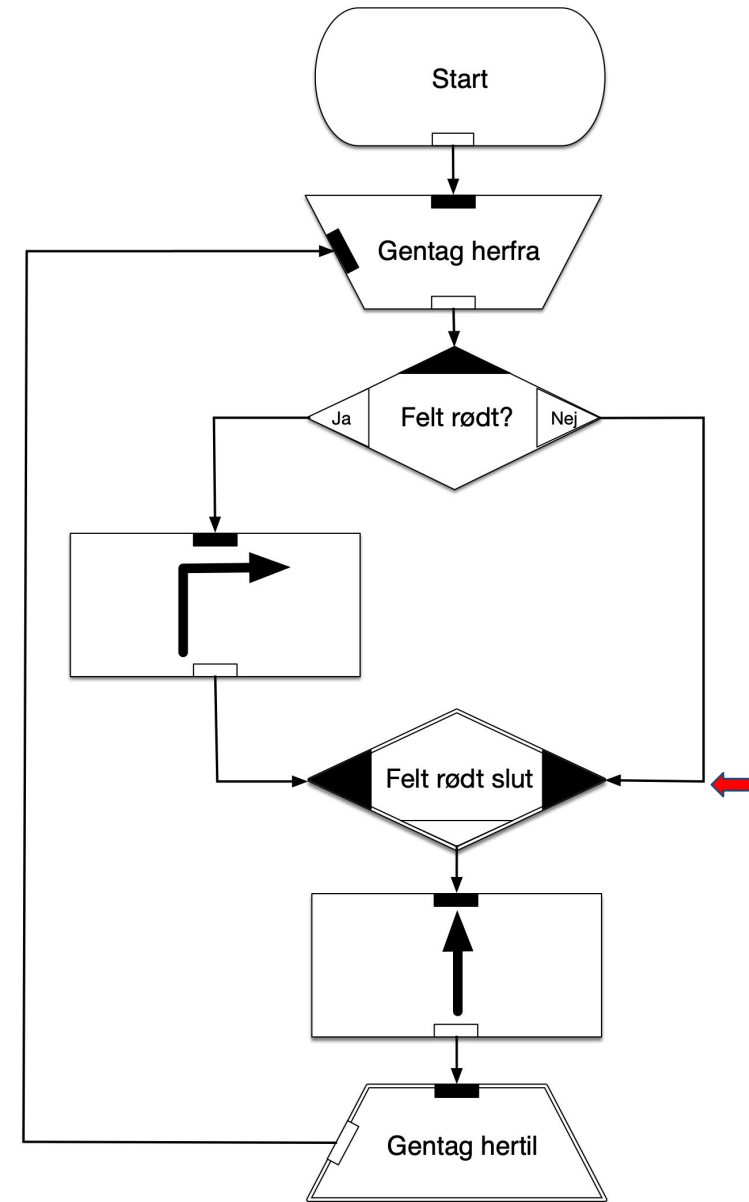
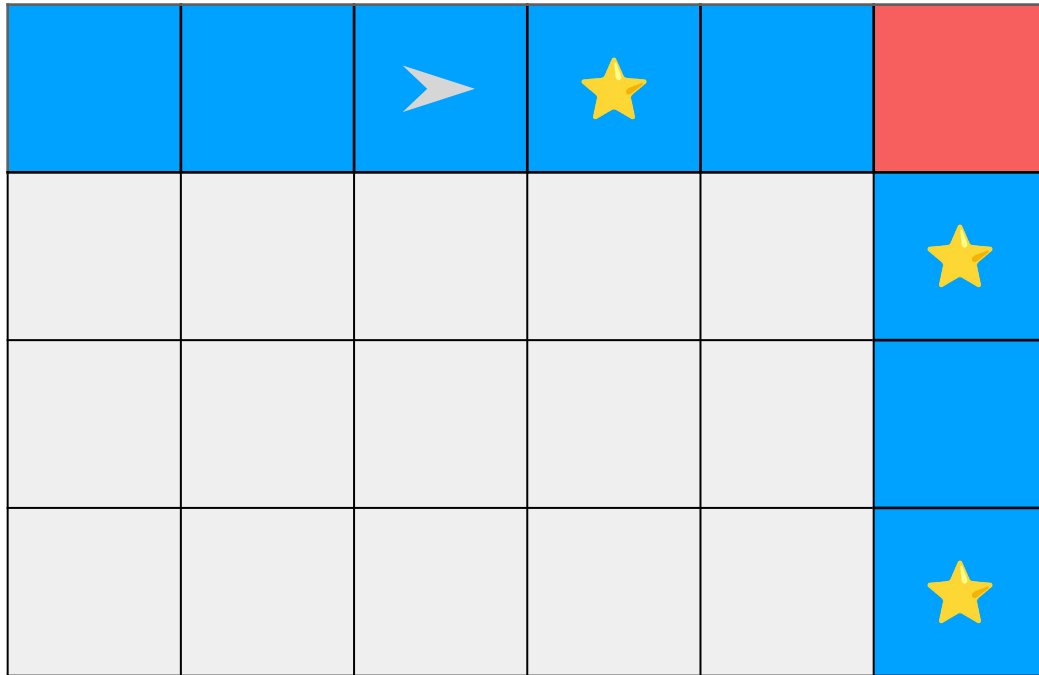
Program med løkker



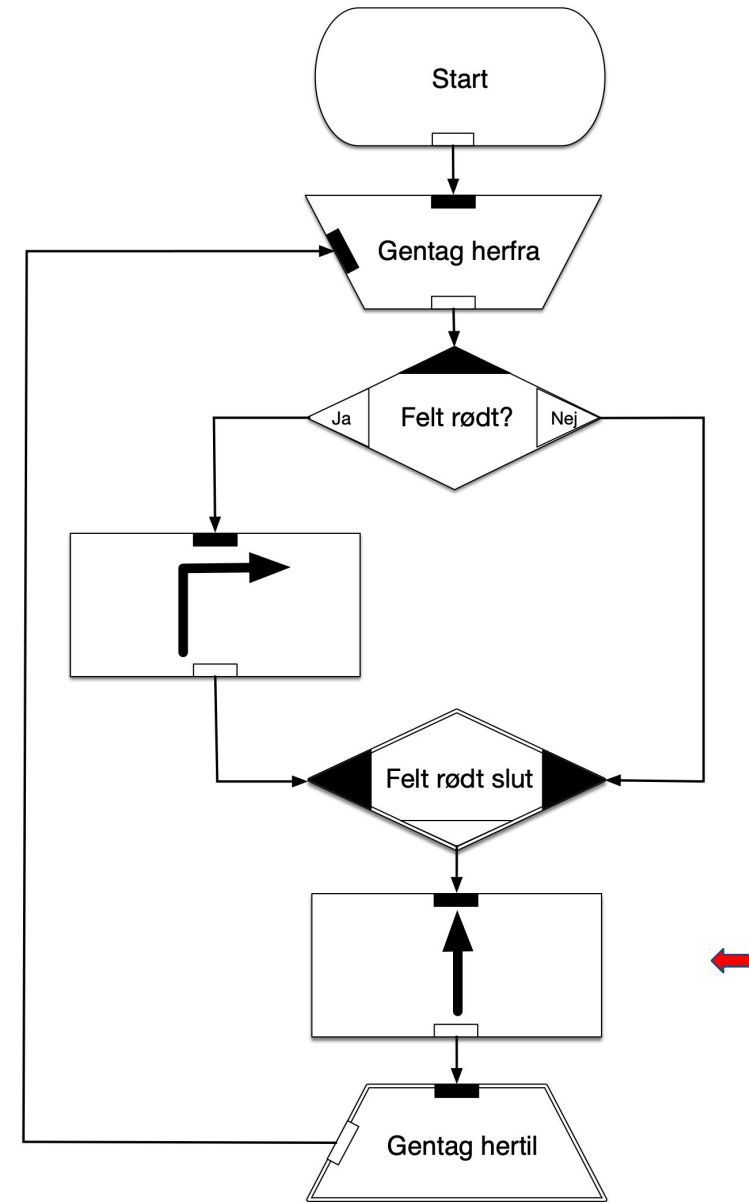
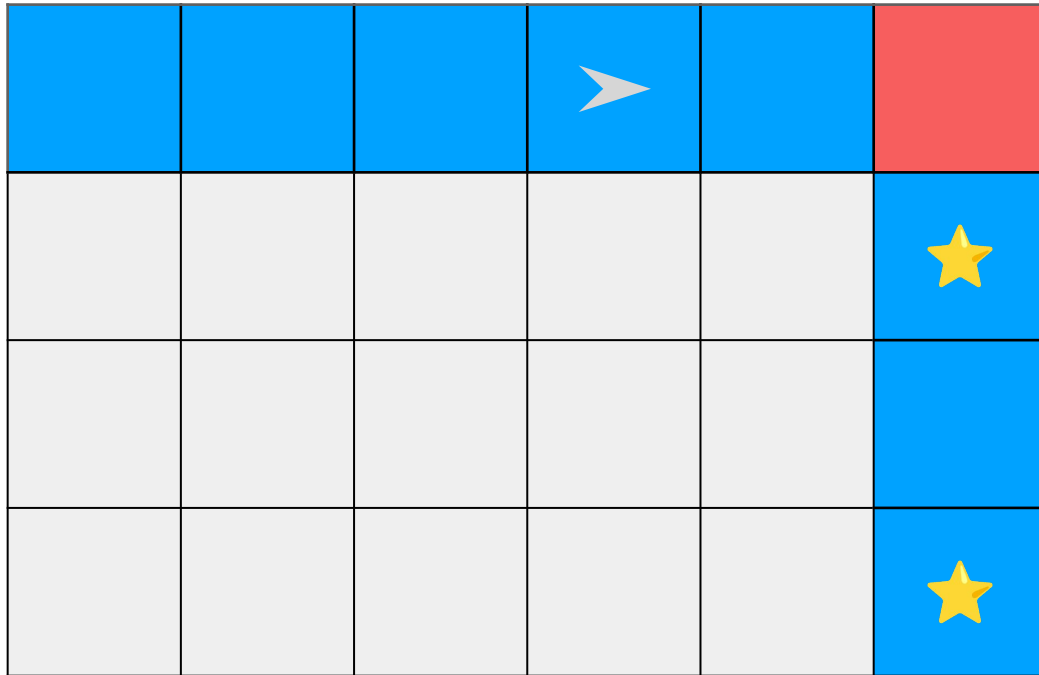
Program med løkker



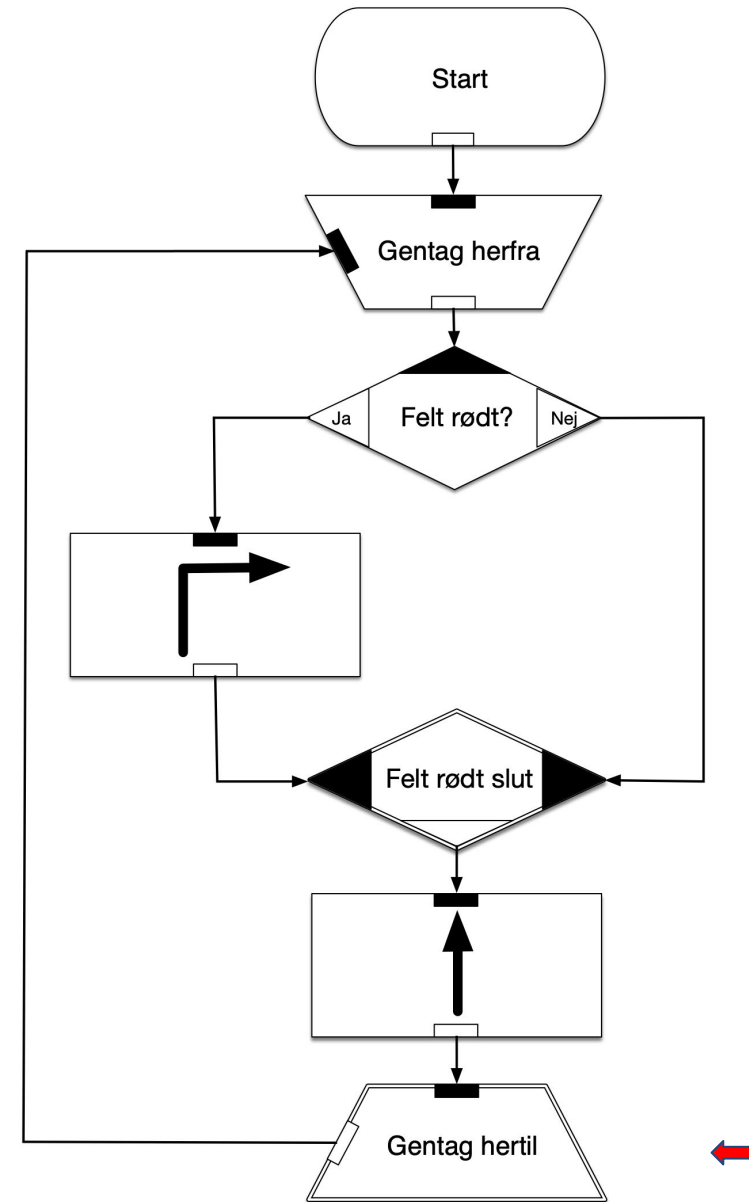
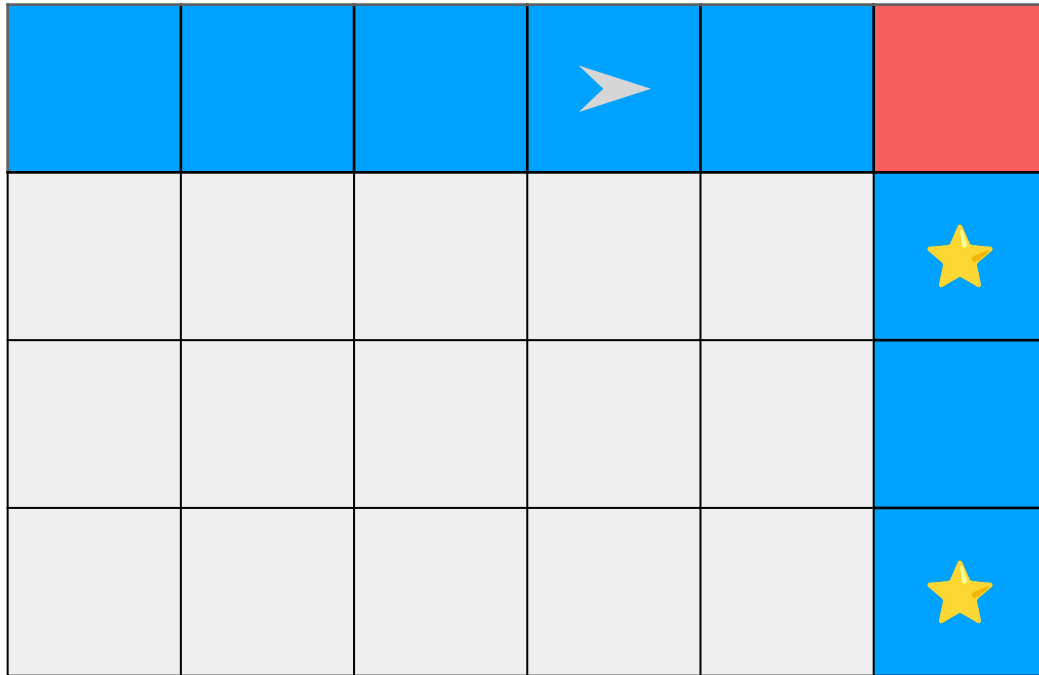
Program med løkker



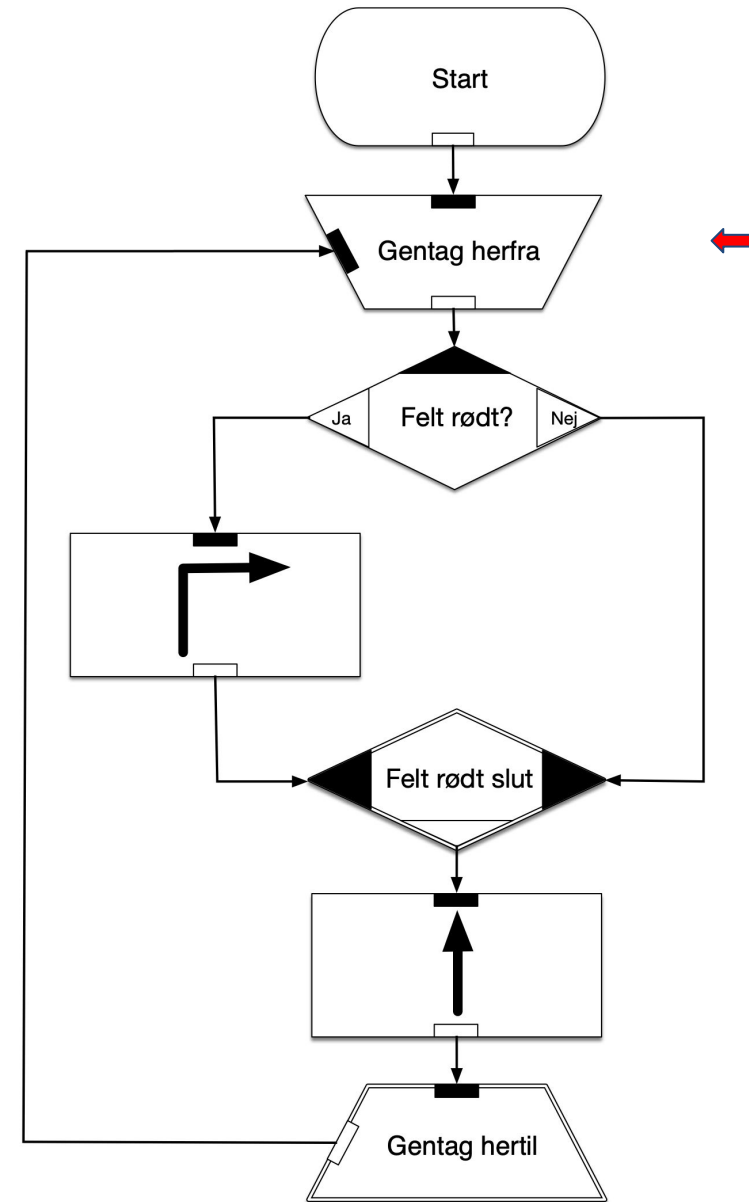
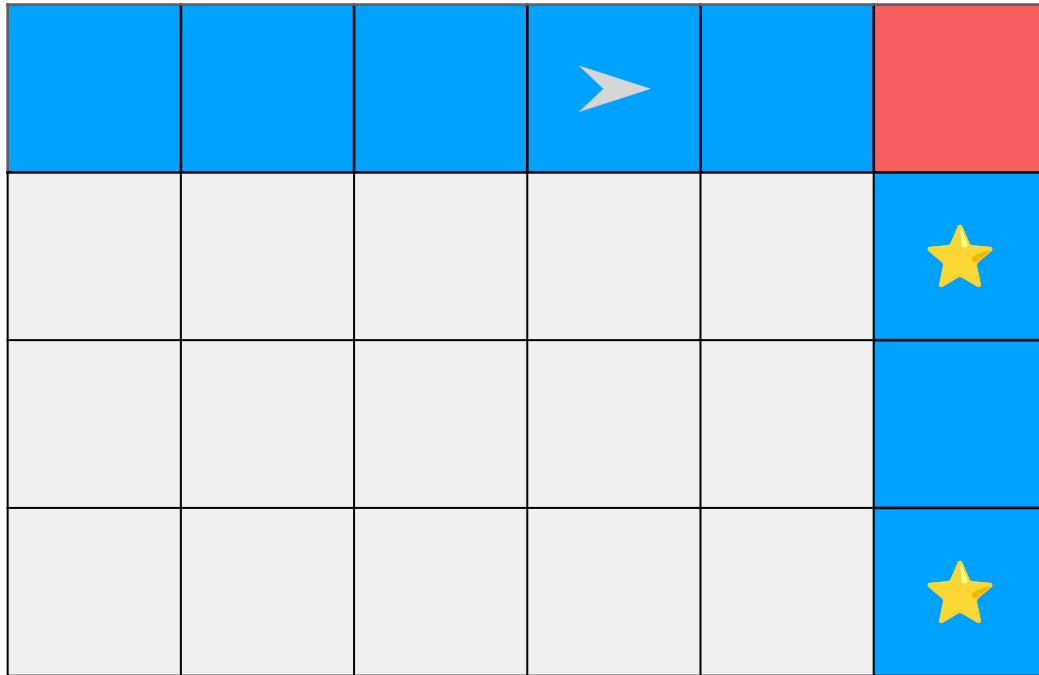
Program med løkker



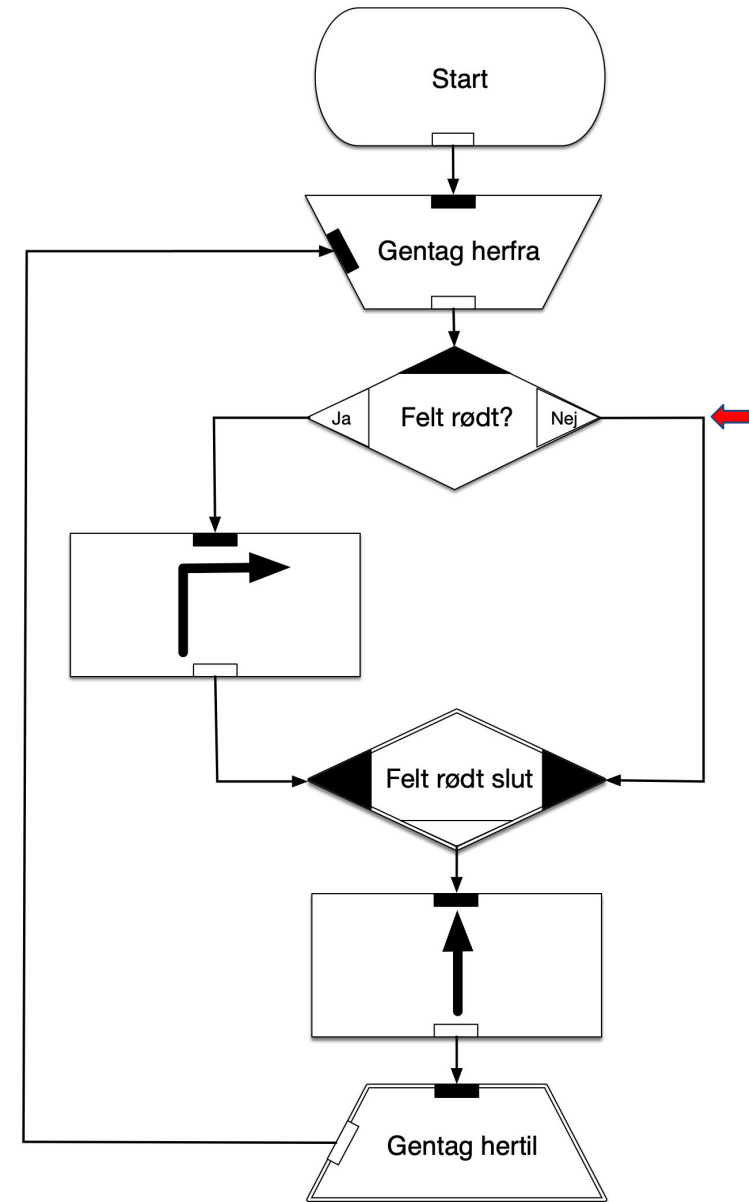
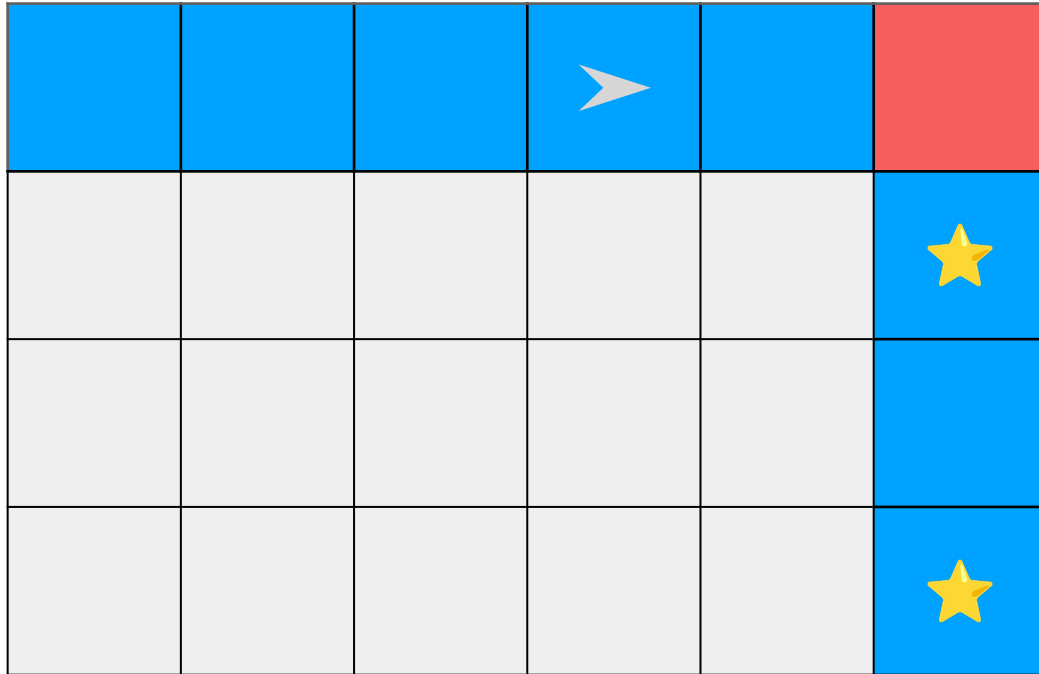
Program med løkker



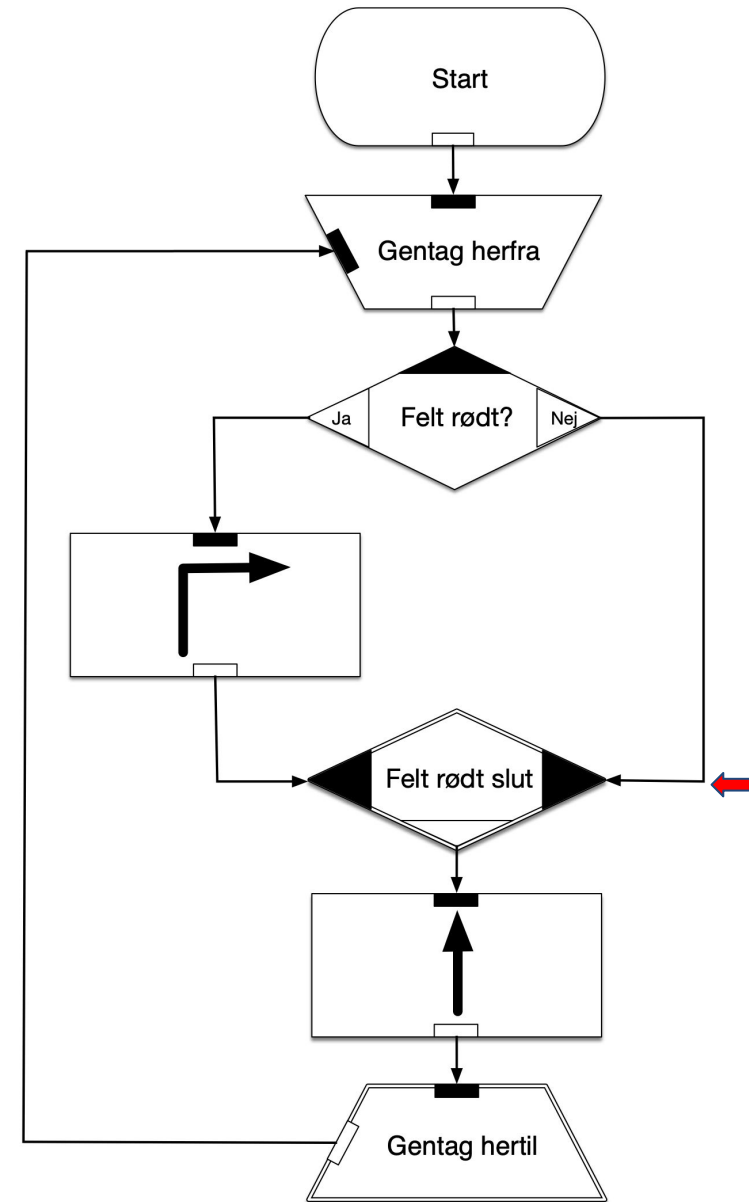
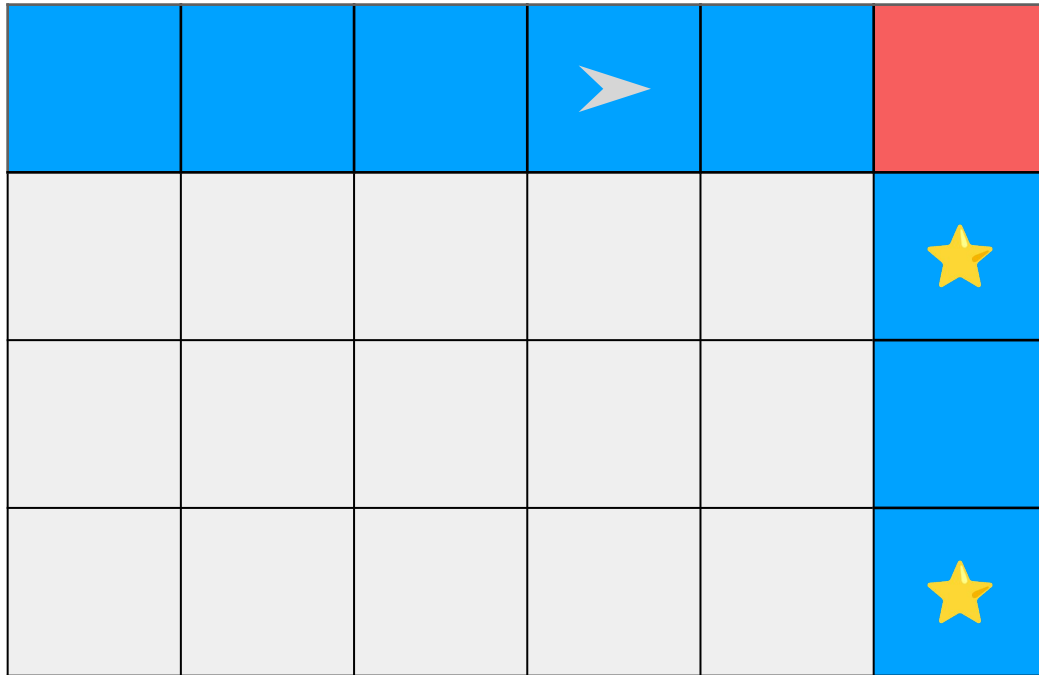
Program med løkker



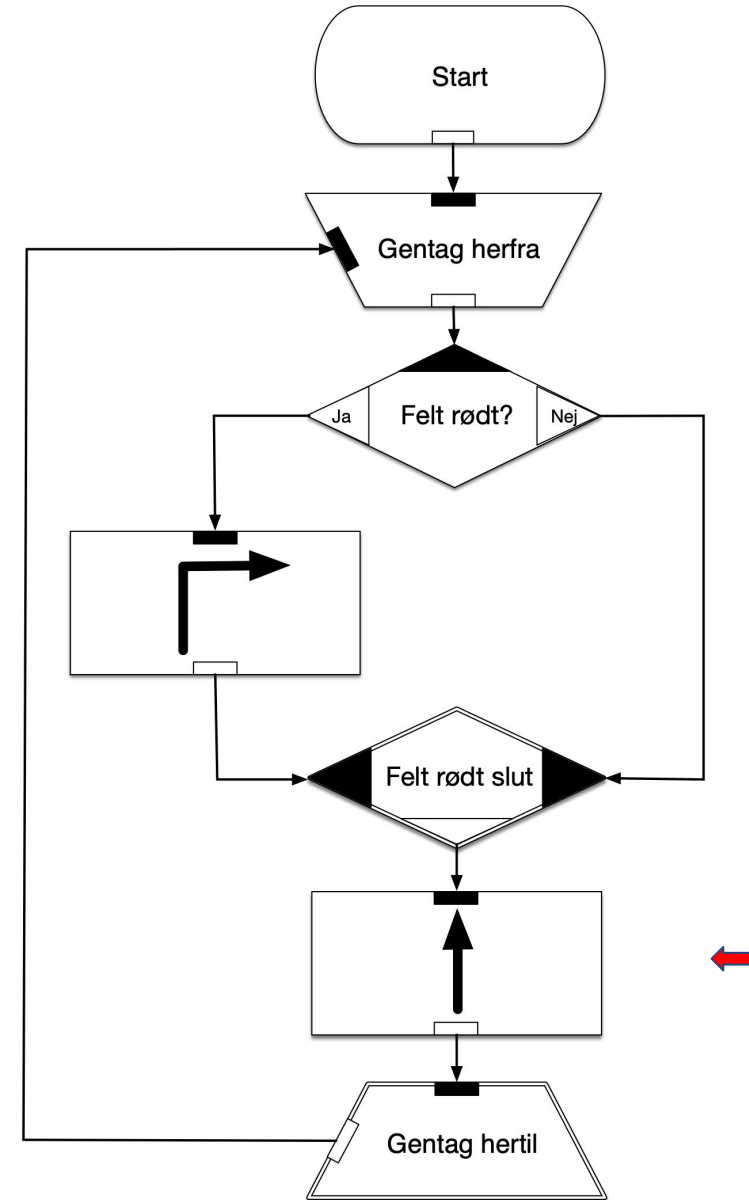
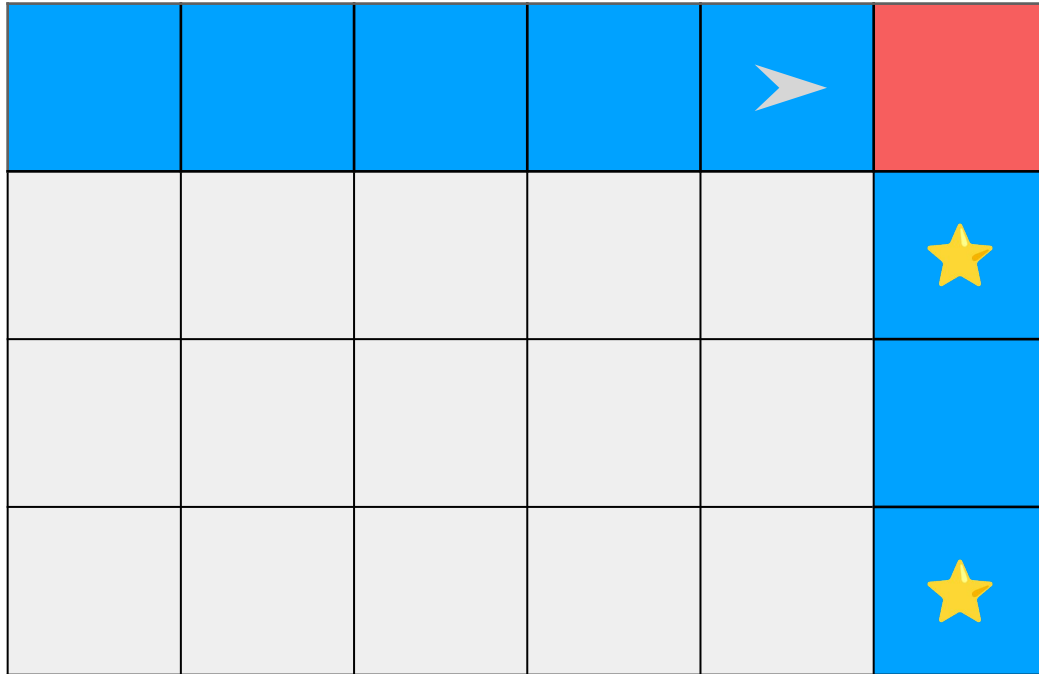
Program med løkker



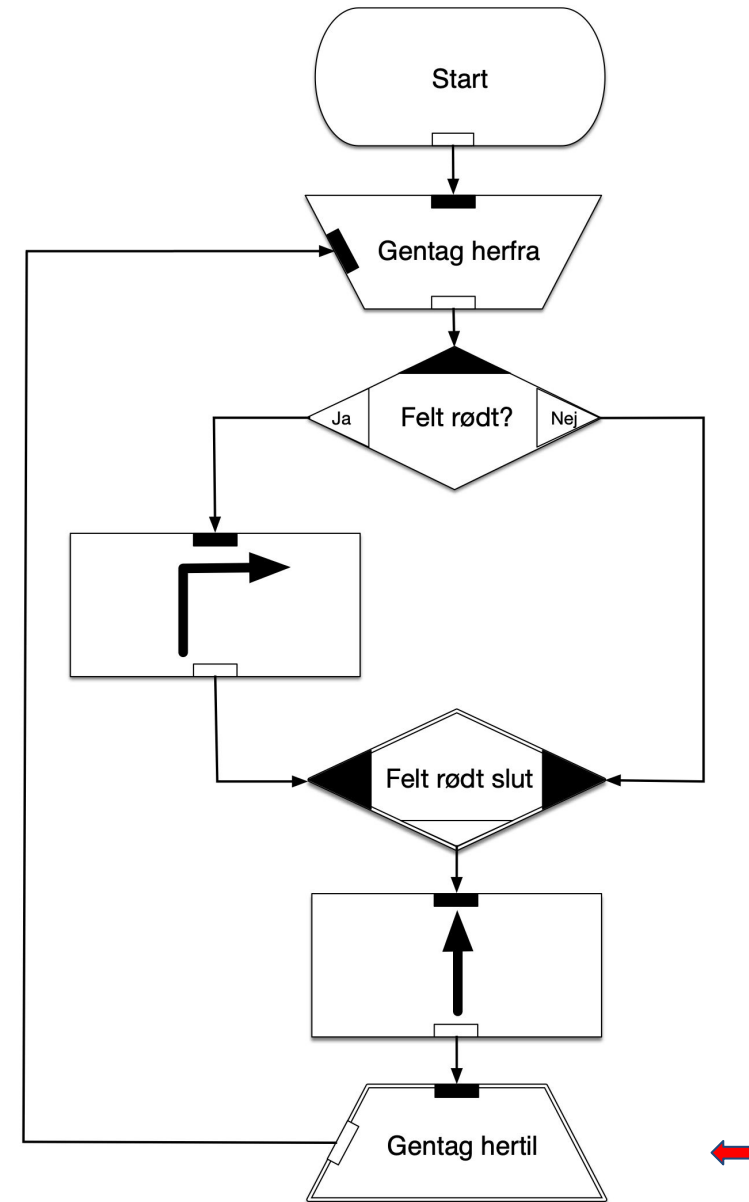
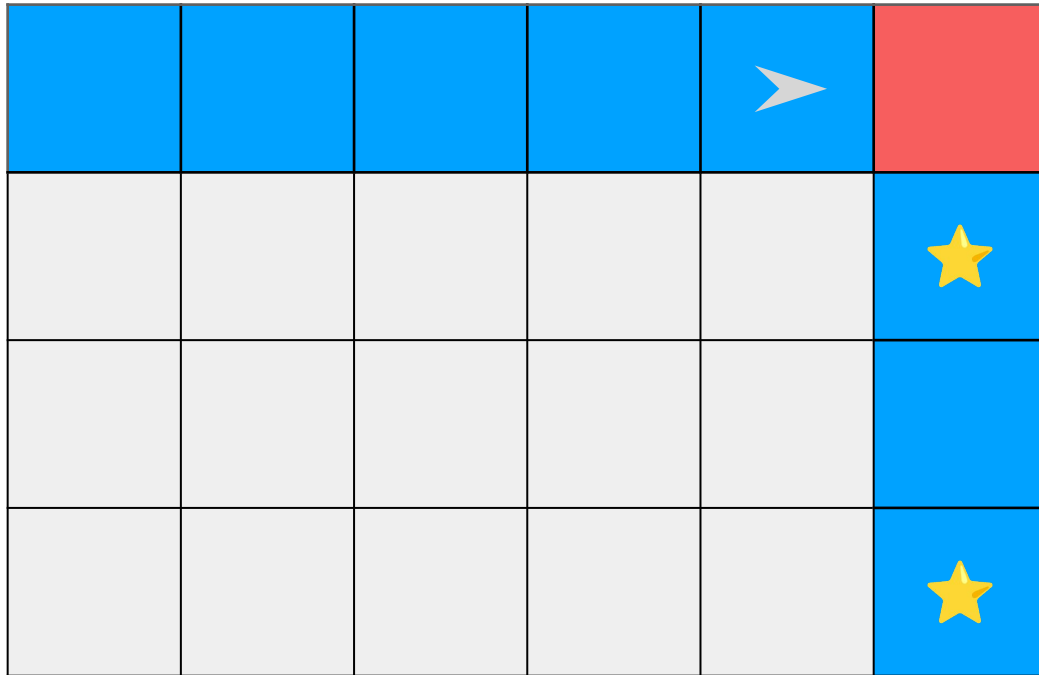
Program med løkker



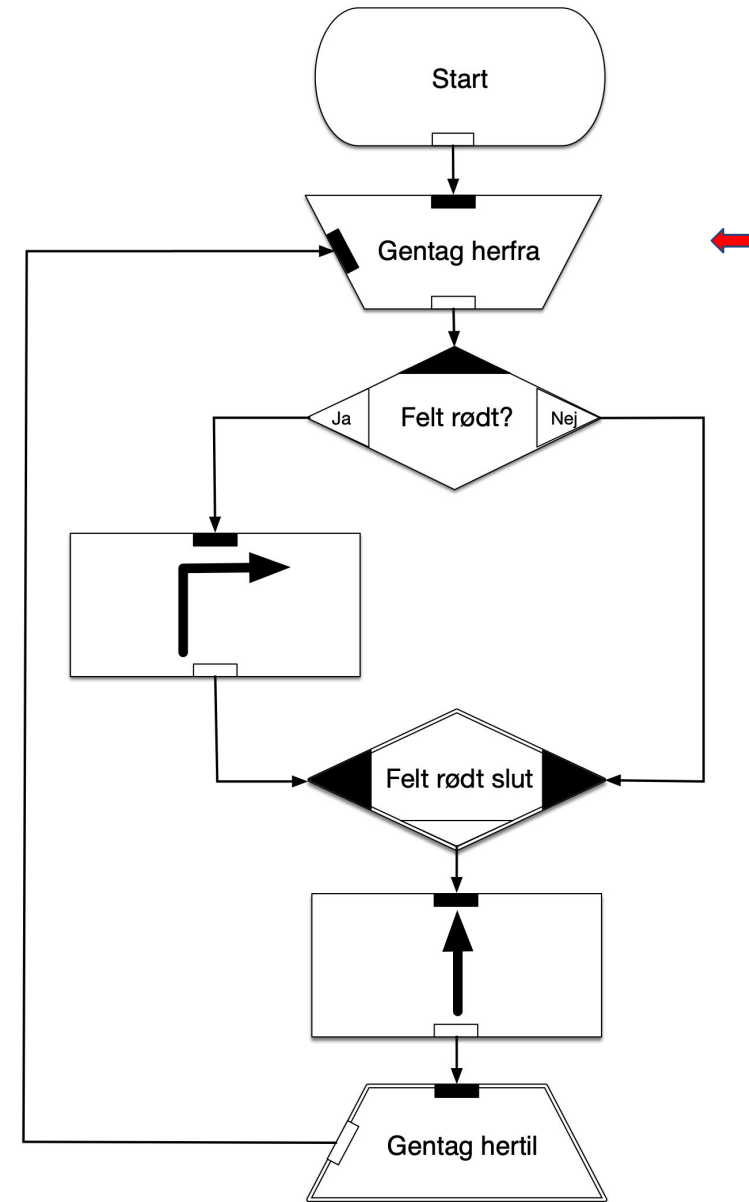
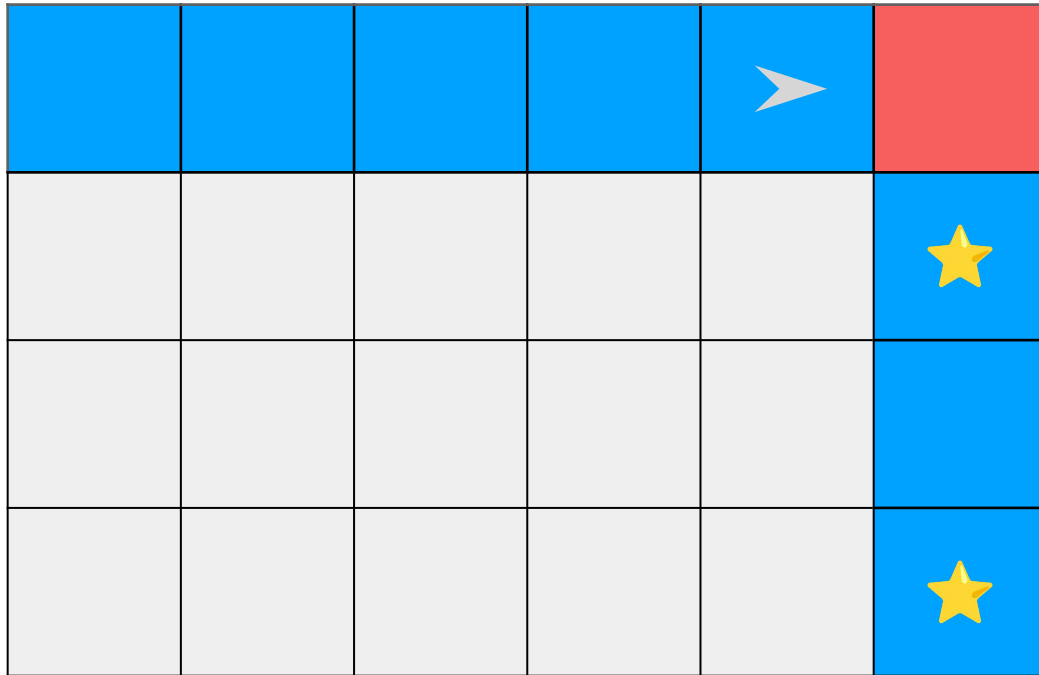
Program med løkker



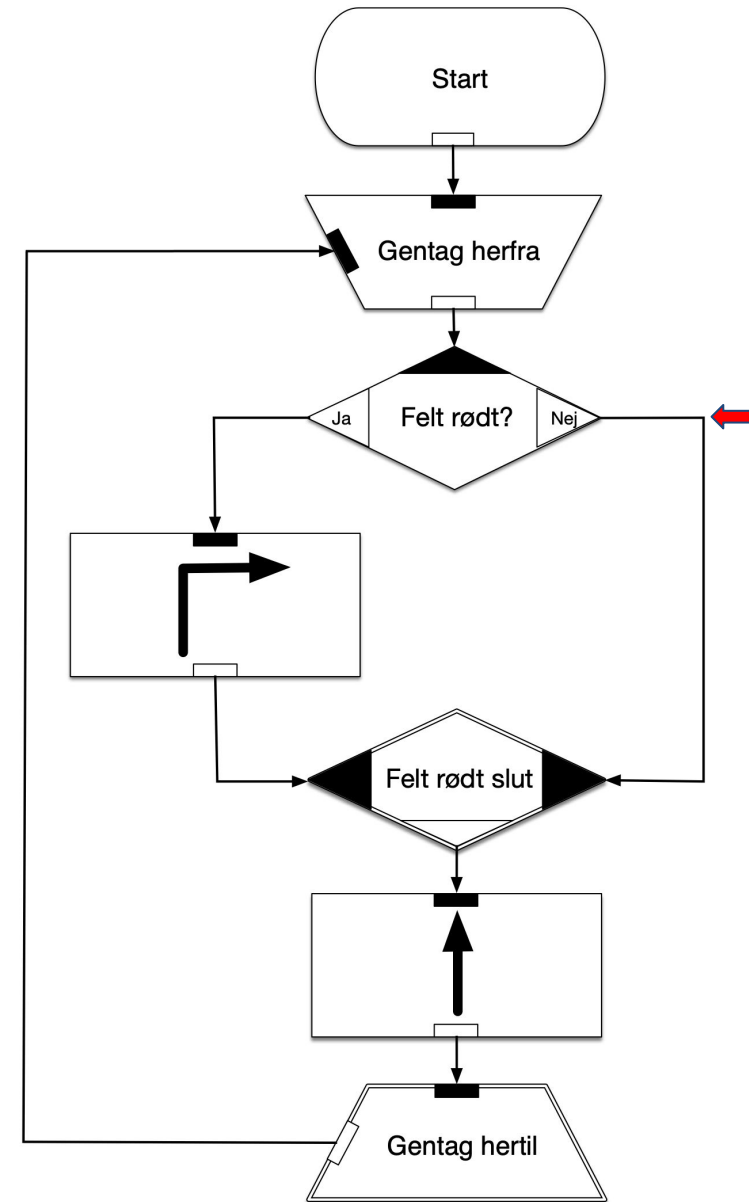
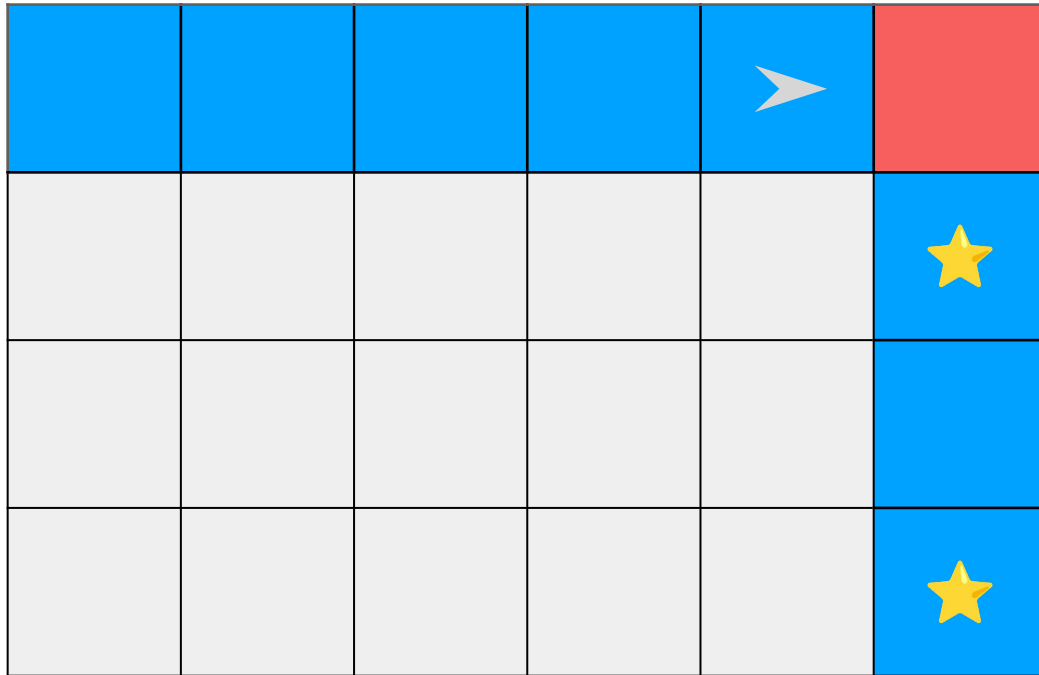
Program med løkker



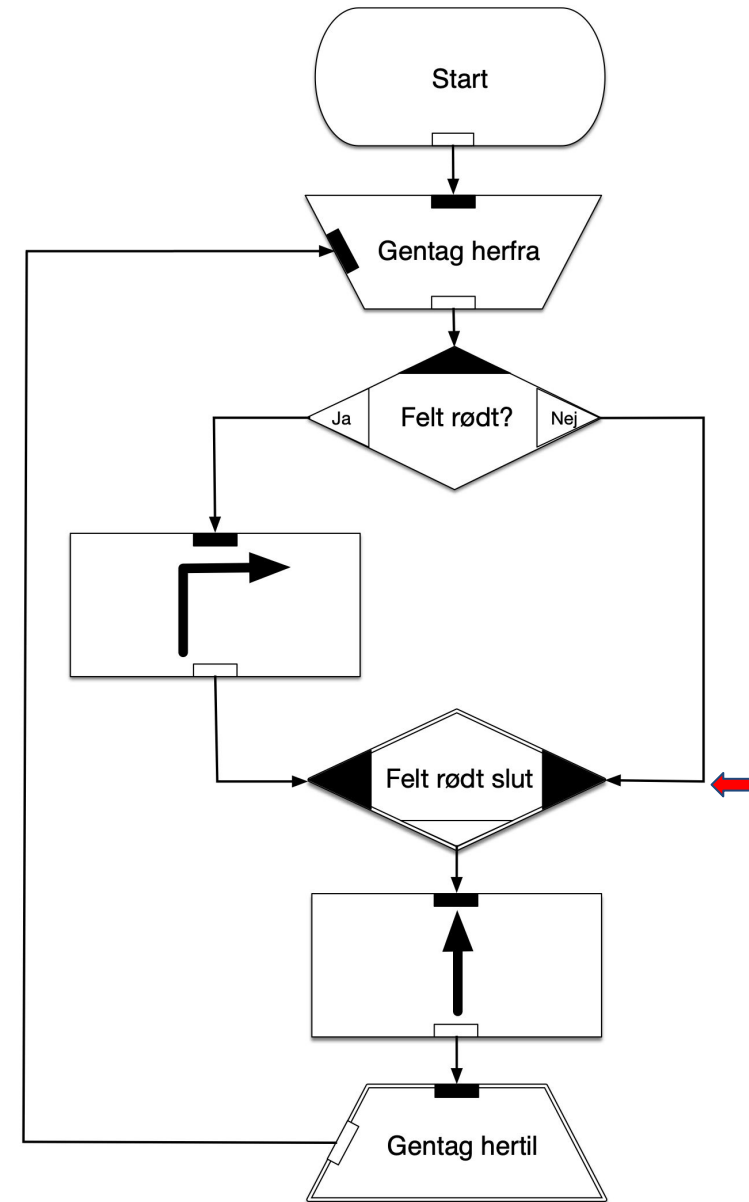
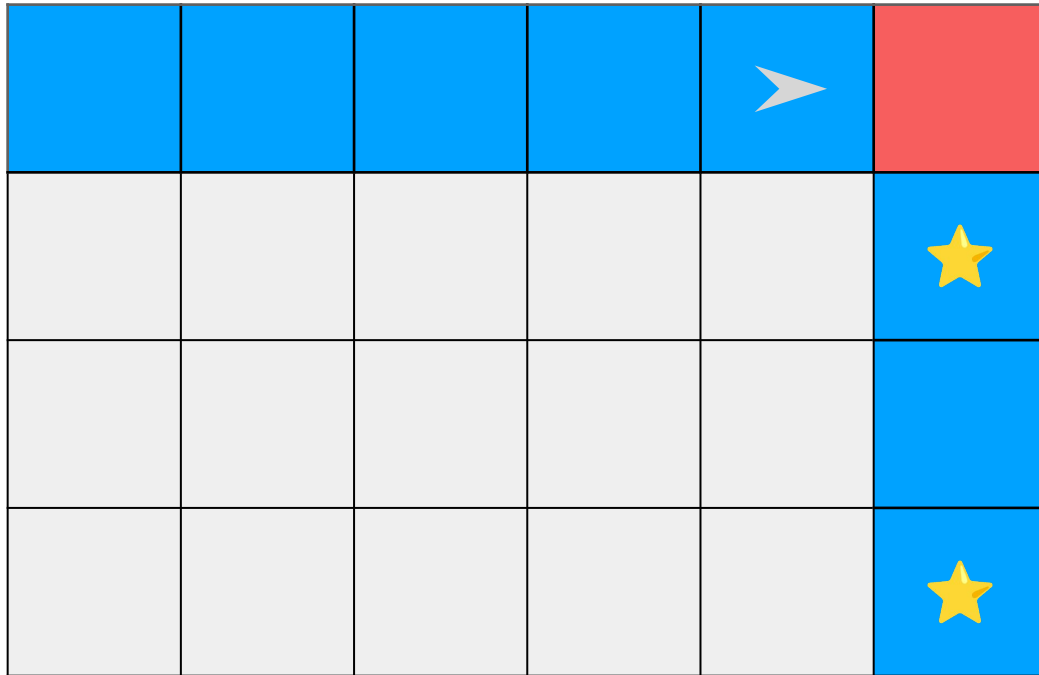
Program med løkker



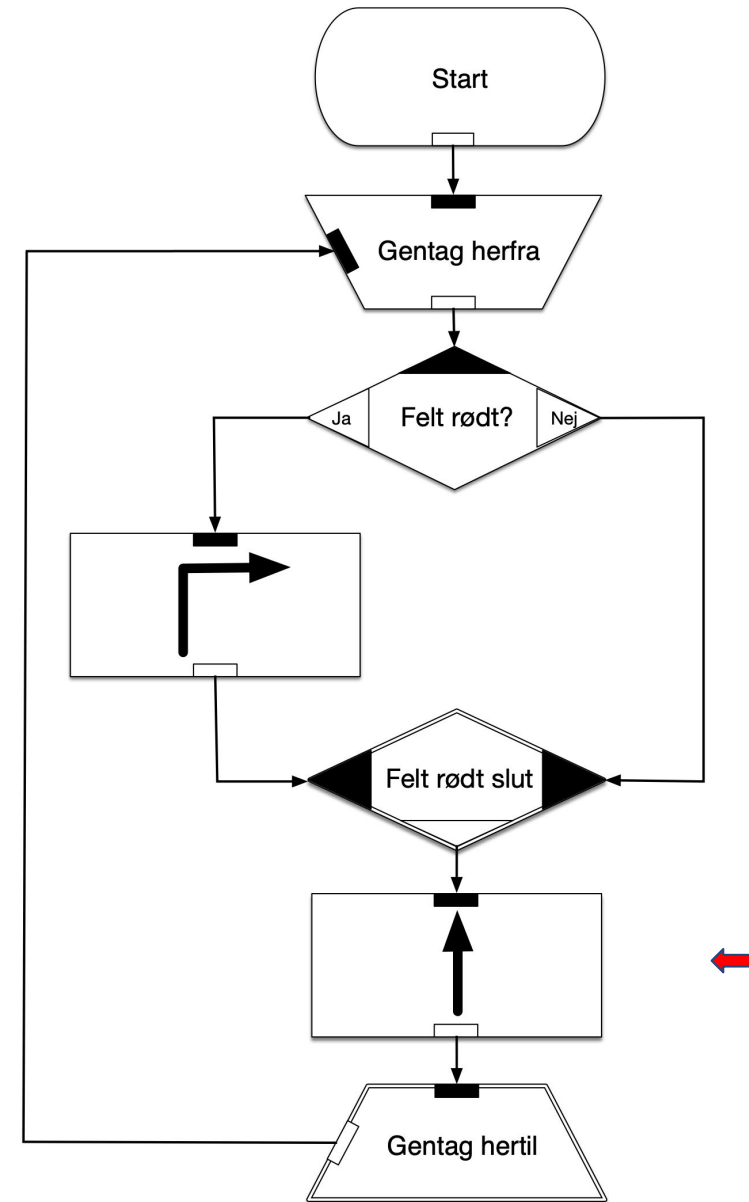
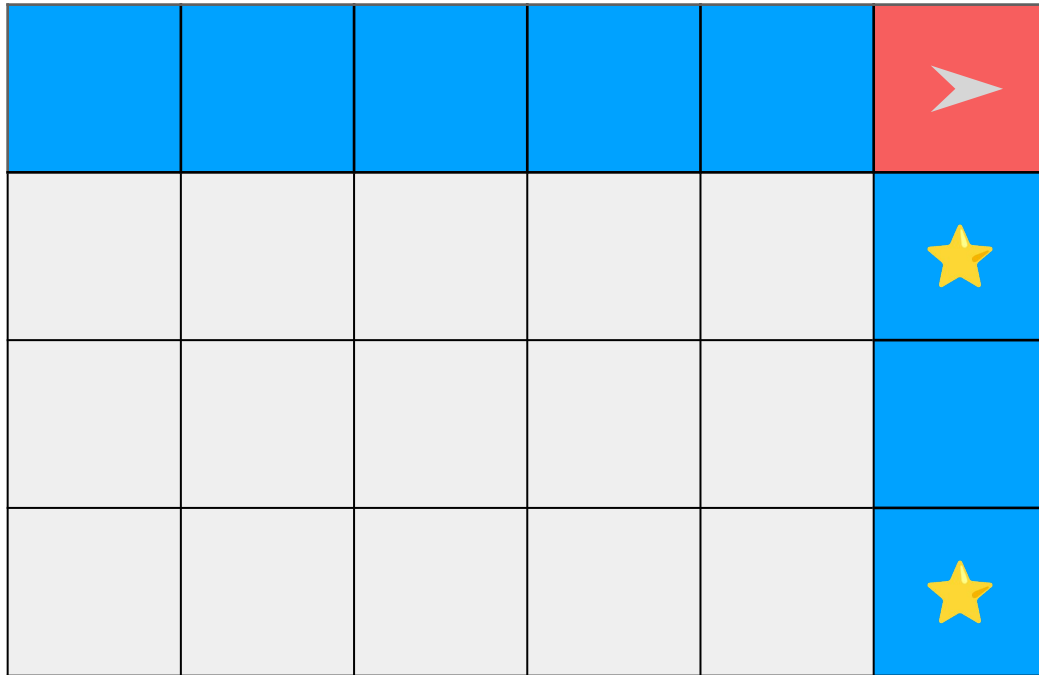
Program med løkker



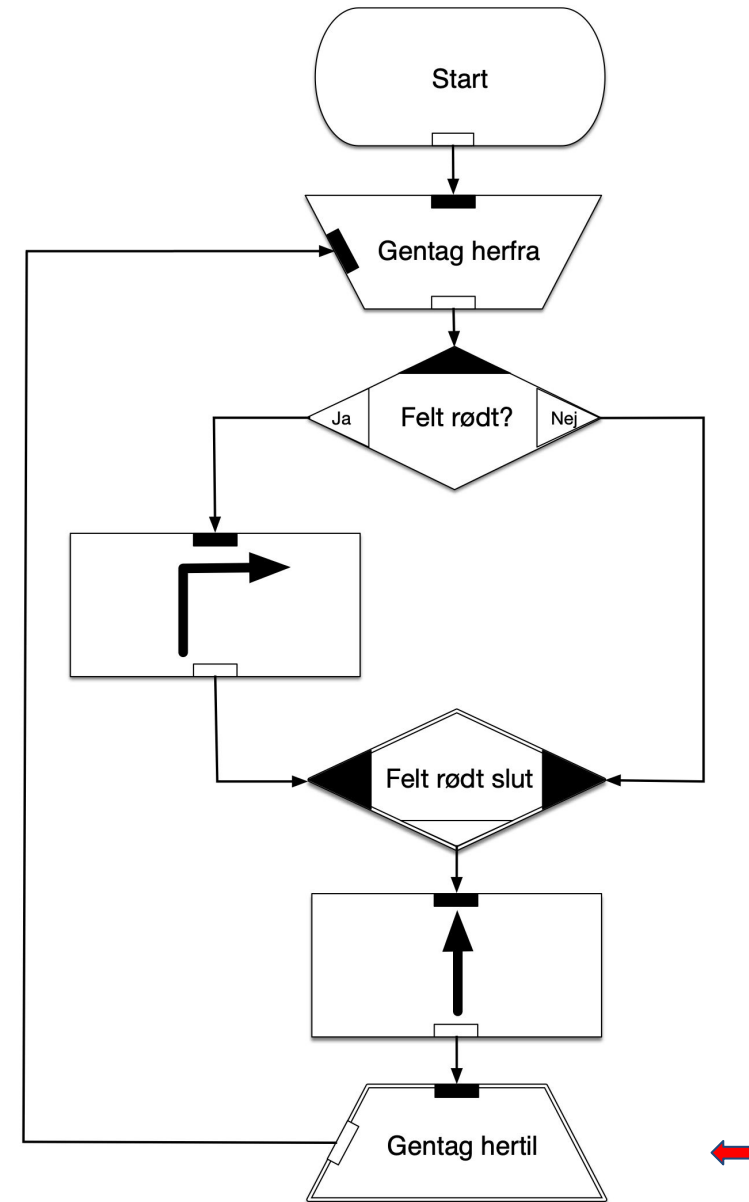
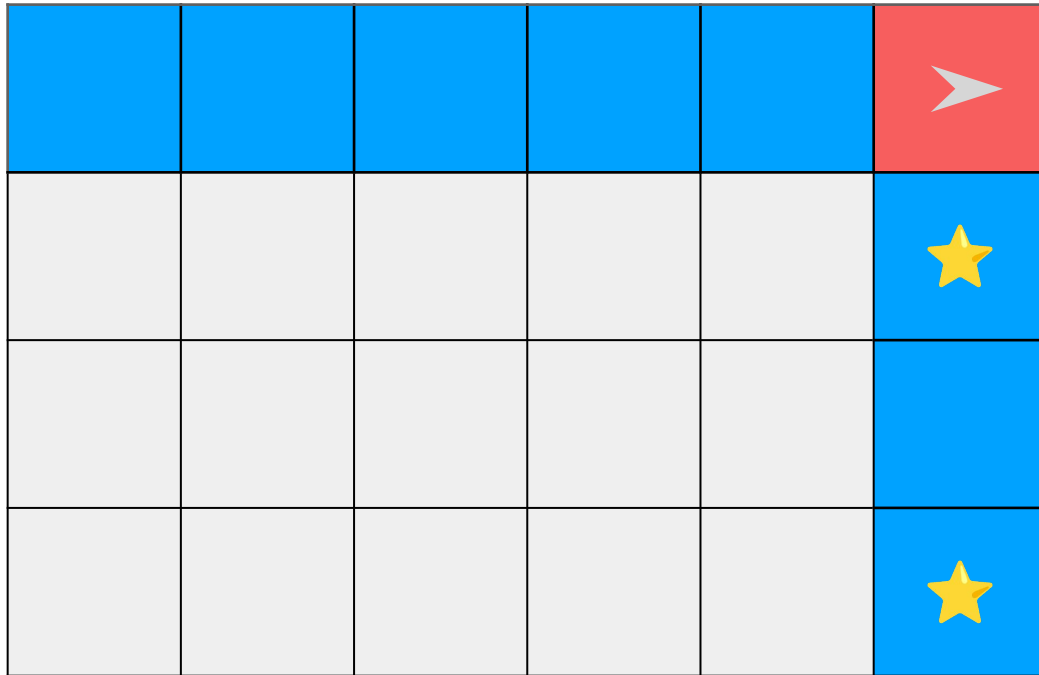
Program med løkker



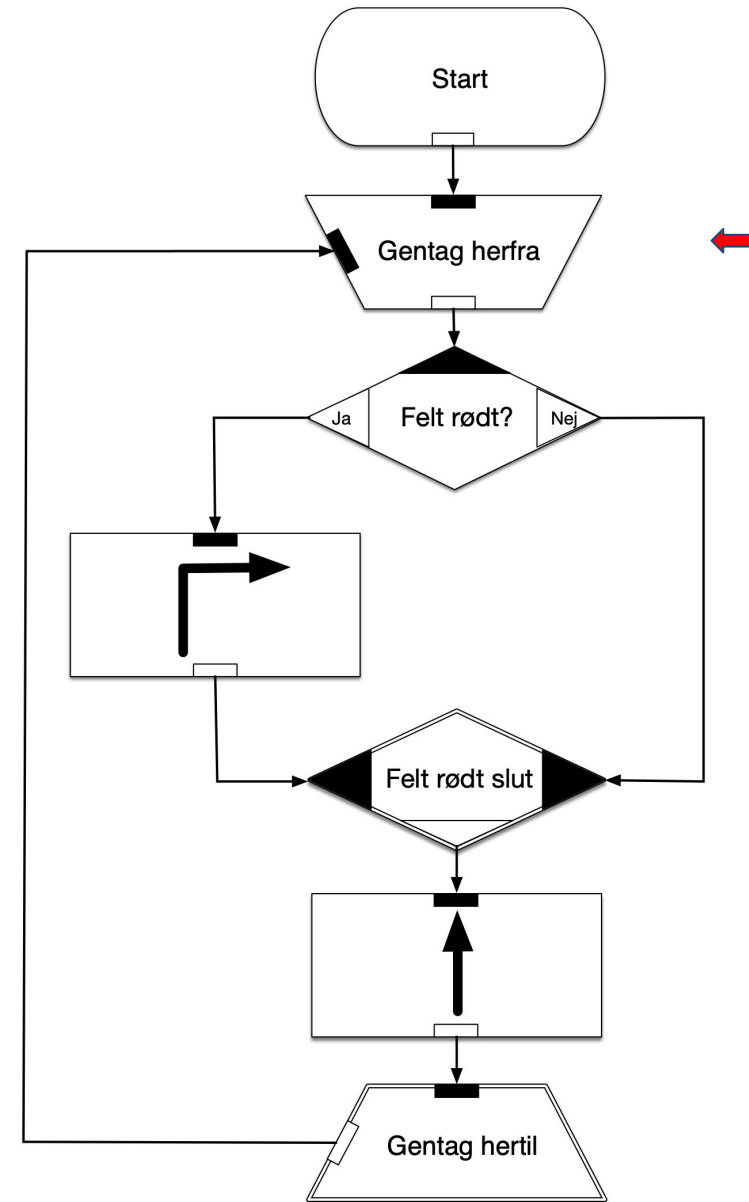
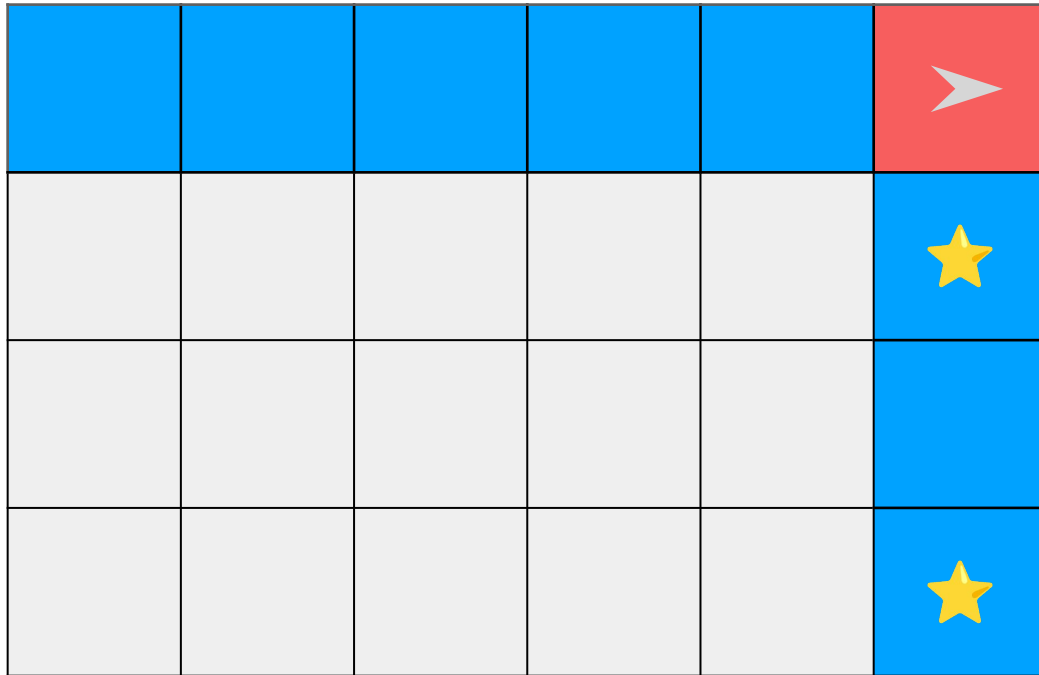
Program med løkker



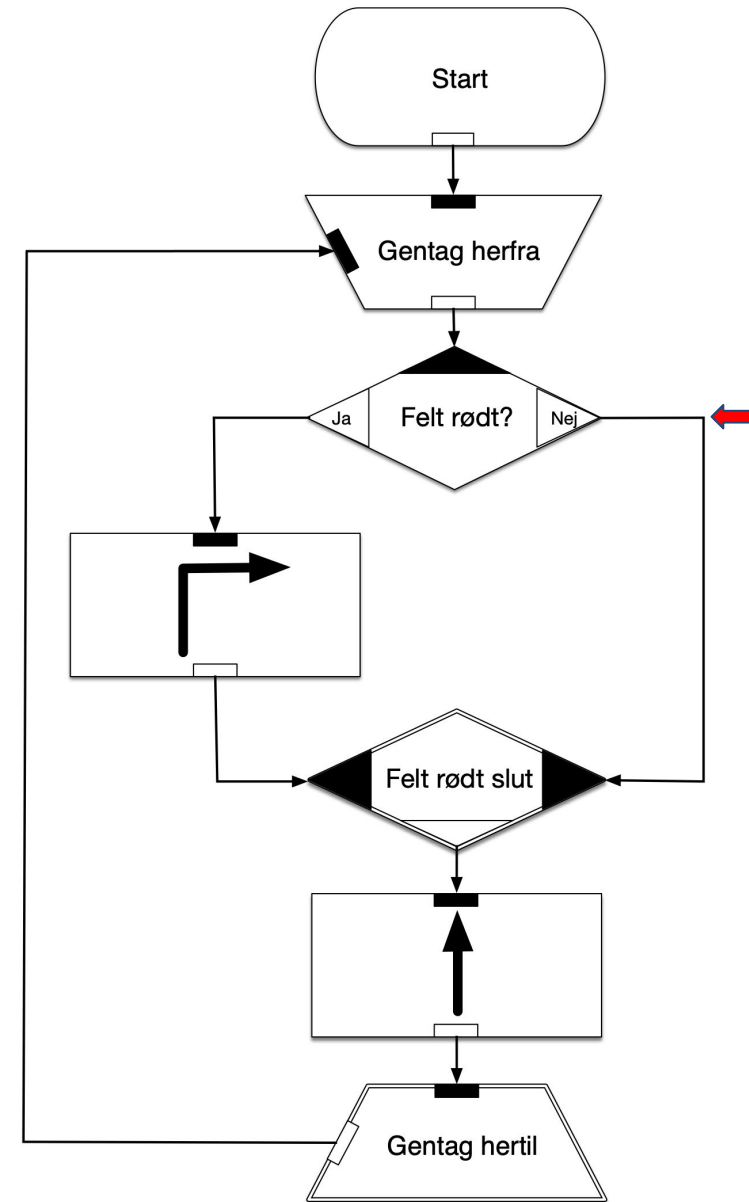
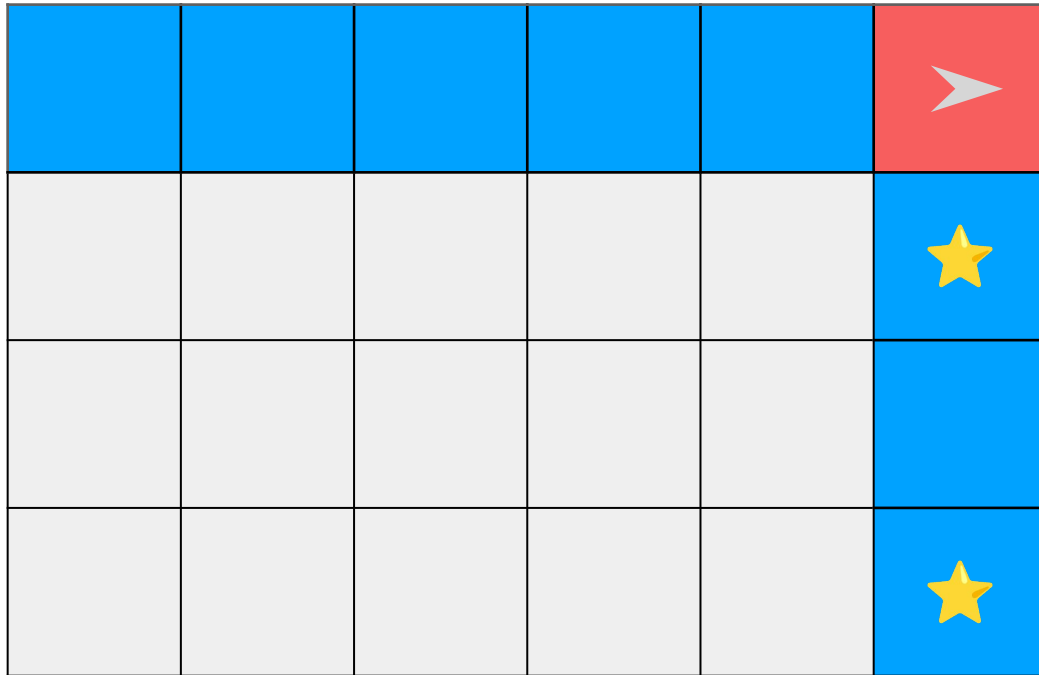
Program med løkker



Program med løkker

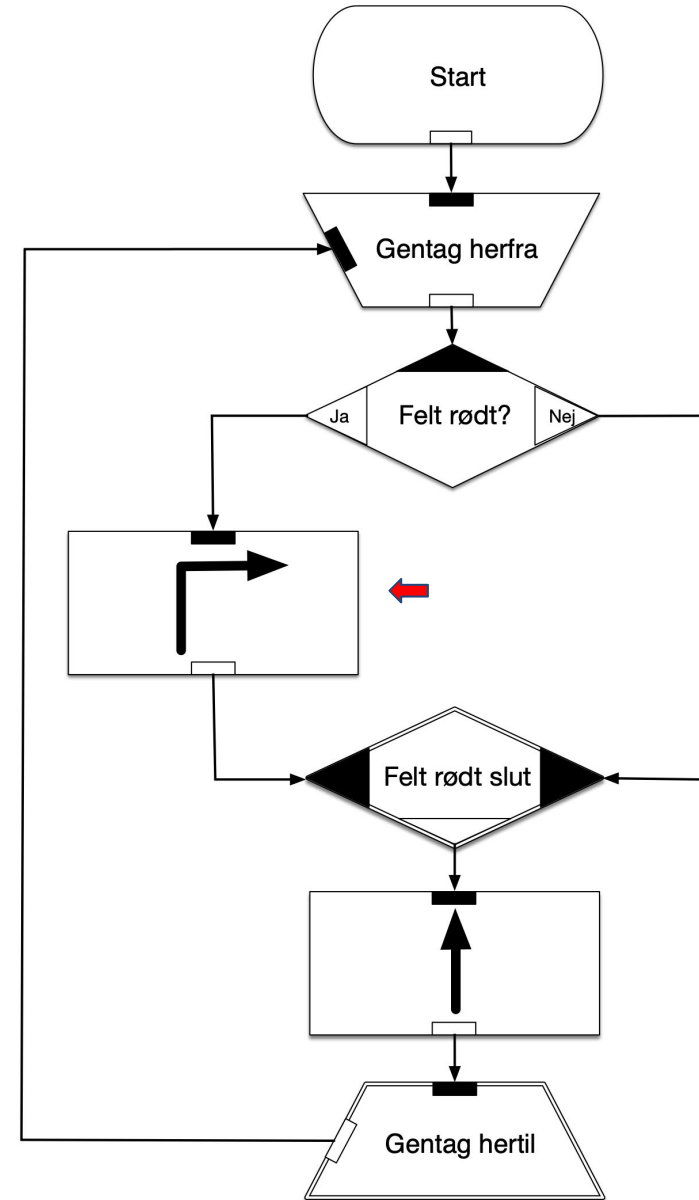


Program med løkker



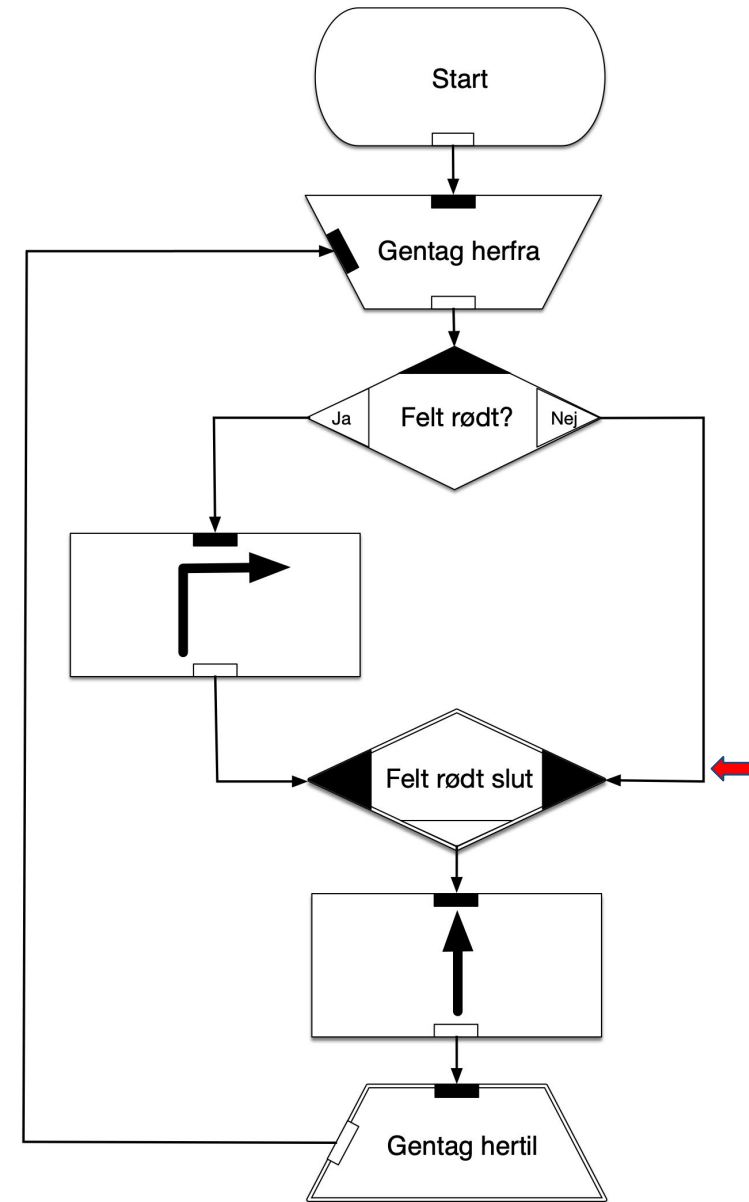
Program med løkker

					⬇
					★
					★

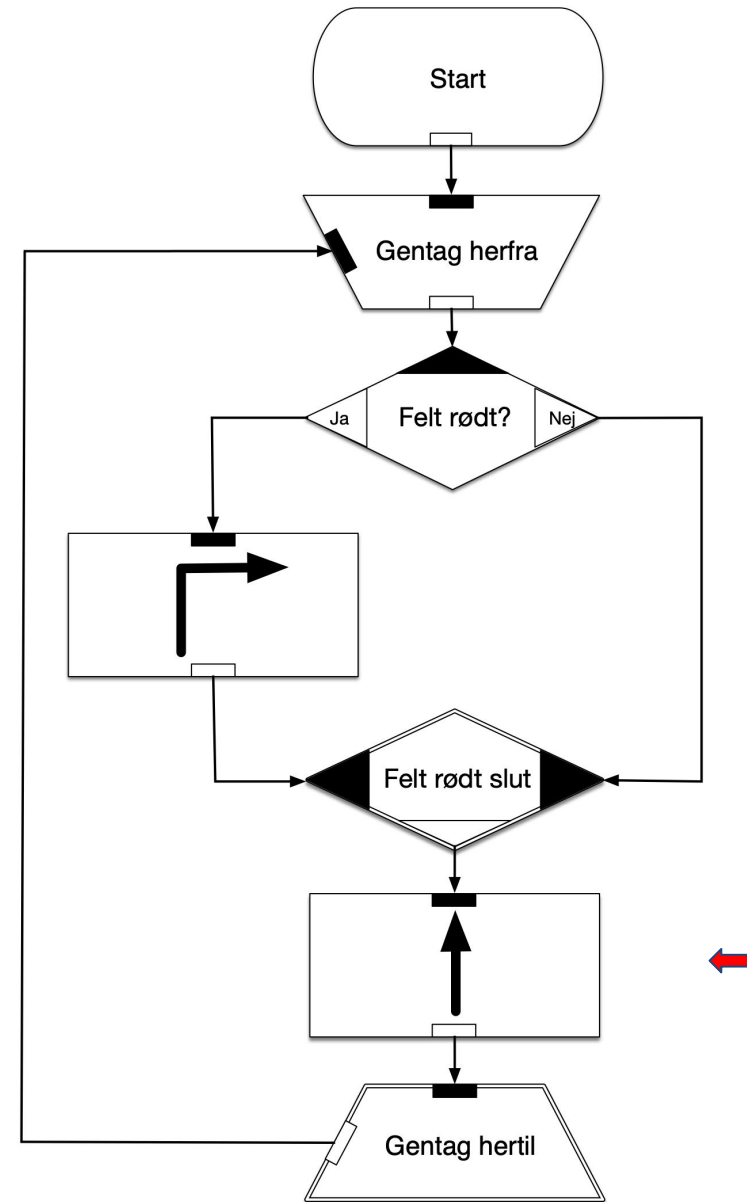


Program med løkker

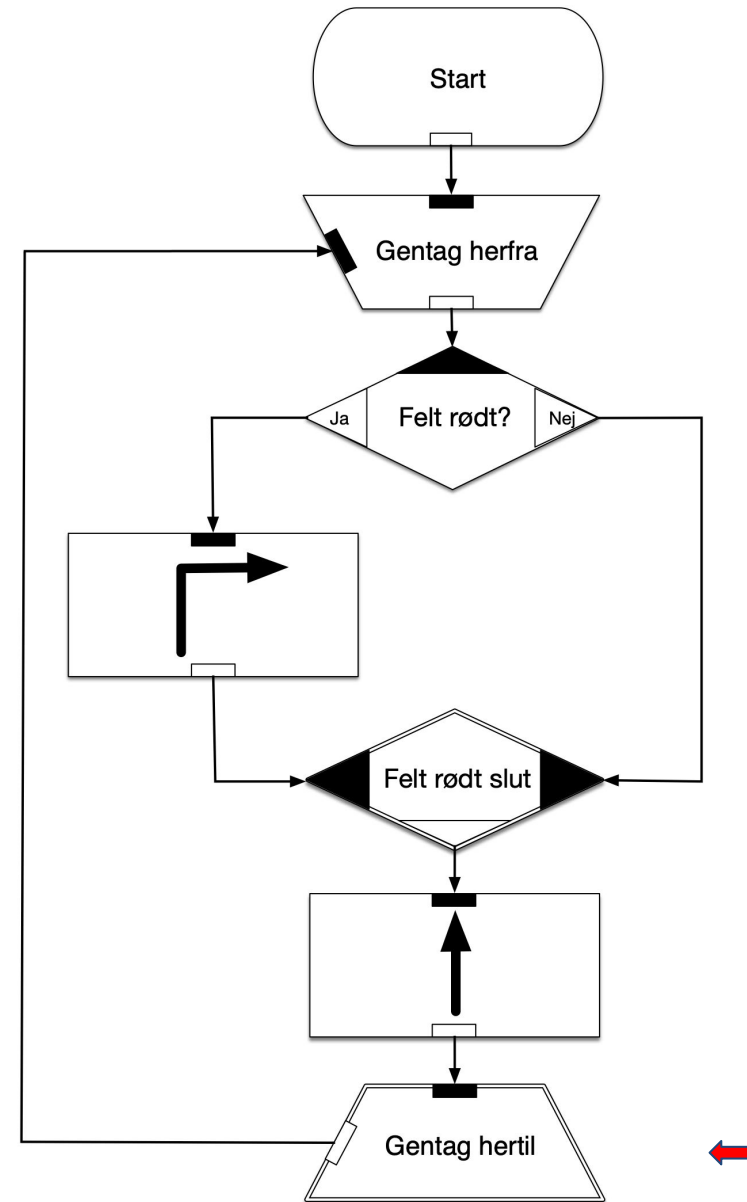
					⬇
					★
					★



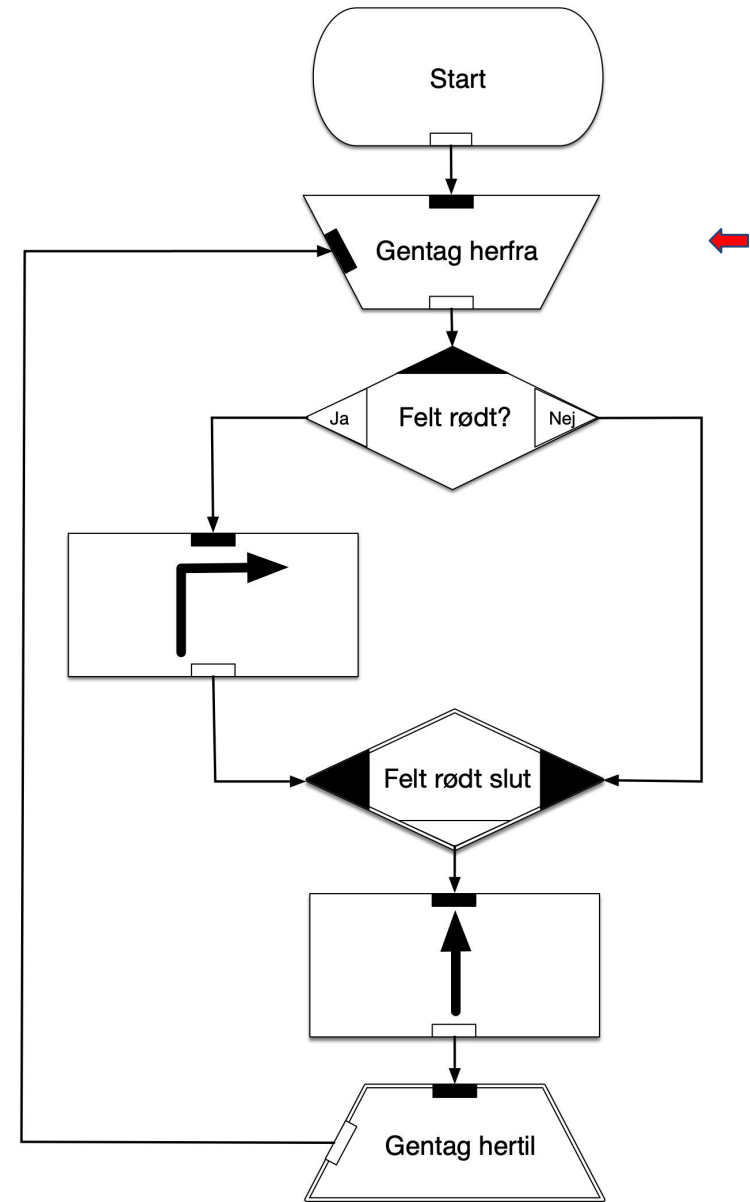
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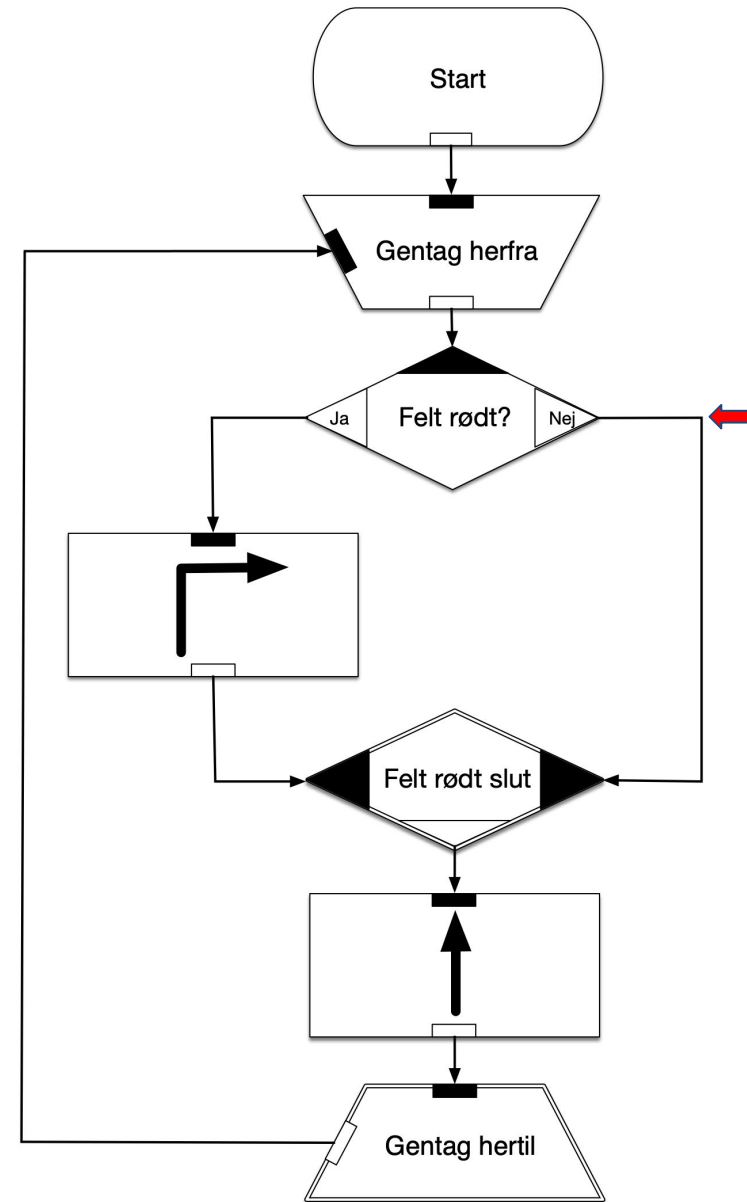
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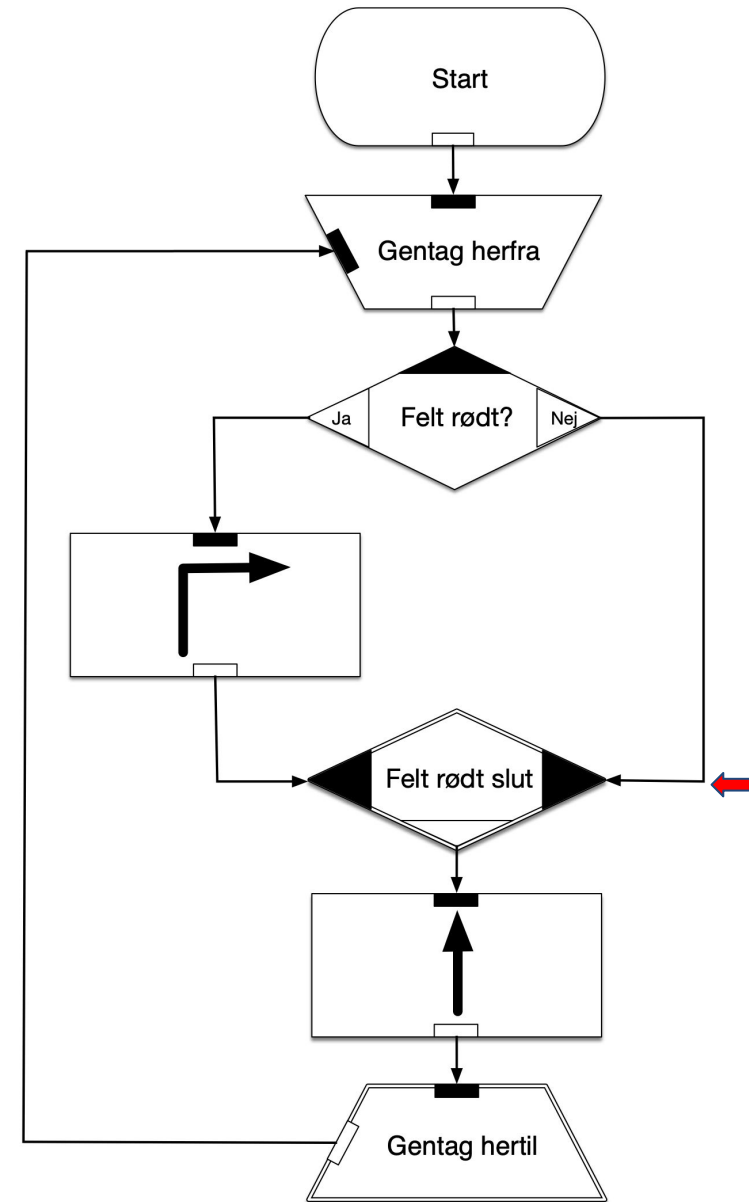
Program med løkker



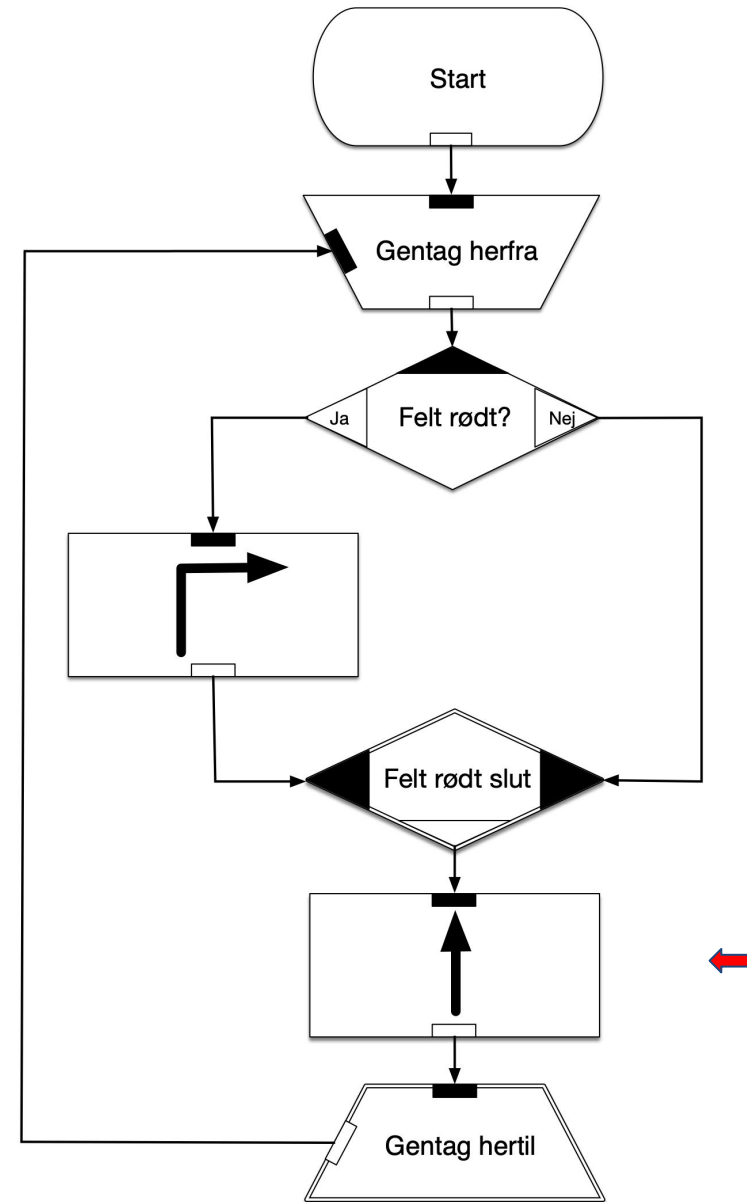
Program med løkker



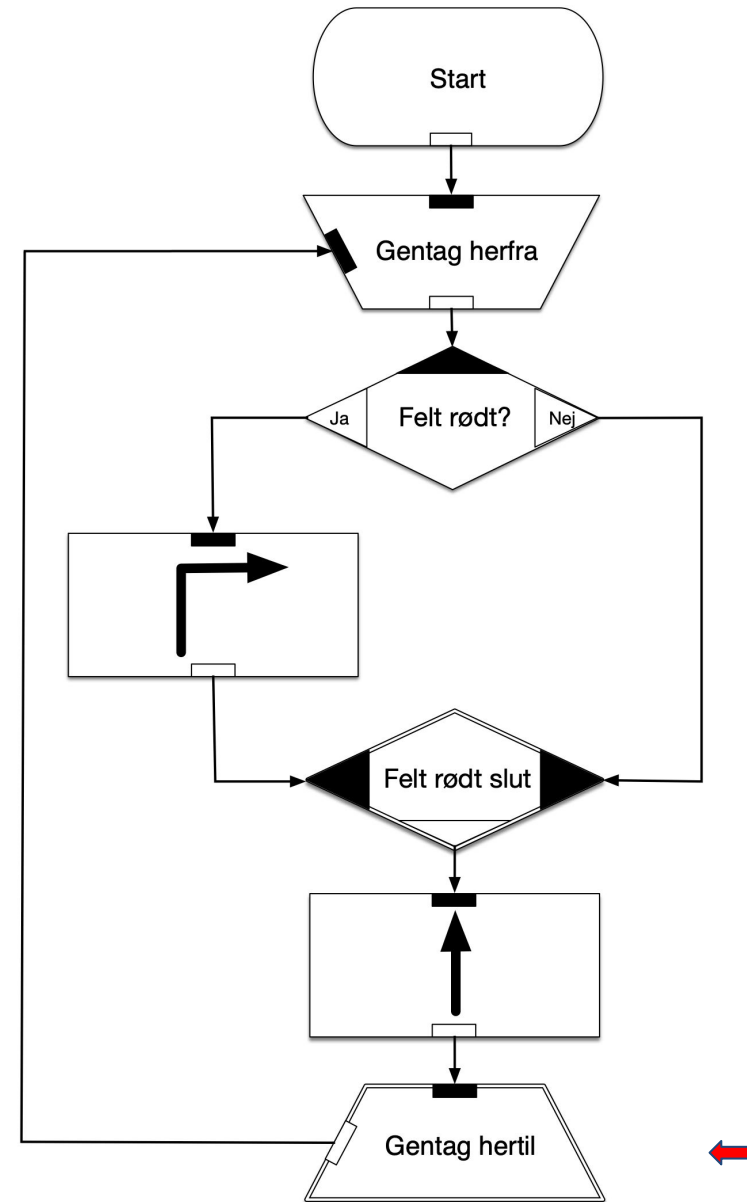
Program med løkker



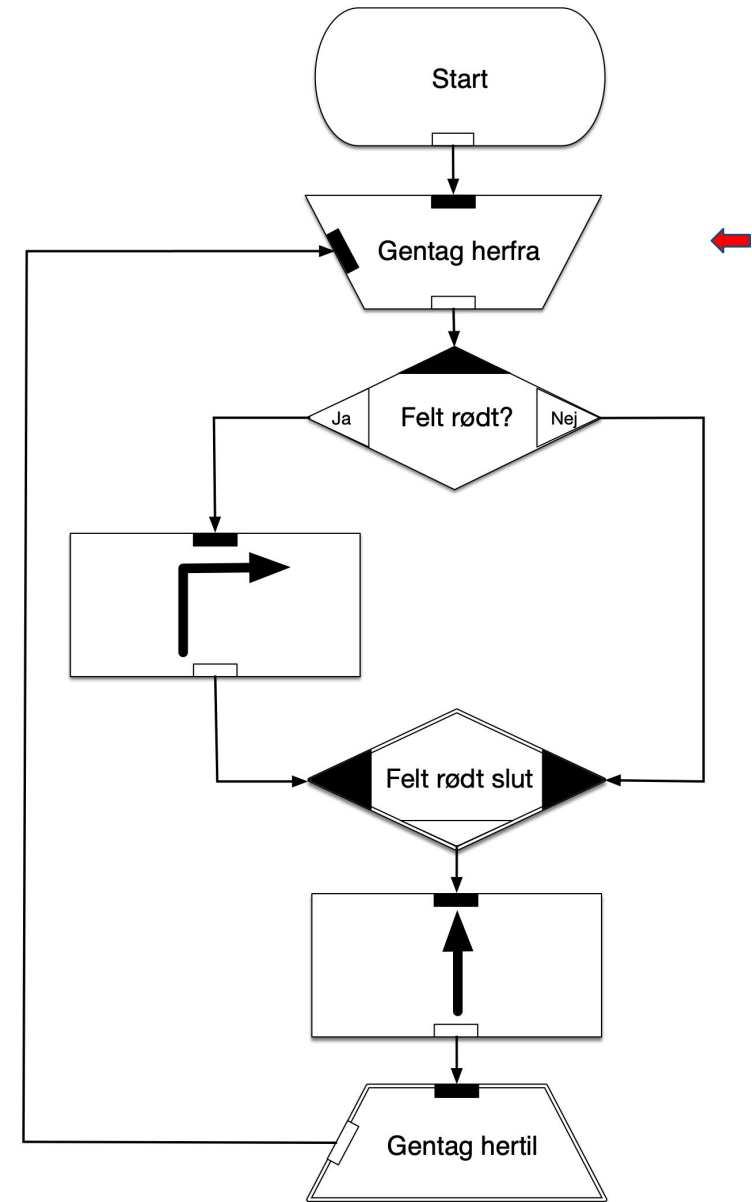
Program med løkker



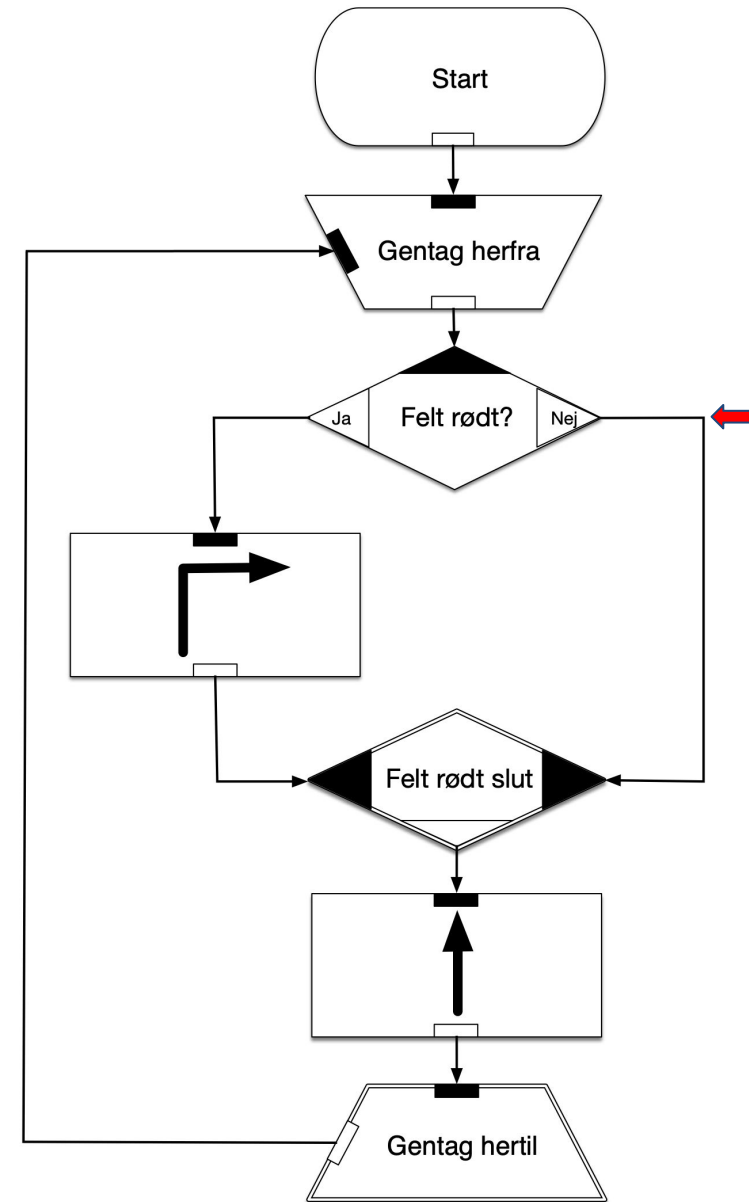
Program med løkker



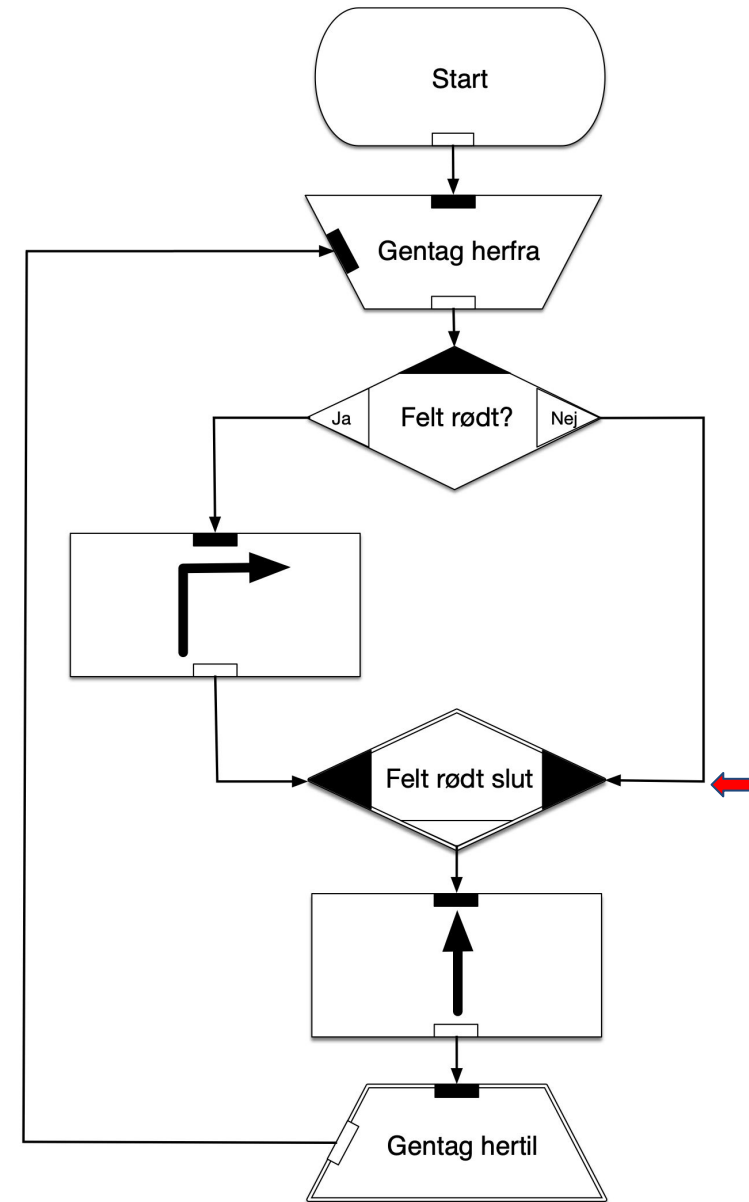
Program med løkker



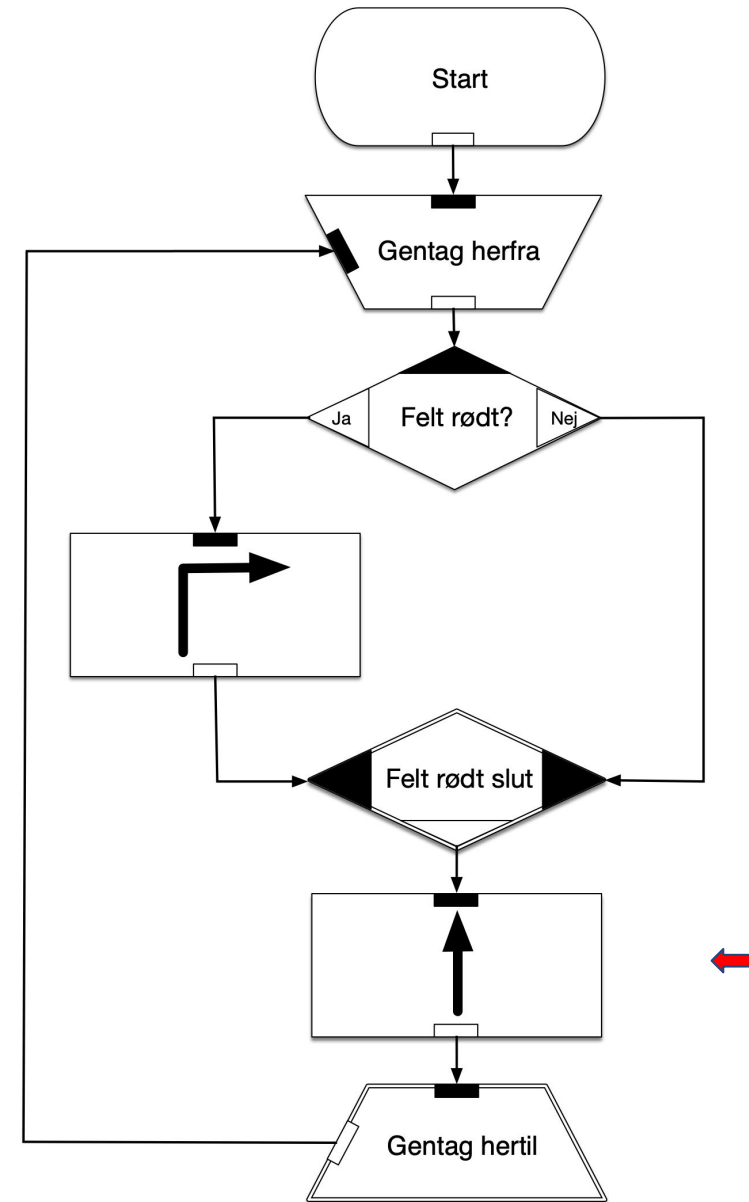
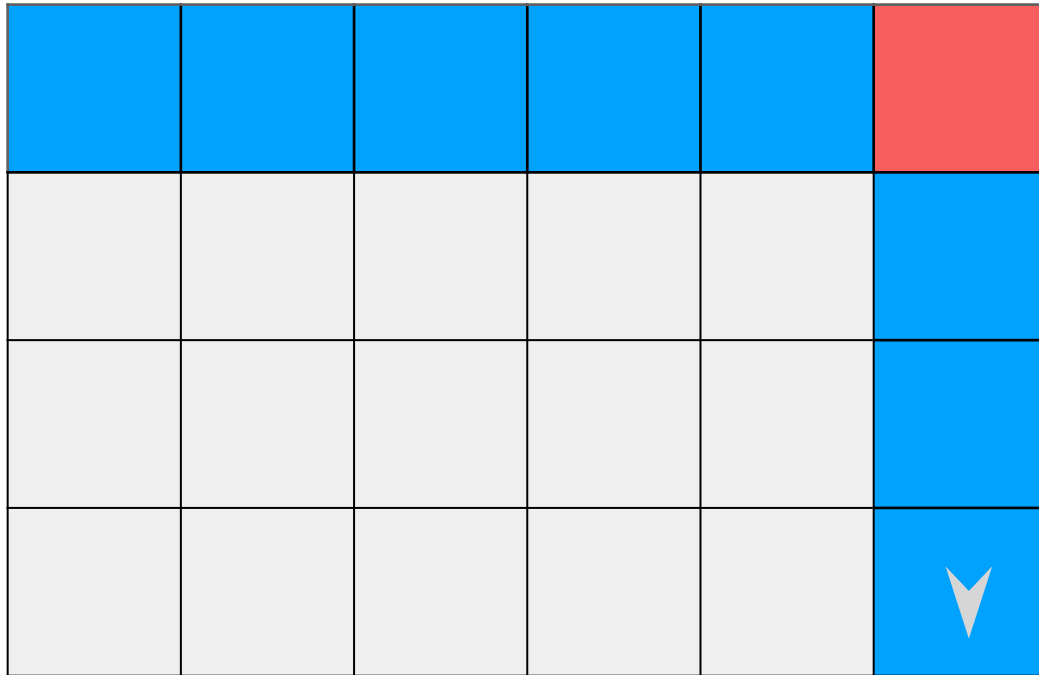
Program med løkker



Program med løkker



Program med løkker



Jeres tur

Oversættelse til kode

med Python

Kommandoer

`move()`

`turnLeft()`

`turnRight()`

`isRed()`

`isGreen()`

`isBlue()`

Kommandoer

`move()`

`turnLeft()`

`turnRight()`

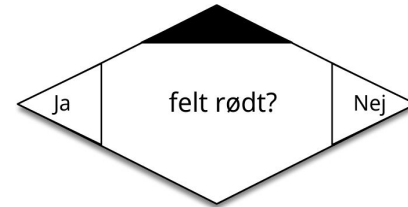
`isRed()`

`isGreen()`

`isBlue()`



```
while True:  
    <kode>
```



```
if isRed():  
    <kode>
```



```
(eller isGreen() / isBlue())
```

Kommandoer

`move()`

`turnLeft()`

`turnRight()`

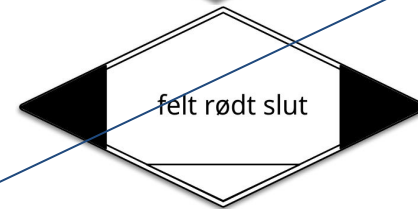
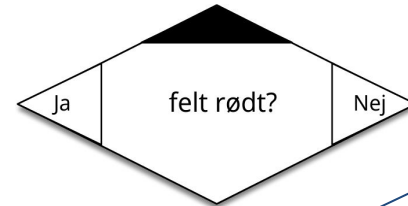
`isRed()`

`isGreen()`

`isBlue()`



```
while True:  
  <kode>
```



```
if isRed():  
  <kode>
```

(eller `isGreen()` / `isBlue()`)

fire mellemrum!

Kommandoer

`move()`

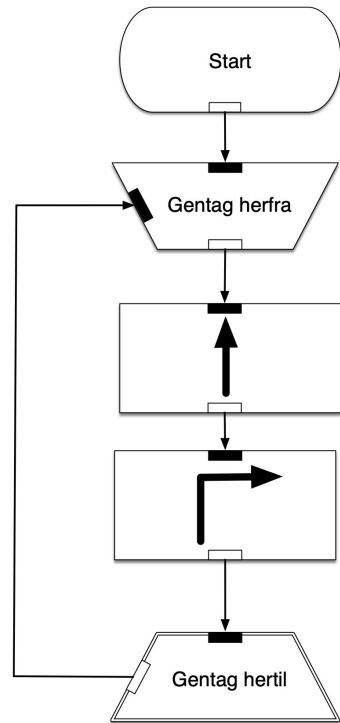
`turnLeft()`

`turnRight()`

`isRed()`

`isGreen()`

`isBlue()`



```
while True:
    move()
    turnRight()
```

Kommandoer

`move()`

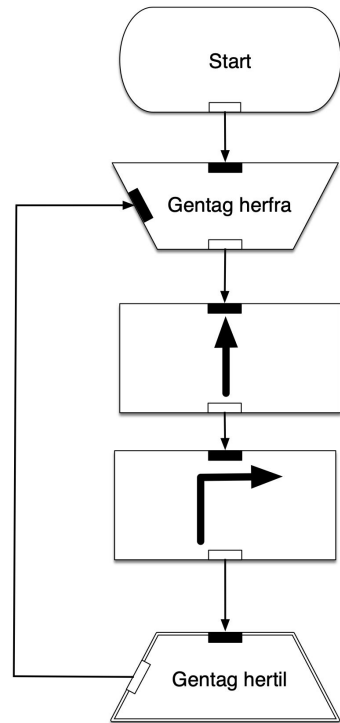
`turnLeft()`

`turnRight()`

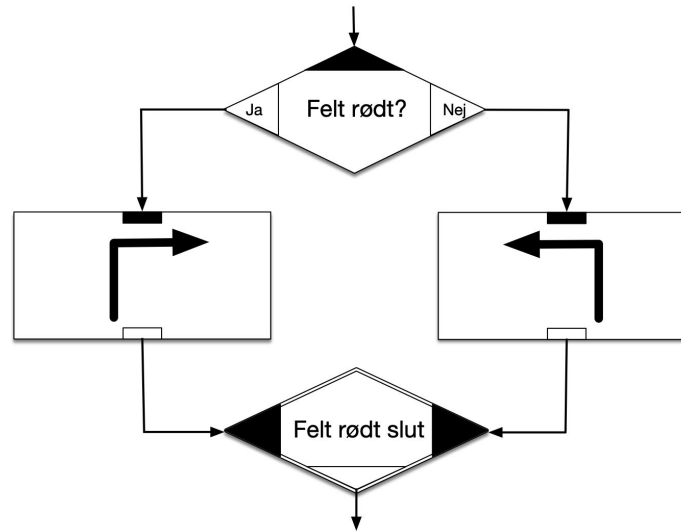
`isRed()`

`isGreen()`

`isBlue()`



```
while True:
    move()
    turnRight()
```



```
if isRed():
    turnRight()
else:
    turnLeft()
```


Kommandoer

move()

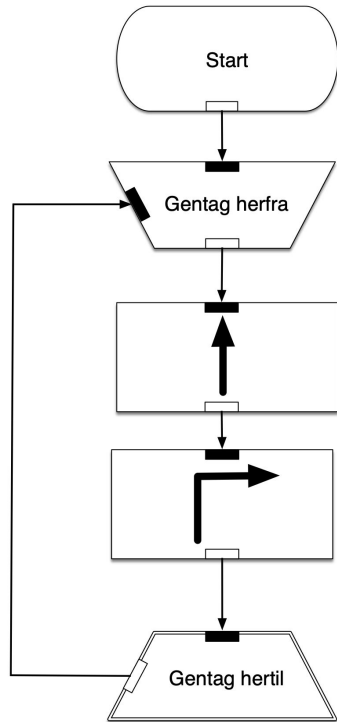
turnLeft()

turnRight()

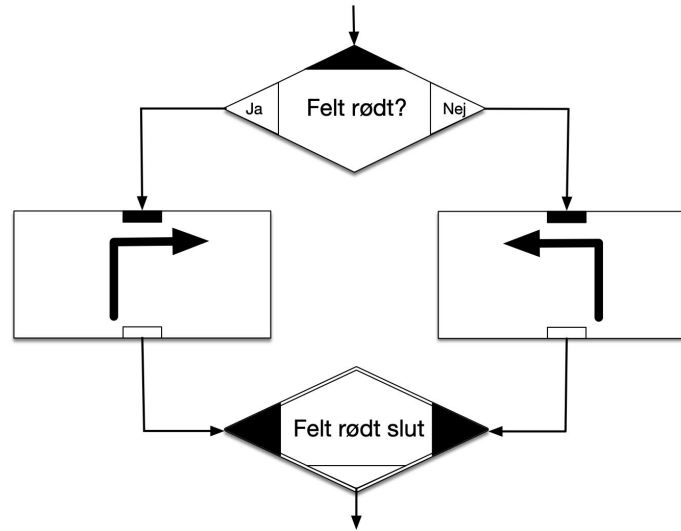
isRed()

isGreen()

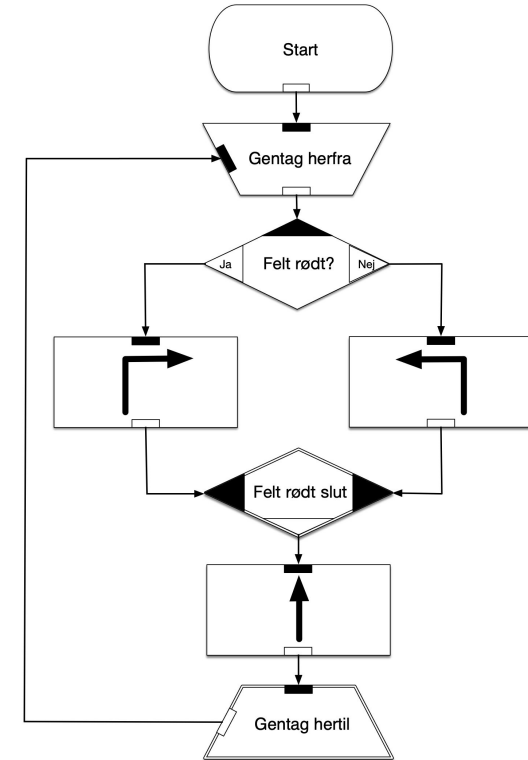
isBlue()



```
while True:
    move()
    turnRight()
```



```
if isRed():
    turnRight()
else:
    turnLeft()
```



```
while True:
    if isRed():
        turnRight()
    else:
        turnLeft()
    move()
```

Jeres tur:

tjstordalen.github.io/svinesti

move()

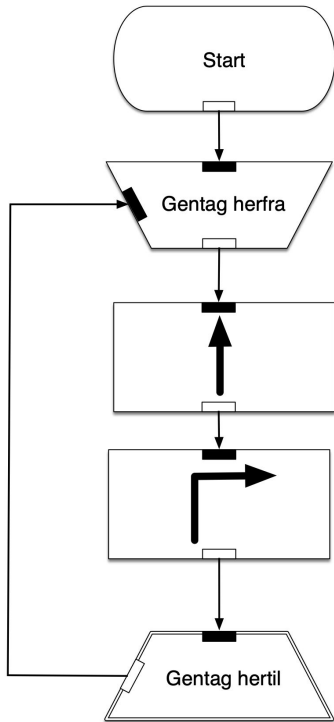
turnLeft()

turnRight()

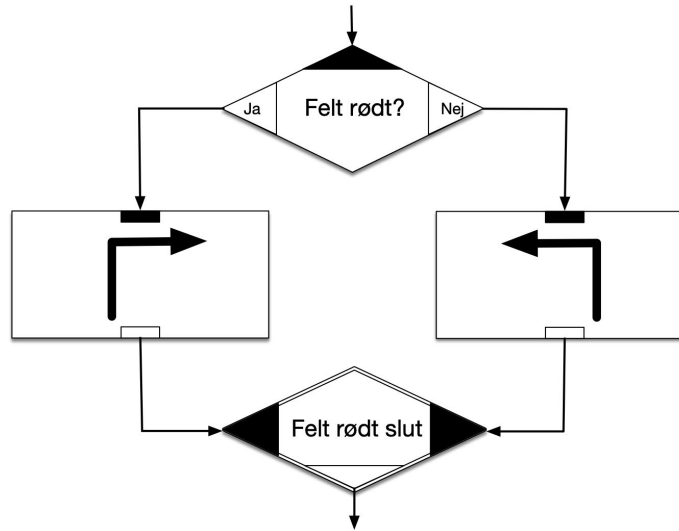
isRed()

isGreen()

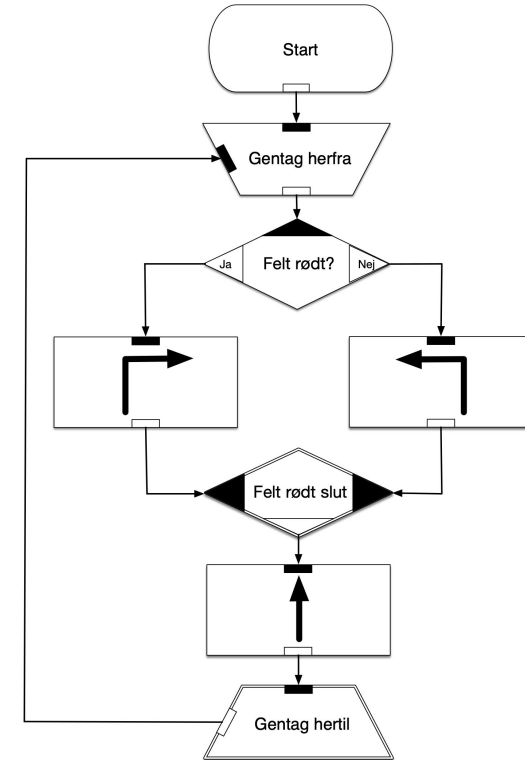
isBlue()



```
while True:
    move()
    turnRight()
```

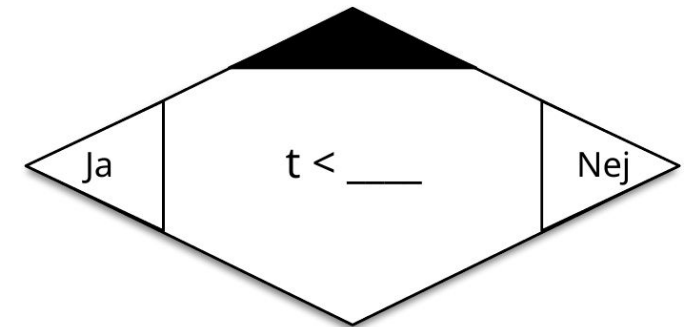
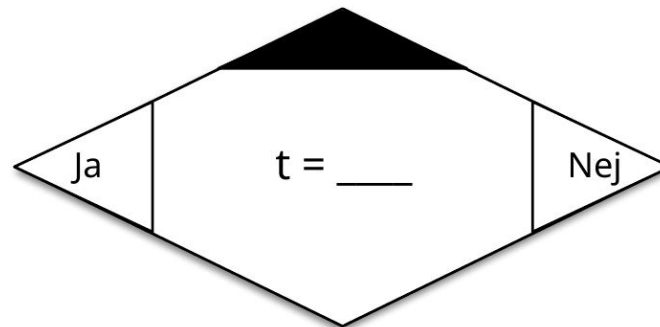
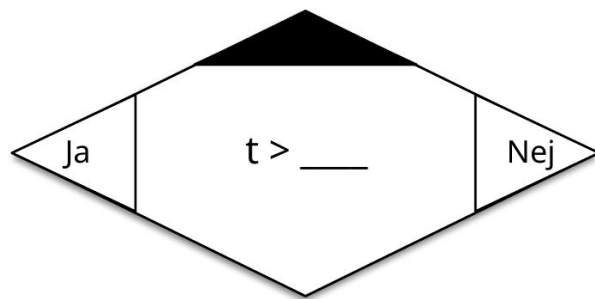
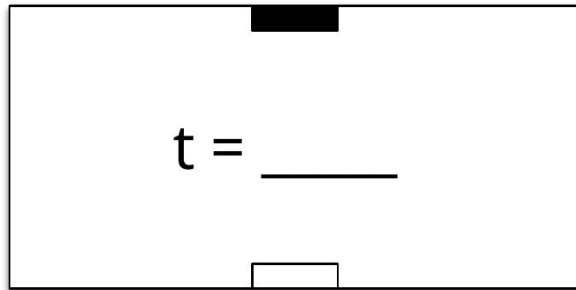


```
if isRed():
    turnRight()
else:
    turnLeft()
```

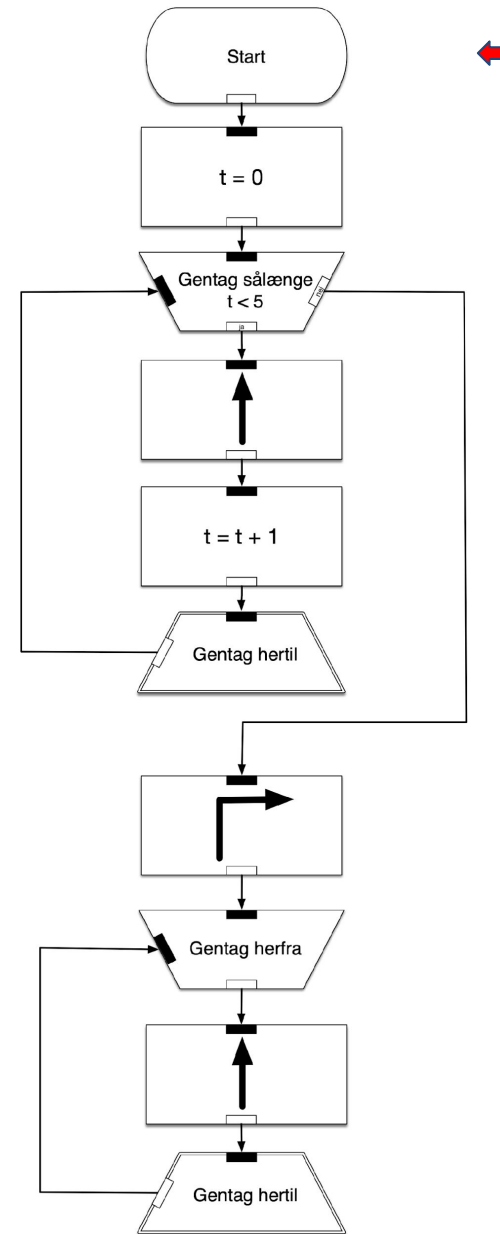
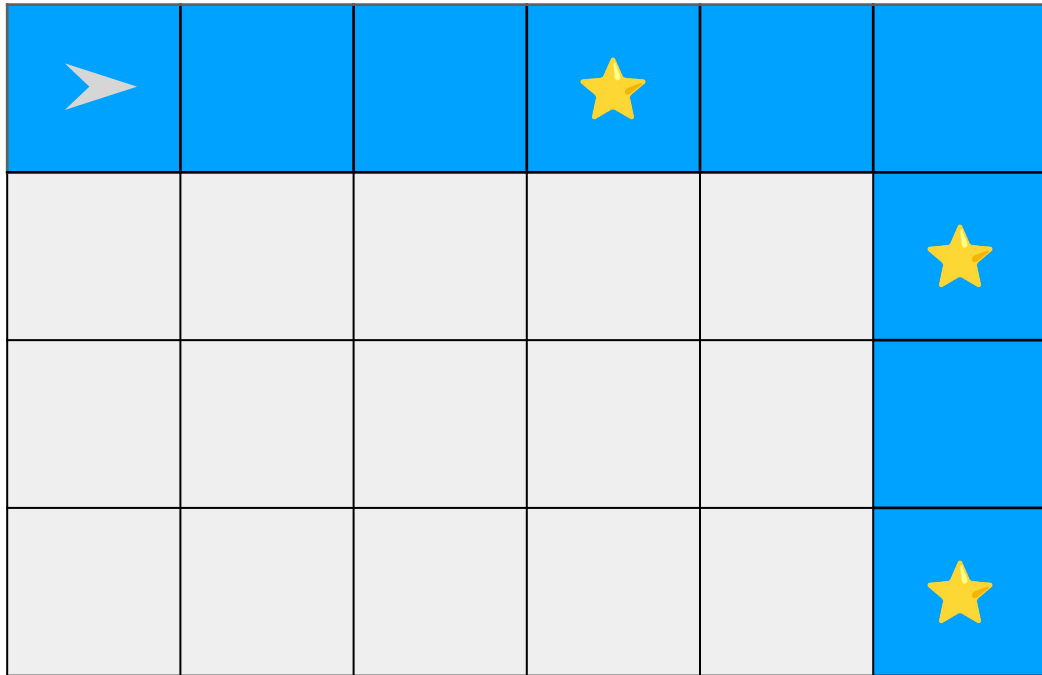


```
while True:
    if isRed():
        turnRight()
    else:
        turnLeft()
    move()
```

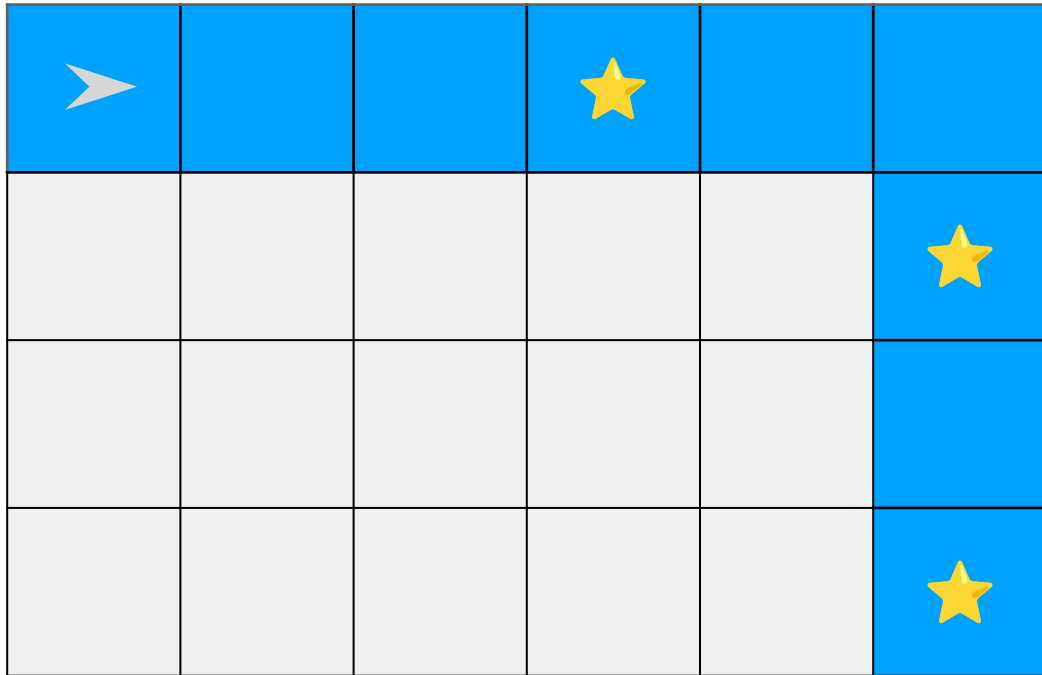
Tællere: løkker og betingelser



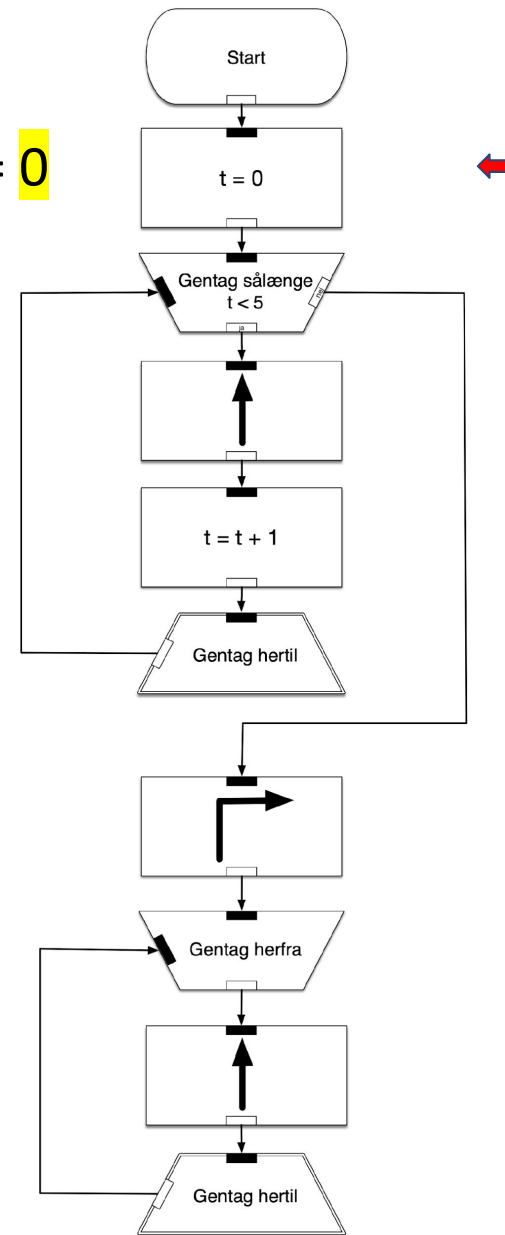
Tællere: eksempel



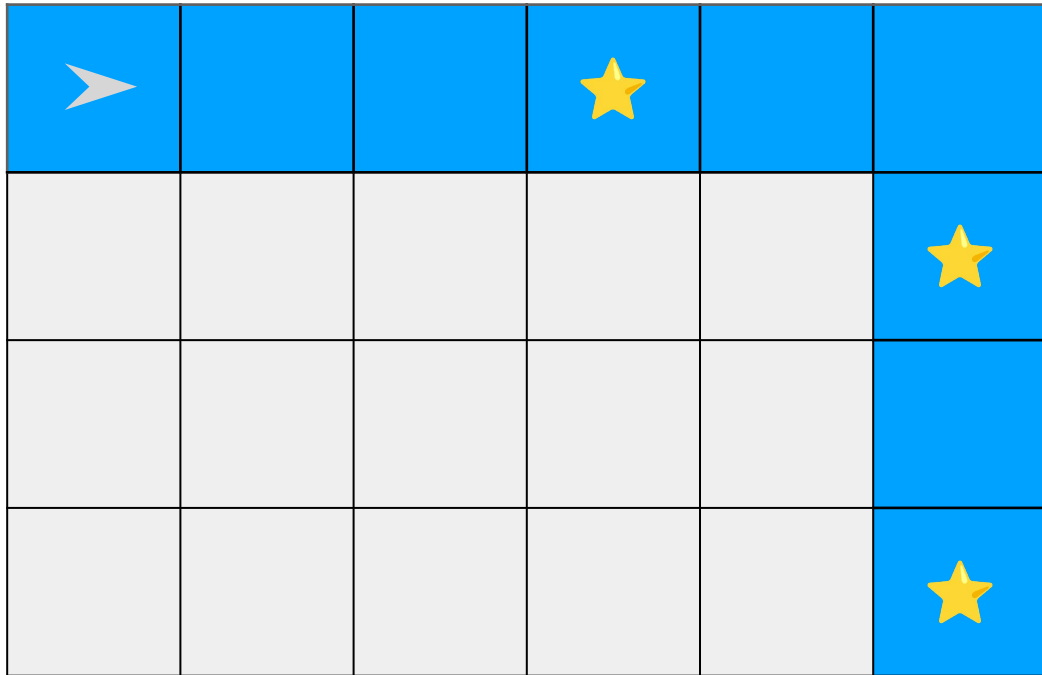
Tællere: eksempel



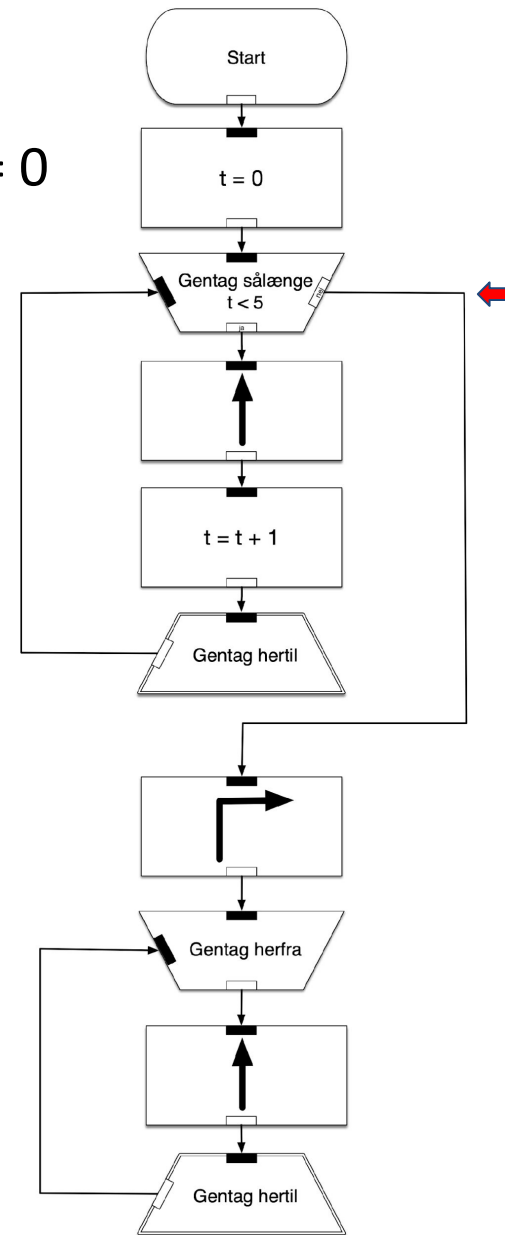
$t = 0$



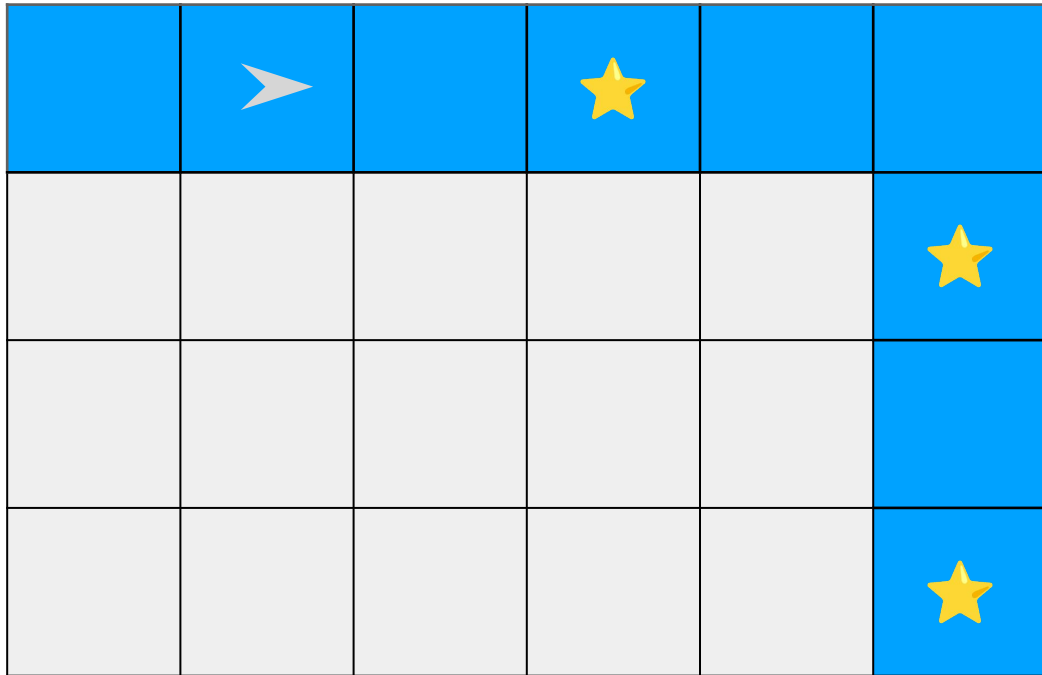
Tællere: eksempel



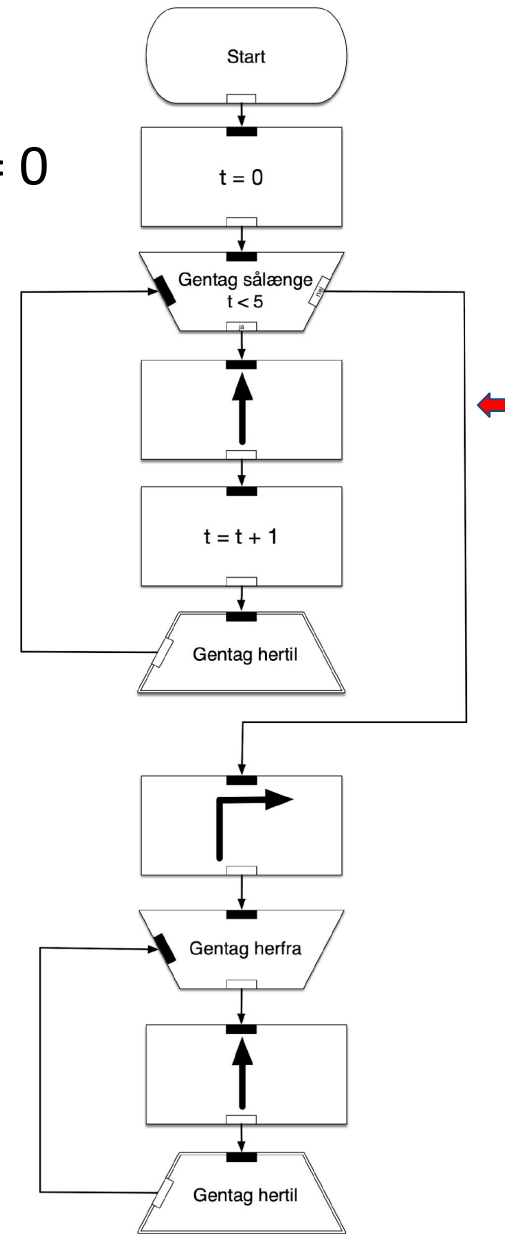
$t = 0$



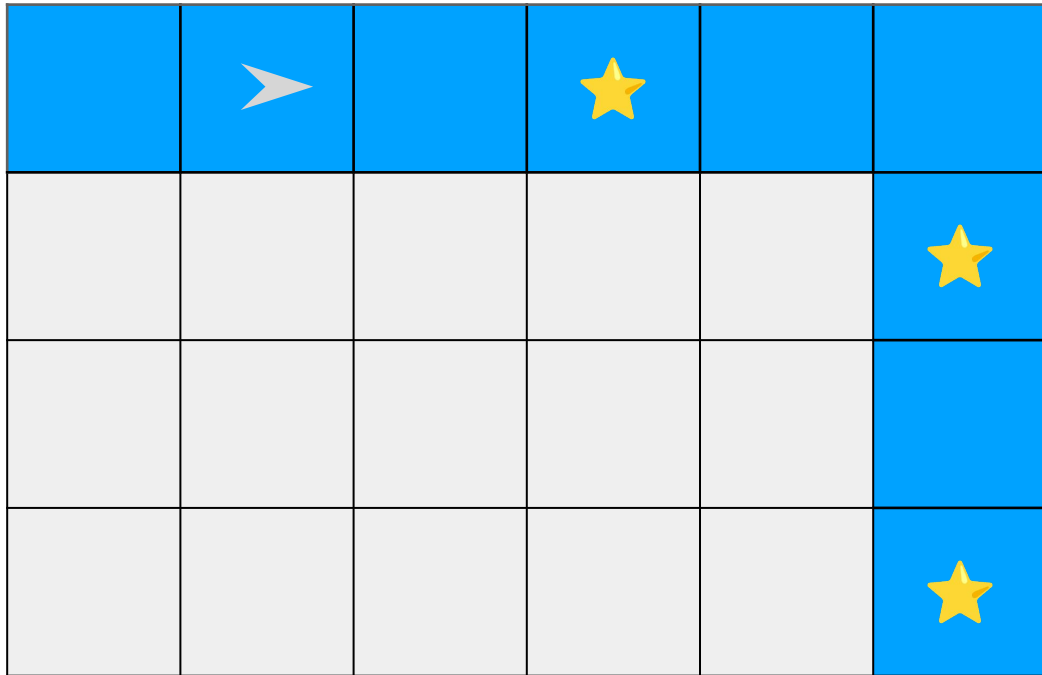
Tællere: eksempel



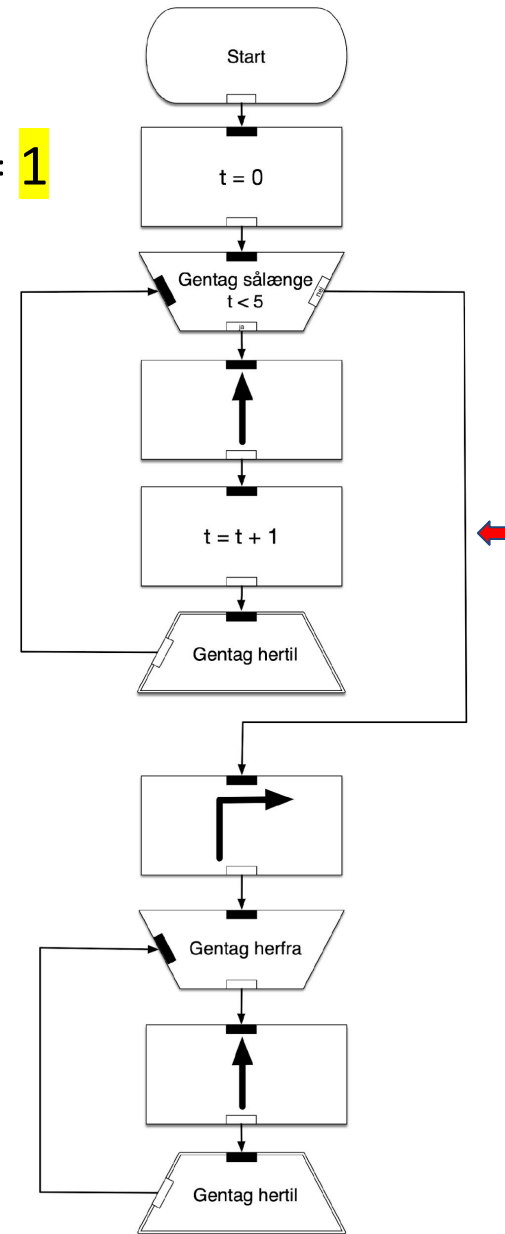
$t = 0$



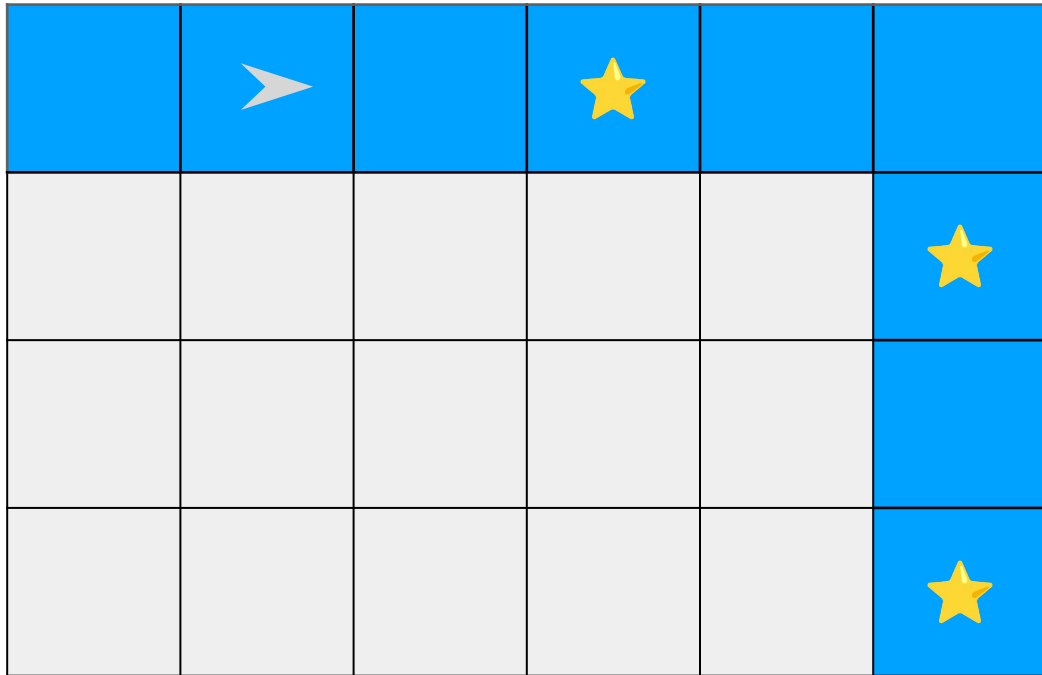
Tællere: eksempel



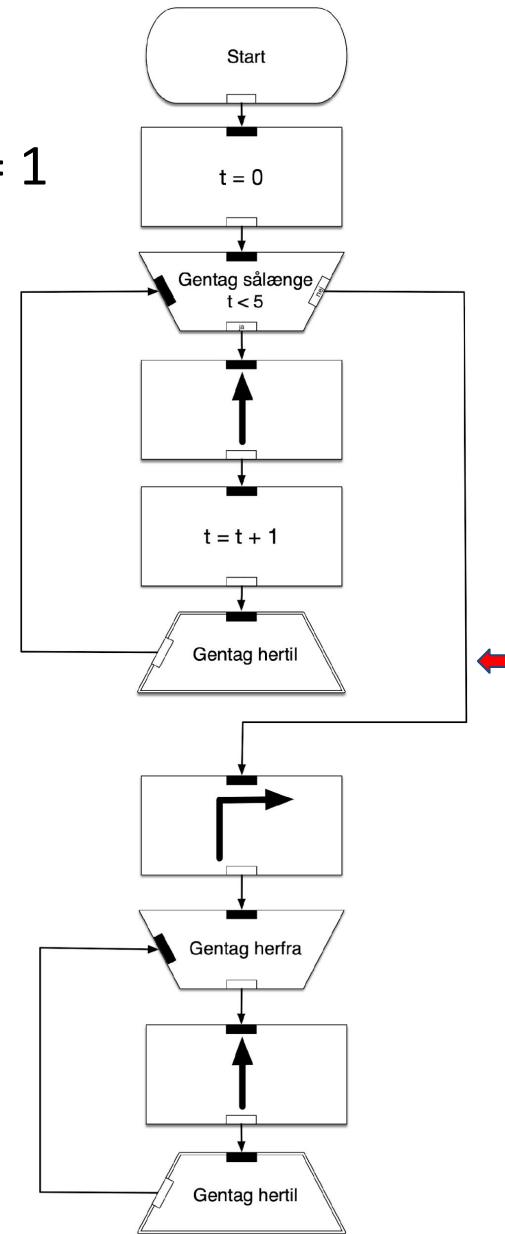
$t = 1$



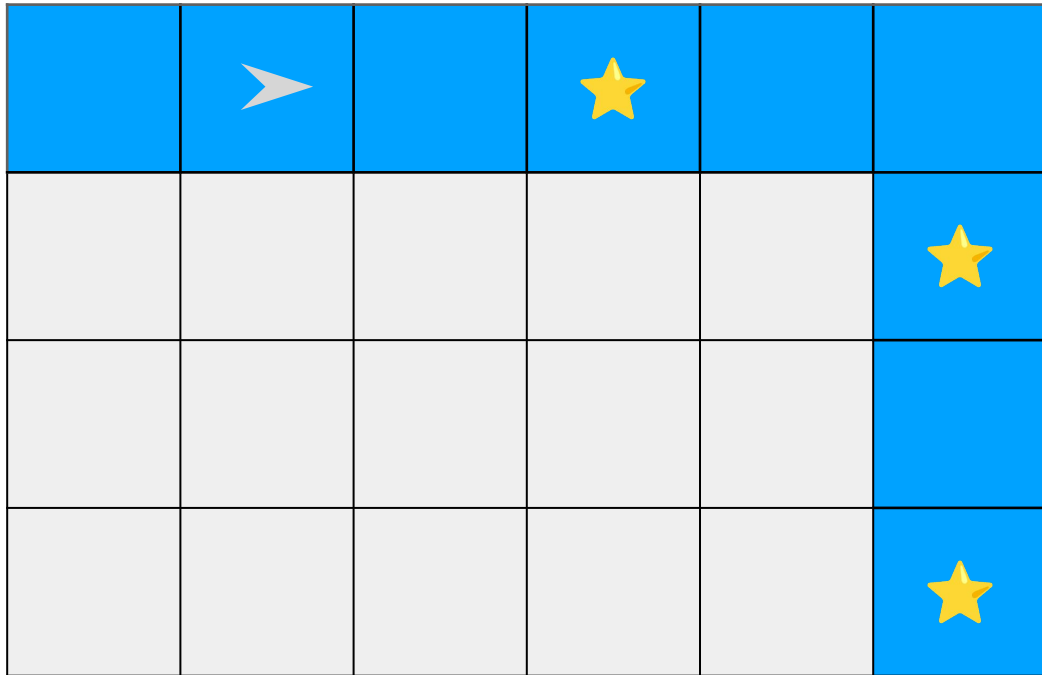
Tællere: eksempel



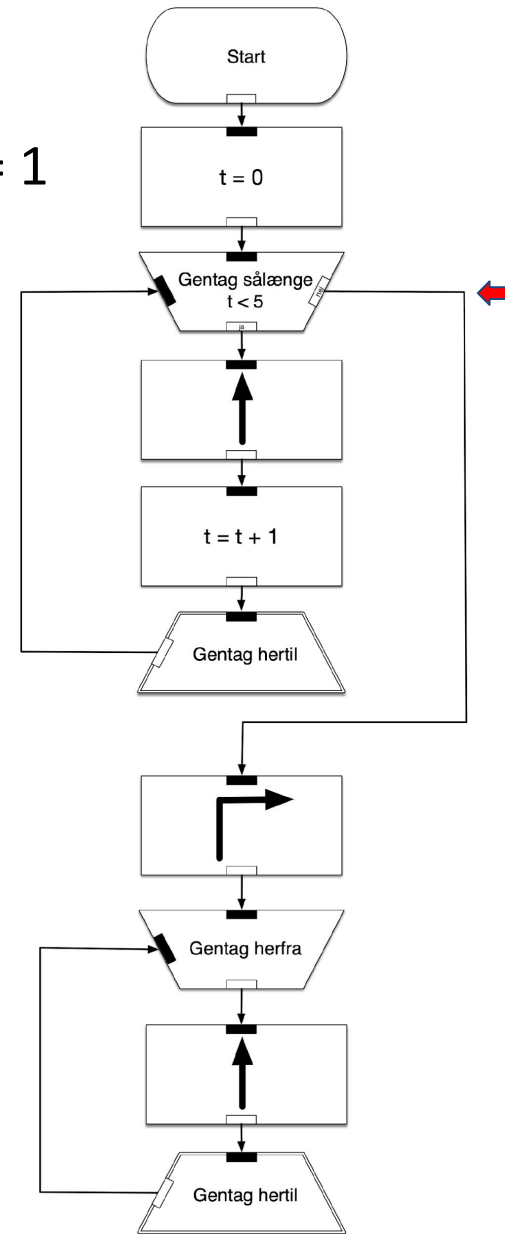
$t = 1$



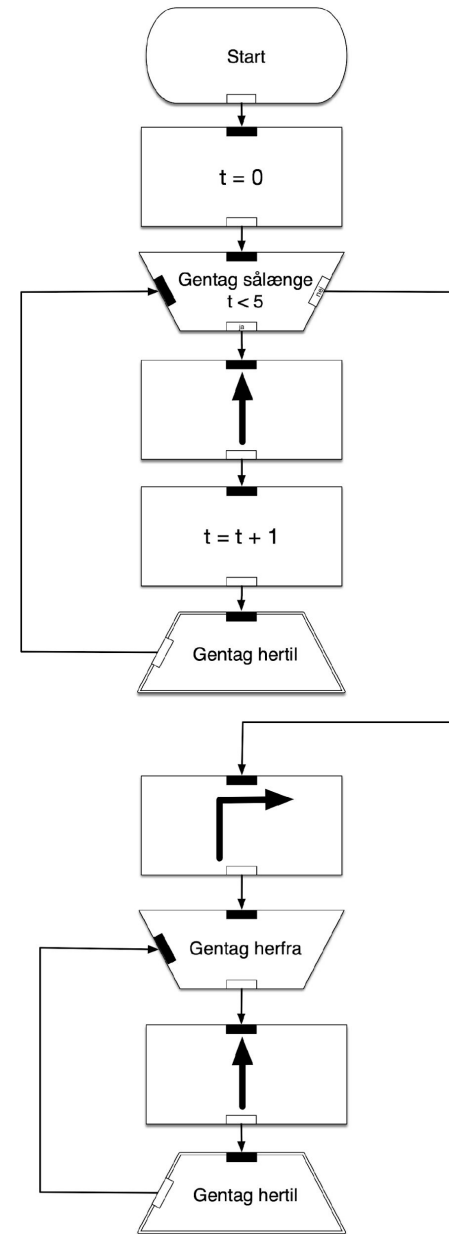
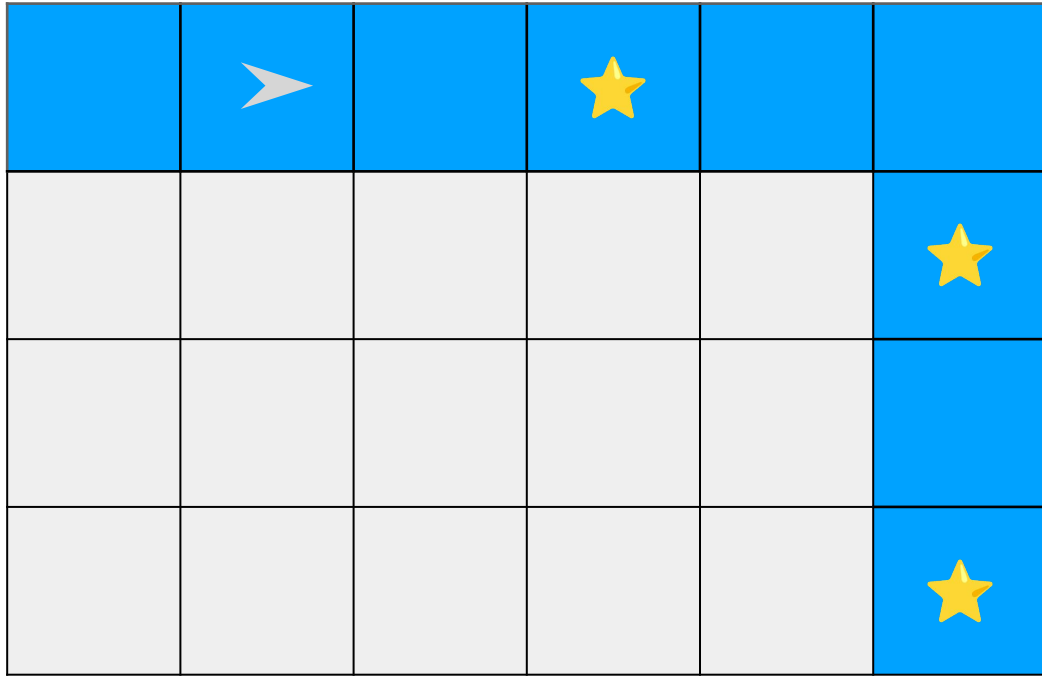
Tællere: eksempel



$t = 1$

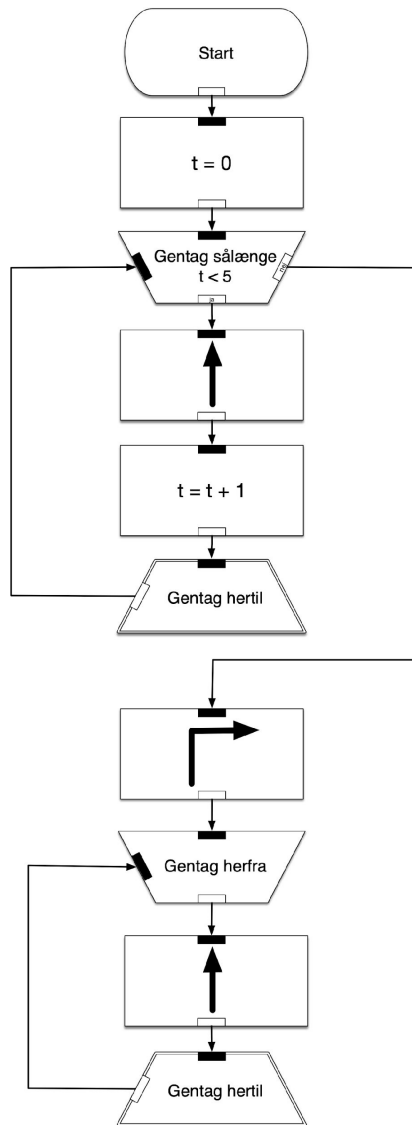


Tællere: eksempel



```
t = 0
while t < 5:
    move()
    t = t + 1
turnRight()
while True:
    move()
```

Eksempler:



```
t = 0
while t < 5:
    move()
    t = t + 1
turnRight()
while True:
    move()
```

```
a = 3
if a > 7:
    move()
    move()
    turnLeft()
else:
    turnRight()
    move()
```

```
b = 3
if b == 1:
    move()
if b == 2:
    turnLeft()
    move()
if b == 3:
    turnRight()
    move()
```

Mærk:

et lighedstegn - vi SÆTTER verdien af b
to lighedstegn - vi TJEKKER verdien af b